Smart Home Security System in Honeywell

Group 6

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Project Executive Summary

The primary objective of this project was to simulate a corporate research and development cycle for Honeywell, targeting the Smart Home Security market. The team aimed to identify market gaps and develop a feasible product concept that integrates hardware sensors with software automation to address the critical societal need for reliable, false-alarm-free home security.

Phase I Company and Industry Analysis

- The team selected Honeywell based on its strong capabilities in hardware-software integration and its existing sensor portfolio. Using Porter's Six Forces and Competitive Strategy Matrices, the team analyzed the Building Automation industry.
- The analysis determined that Honeywell employs a Differentiated Strategy, competing in a market characterized by intense rivalry against major incumbents but low threats of new entrants due to high barriers.
- The team concluded that Honeywell is positioned to succeed by developing a "Smart Home Security System" that utilizes its scale and technological expertise to differentiate through reliability and integration rather than low cost.

Phase II House of Quality

- The team reverse-engineered three products: SimpliSafe Smart Lock, Ring Alarm Pro, and Ecobee Smart Security, using Function Analysis System Technique (FAST) to map the functional logic of these competitors.
- Using the benchmarks from the dissection, we established customer requirements and technical metrics using a House of Quality. The House of Quality analysis revealed to us that **Wireless Signal Reliability** and **Sensor Actuation Accuracy** were the highest-priority technical metrics (Target: <1% packet loss, 2cm actuation). As such, false alarm minimization was identified as the critical competitive advantage.
- The team concluded that maximizing sensor accuracy and wireless reliability were the most critical engineering characteristics required to support the proposed system.

Phase III: Conceptual Design

- The final phase focused on generating and selecting product concepts using a Function Structure and Morphological Matrix.
- Six alternative concepts were developed by systematically combining different solution principles, such as varying sensor types and actuation methods.
- A weighted Utility Function was applied to evaluate concepts against criteria, including critical Safety, core performance, and stability.
- Concept 6, the "Emergency Ready Lock," was selected as the optimal design, achieving the highest utility rating. This concept features MEMS vibration sensors for anti-tamper detection, stepper motor actuation, and redundant power systems, paired with a mobile application for geofencing and real-time monitoring.

Project Proposal: Customer Needs Identification

1. Project Proposal Overview

In this report, our group outlines the strategic product development initiative that we undertook. Acting as a product development team within a large high-tech ecosystem, our goal is to identify gaps/necessities in the current market where a combination of hardware and software integration can help solve critical societal needs. This proposal outlines our target company selection process, the methodology for generating hybrid product concepts, and the final ranking of these concepts based on feasibility and impact.

2. Objectives

The primary goal of this project is to begin simulating a corporate research & development cycle by adhering to the following objectives:

- Identify three high-tech US-based companies with the following constraints:
    - The company must earn at least $1 billion in revenue.
    - The company must have a hardware and software presence.
    - The company must be United States-based.
- Generate three new product ideas for each of these companies, for a total of nine ideas.
- Provide a societal or customer need for each idea.
- Rank all nine ideas in order of preference, where #1 is the highest-ranked idea and #9 is the lowest-ranked idea among the group.

3. Approach

Team Schedule & Organization

- To ensure effective and efficient team collaboration, the group established a weekly workflow. We established meeting times in-person on Tuesdays at 5:30 PM, with supplementary online meetings taking place over Zoom as needed. The team operated under the workflow of brainstorming ten eligible companies, applying constraints to select the final three, and individually conceiving concepts for products before collaboratively ranking them with an organized voting process.

Brainstorming Methods

- Our brainstorming process followed a four-step cycle:

- Each member researched potential companies and their technological focus areas.
- Individual product ideas were drafted to address customer or societal needs.
- The group discussed the feasibility and innovation potential of each idea.
- Reviewed ideas were consolidated into a single table and ranked accordingly.

4. Implementation of the Plan

Company Selection

- After brainstorming an initial list of ten potential candidates (as shown in the Appendix), the group discussed and voted on the final selection.
- We prioritized companies based on their familiarity with the group, innovation potential, and whether their creation of a hybrid product was realizable. The selected companies are as follows:
  - Honeywell
  - Johnson & Johnson
  - Zebra Technologies

Product Ideation & Ranking

- The team developed three hybrid concepts per company. These were ranked based on innovation, feasibility, and societal impact.

Company	Societal/Customer Need	Product Idea	Rank
Honeywell	To be used by labor workers. Things like heart rate, location, temperature, and air quality are monitored. Improves worker safety by having real-time monitoring of workers to ensure safety and accident response	Wearable device equipped with health and environmental sensors for labor workers	1
Honeywell	Many smart door locks contain too many features that interfere with their main purpose: reliably locking the door. This creates frustration and a lack of feeling safe for customers.	A smart door/window lock with built-in sensors that connects with a mobile app. It should alert homeowners when an entry point is disturbed. Also, allowing the option to contact law enforcement	2
Johnson & Johnson	In hospitals, post-operative infections often occur because wound healing is not continuously monitored between checkups.	A small, disposable patch that sticks near a wound and monitors for signs of infection, such as changes in temperature, pH, or oxygen levels. It	3

		connects to an app that sends alerts if something looks off, so patients can contact their doctor early instead of waiting until the next appointment.	
Johnson & Johnson	People who take multiple medications (like seniors or those with chronic diseases) often forget doses or take them at the wrong time, leading to health problems or hospital visits.	A smart pillbox that tracks when each compartment is opened and sends reminders through a connected app. The app can alert family members or caregivers if doses are missed. Over time, it uses simple analytics to show patterns and help users stay consistent with their medication routine.	4
Honeywell	Makes the use easier for elderly individuals. Most of the at-home products (eg, air purifier, fans, etc) should also be easy to set up and not require a professional. Any associated software should allow other trusted family members to remotely monitor.	Simplified smart-home devices with remote access for family members	5
Johnson & Johnson	After surgery or an injury, a lot of patients struggle to keep up with physical therapy at home. Many people don't know if they're doing the exercises correctly, which can delay recovery.	A wearable brace that tracks movement using built-in motion sensors. It connects to a mobile app that shows real-time feedback on posture and motion, helping users do exercises correctly. The data can also be shared with doctors or therapists, so they can check progress remotely and adjust recovery plans.	6
Zebra Technologies	Removes the need for labels. Further increases automation and reduces human error for shipping packages. Allows other people involved in logistics to print on the package while reducing environmental impact by cutting down on paper usage.	A thermal printable package surface that eliminates paper labels.	7
Zebra Technologies	Healthcare workers carry big tablets that contain the patient's medical information everywhere, which is bulky and unhygienic. Having a device that allows them to receive alerts about patients and any urgent notifications immediately. This	A smart wristband that healthcare workers wear lets them scan patient IDs and get urgent notifications immediately.	8

	can be a device that healthcare workers wear that lets them scan patient IDs and get notifications on it. This can potentially be faster, efficient, and safer.		
Zebra Technologies	Efficiently track vehicles on the road for data, vehicle violations, and more. It will increase safety among communities. A network of readers would use software to communicate with each other.	Smart license plate readers that can communicate through Zebra software networks. A network of readers will allow efficient tracking of vehicles.	9

5. Conclusion of Project Proposal

- During the initial discussions, we considered the most well-known companies to have a smaller scope for innovation. We came up with our ideas by putting ourselves in the shoes of the consumer and based our proposals on what societal needs we felt should be addressed.
- After spending some time drafting up ideas for initiatives based around the most well-known companies, we found that there was more room for innovations for companies that were not as well-known, or companies that mainly focused on multiple niches, rather than one overarching product that addressed a singular issue.
- Our focus was on companies with the capability to create products that integrate both software and hardware, especially beyond the scope of Silicon Valley. We reasoned that Silicon Valley companies built their success through specialization in one market, whereas the companies we selected target a wide variety of markets and address a range of issues.
- From this, we were able to conclude that the companies that focused on multiple niches/markets typically have greater potential to expand their product line due to their widespread reach and variety. With more room to innovate, the companies that we selected would realistically go forward with the initiatives we proposed to expand their product line.

Project Phase I: Company and Industry Analysis

1. Phase I Overview

The initial phase of work is focused on executing a detailed **Company Analysis** and **Industry Analysis** for **Honeywell**. The project's primary objectives are to streamline team operations by agreeing upon a shared process and clearly identifying individual team member roles. The key deliverables resulting from this analysis include an *Enhanced Company Analysis Report* and an *Enhanced Industry Analysis Report*.

2. Process Development

- The pre-meeting discussion was critical for establishing a unified approach to ensure seamless integration of individual research.
- Before our meeting, the team agreed on the following structured template for individual contributions. This ensures consistency and facilitates a unified, evidence-based analysis by structuring the research mindset
    - Template:
        - Identify the problem statement.
        - Identify the resources available to solve the problem statement.
        - Hypothesis about the industry structure and company strengths.
        - Write key metrics and data collected.
        - Verify the accuracy of the analysis performed.
        - Summarize the results into a concise paragraph.
- The questions to be answered for the **Company Analysis** that will help us understand Honeywell's strategies are:
    - Vision Statement: What is Honeywell's stated long-term purpose?
    - Mission Statement: What is its scope (product, market, technology)?
    - Business goals: What are the financial and operational targets (growth, margin, recurring revenue)?
    - Competitive strategy: What is the overall competitive approach (Cost, Differentiation, or Focus)? How does it create value?
    - Technology strategy: What are the core technologies that give the company a technological advantage over its competitors? What is the company's approach to product and technology development?
    - Product/Market strategy: What differentiates the company's products from its competitors? What markets does the company serve?
    - Developmental goals: What products should the company develop over the next year?

- **Industry Analysis** requires the following, which will help us assess the attractiveness of the relevant industry that Honeywell primarily operates in:
    - Structural I/M Map: Porter 5 (6) Forces analysis
    - Porter 5 (6) Forces diagram (with companies)
    - Competitive Strategy Matrix

3. About Honeywell





Within the building automation industry, Honeywell maintains a diversified presence, particularly in the consumer market. The company is widely known for its robust line of consumer appliances, including thermostats, power generators, air filters, air conditioners, refrigerators, smart-door locks, and many more. This showcases a robust line of consumer products built with Honeywell technologies.



Honeywell's consumer products are readily accessible to a mass market, distributed through a variety of major retailers, including Home Depot, Costco, Walmart, Amazon, and Lowe's. This large retail network ensures maximum market penetration and brand visibility. The prices of these products range widely from $25 to well over $6,000. It confirms the company's capability to cater to both entry-level consumer needs and higher-end residential and commercial solutions.

A critical technological commonality across many of the products above is the reliance on advanced sensor technology. For example, modern thermostats and security devices require complex inputs to function, such as motion sensors, temperature sensors, particle sensors, and gas monitoring systems.

Honeywell's expertise in developing, manufacturing, and integrating these various sensor types shows how its products are highly valuable. The development of these sensors will allow us to propose a high-tech product that utilizes existing technology within the company, minimizing development risk and maximizing integration within the existing building management systems ecosystem.

4. Company Analysis

- The company analysis will allow us to understand Honeywell's strategies as a company.
- We will identify Honeywell's vision statement, mission statement, business goals, technology strategy, product/market strategy, and developmental goals.

Strategy	Key Questions	Information
Vision Statement	What is Honeywell's stated long-term purpose?	Create a smarter, safer, and more sustainable world and empower communities.
Mission Statement	What is its scope (product, market, technology)?	Solves the world's complex challenges in automation, aviation, and sustainability using hardware and integrated software platforms.
Business goals	What are the financial and operational targets (growth, margin, recurring revenue)?	Financial goals include increasing sales to 40 billion from 38 billion dollars, organic growth of 5%, and increasing shareholder earnings to $10.65 per share.
Competitive strategy	What is the overall competitive approach (Cost, Differentiation, or Focus)? How does it create value?	Honeywell strategically realigns its business to focus on three major trends: automation, the future of aviation, and energy transition. The company embeds AI across all platforms, from control systems to warehouse tools and healthcare diagnostics. Significant investments in innovation to develop cutting-edge products. Aligns its offerings with customer demand for sustainable solutions and digital transformation <u>Competitive strategy for Building Automation Industry:</u> focusing on high-growth markets like data centers, hospitals, and airports; investing in R&D for products that focus on integrating AI/ML to improve its offerings and decision-making (energy management)
Technology strategy	What are the core technologies that give the company a technological advantage over its competitors? What is the company's approach to product and technology development?	Honeywell integrates AI, IoT, and cloud technologies to offer solutions, like the Honeywell Forge platform for industrial operations. Invests R&D and strategic capital in solutions for three powerful global trends: Automation, Aviation, and Energy Focuses on digital transformation, creating software and solutions that bridge the physical and digital worlds
Product/Market strategy	What differentiates the company's products from its competitors?	Honeywell's product and market strategy relies mostly on creating a variety of products, applied through a *Differentiated Strategy***.** They have not only created a wide variety of unique

	What markets does the company serve?	products and software (sensors, scanners, sustainable technologies) across multiple markets (aerospace, building automation, performance materials), but also have constructed an ecosystem in which said products can integrate in the form of their Honeywell Forge platform, which introduces major switching costs.
Developmental goals	What products should the company develop over the next year?	Honeywell should continue to develop products in aerospace, industrial automation, build automated detection, and energy & sustainability solutions. The company should focus on developing a wearable device outfitted with health & environmental sensors, an electrical-signature detector as an early warning fire detection solution, and simplifying its consumer electronics.

Conclusion of Company Analysis

Honeywell has four high-tech engineering businesses that it focuses on developing products for. The first industry is aerospace, which targets products for defense, space, and commercial. The second industry is industrial automation, which targets products for healthcare, warehouses, and productivity. The third industry is building automation, which targets security and building management. The fourth industry is energy & sustainability, which targets petrochemical shipment, gas processing, and other sustainability efforts.

Within the four businesses of the company, Honeywell focuses on 11 industries: Aerospace, Commercial Buildings, Data Centers, Defense, Healthcare, Life Sciences, Manufacturing, Oil & Gas, Retail, Utilities, Warehouse & Logistics.

Honeywell's objective across the four industries is to solve complex challenges using hardware and integrated software platforms. The developmental goals are in the industries Honeywell is established and support the company's vision and mission statements.

5. Industry Analysis

- The industry analysis will focus on the building automation sector of Honeywell and how it competes against other similar companies in the space.
- We will identify the new entrants, suppliers, competitors, buyers, complementors, and substitutes that exist within the building automation industry.
- This allows us to perform Porter's Five (six) Forces analysis to determine Honeywell's position and competitive strategy within the industry/market landscape.

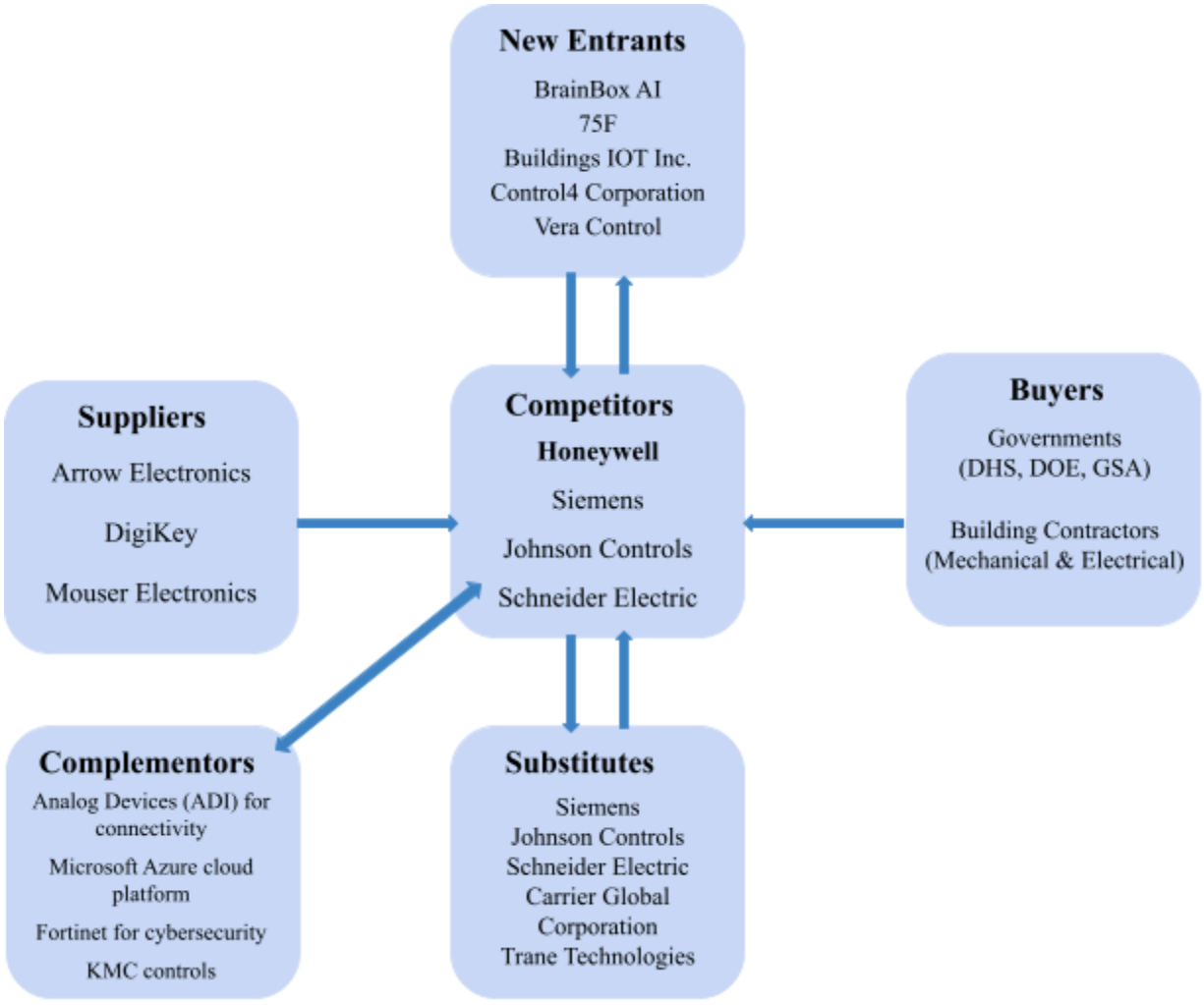
Criteria for Structural Map I/M

- For the Porter Five (or Six) Forces, the team established that a structural map would be beneficial to present our results; this would especially help to convey information concisely.
- We have also established a set of criteria for the structural map:
    - **New Entrants:** No company older than 2010, and can be a branch of a pre-existing established company
    - **Suppliers:** Not limited to one sector of Honeywell; broader scope.
    - **Competitors:** Focused on the highest threatening competitors.
    - **Buyers:** More broad buyers and not sector dependent.
    - **Substitutes:** Strayed away from companies, more idea-dependent.

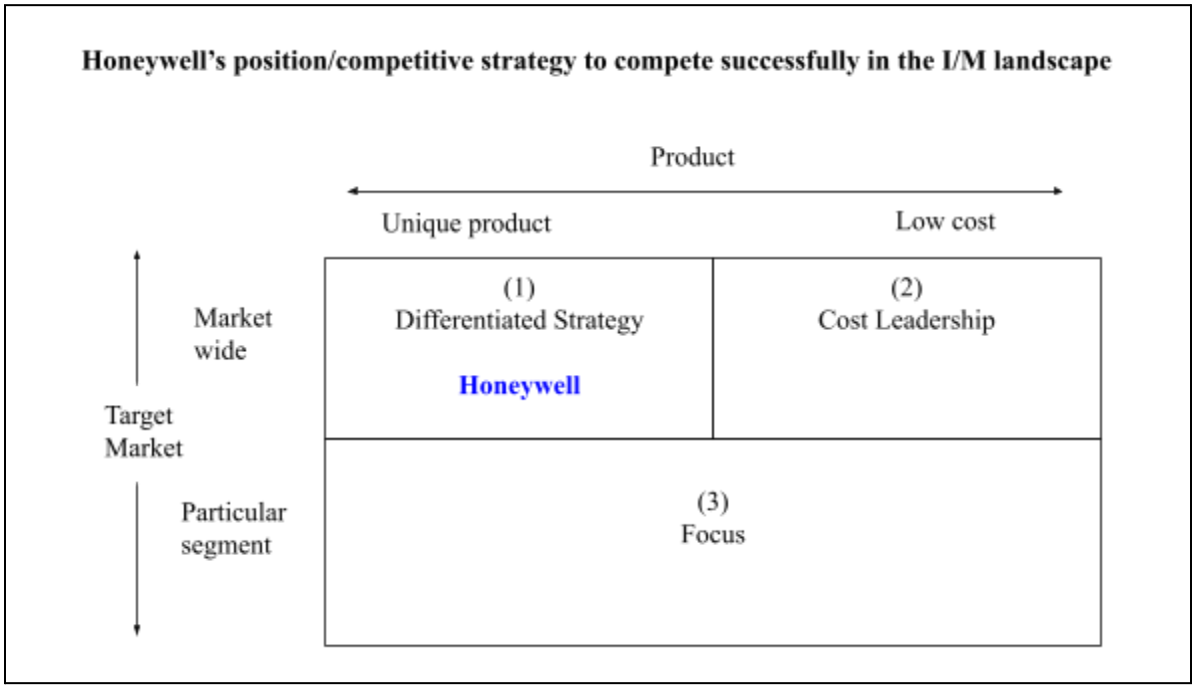
Force	Analysis	Intensity
Competitive Rivalry	Honeywell faces competition across its diverse business segments. Competitors invest heavily in R&D, leading to rapid innovation cycles.	HIGH
Threat of New Entrants	New entrants face high barriers due to large capital investments, strict safety regulations, and the technical expertise needed to design industrial systems	LOW
Threat of Substitutes	Honeywell's technologies could be replaced by direct competitors such as Siemens, Schneider Electric, and Emerson. Open-source technologies and/or IOT platforms could reduce the reliability of physical devices or hardware in general. These replacement solutions tend to be more prominent as they can optimize various building operations without Honeywell's systems.	MODERATE
Buyer Power	Major contractors such as the government, large firms in aviation, and construction are all buyers. Because they purchase in large volumes, bidding becomes competitive, or they can opt for alternatives.	MODERATE
Supplier Power	Honeywell relies on specialized suppliers for electronics, sensors, semiconductors, and cloud systems. Its large scale helps it reduce risk, but shortages and software costs give suppliers moderate influence.	MODERATE
Threat of Complements	Honeywell depends on complementary technology platforms that strengthen their position. However, this mutual dependency limits price flexibility in digital spaces.	MODERATE

Attractiveness Assessment
Although competition in the industry is high, the market remains profitable and strategically attractive for Honeywell. Being nearly 120 years old, Honeywell's longtime presence in the market has allowed it to introduce high entry barriers and limited substitutes to help maintain stability, while the company's scale, brand reputation, and advanced technologies position it to sustain a strong competitive advantage and imply continued growth in automation.

Porter Five (or Six) Forces Diagram



Competitive Strategy Matrix



Conclusion of Industry Analysis

The analysis matches Honeywell's real position in the Building Automation industry and follows Porter's Six Forces framework. Each force was evaluated logically, for example, intense rivalry due to major competitors like Siemens and Schneider Electric, and low threat of new entrants because of high startup costs and regulations. The team assumed Honeywell's Building Automation division follows the same goals as the overall company (automation, energy transition, and sustainability.

The results align with Honeywell's scale, strong reputation, and wide product range. High rivalry and moderate supplier/buyer power make sense since Honeywell competes in a fast-moving, tech-focused market, but still holds an advantage through its size and partnerships.

Honeywell operates in a competitive but stable market. Its strong brand, advanced technology, and high entry barriers help it maintain a leading position despite intense rivalry. Our Porter Five (Six) Forces diagram clearly reveals the I/M landscape regarding the building automation industry in relation to Honeywell, offering ample information to proceed with the next phase of our project.

6. Review and Validation

By reviewing the primary data sources, we ensure that our company and industry analysis are accurate and consistent with our findings. From this, we can determine whether our qualitative observations can be validated.

- We cross-verified figures and data through official Honeywell press releases and investor relations pages.
- Each established framework (eg, Company Industry Analysis) was checked to ensure that it properly conveys the I/M landscape.
- Team members peer-reviewed all the sections to confirm flow and clarity.

Outcome

- After review, our team has concluded that the analyses above are an accurate representation of Honeywell's position within the industry and landscape, as shown to be consistent across all of our included frameworks.

7. Insights and Implications

- Main Takeaways
    - Honeywell utilizes a "Differentiated Strategy" by offering a wide variety of unique products and software in the building automation industry.
    - The company offers a wide range of products and integrated solutions for HVAC, fire safety, access control, and security in commercial and industrial buildings.
    - Honeywell's long-standing presence, brand reputation, and scale offer stability and help sustain a strong competitive advantage.
- Strategic Learning
    - Through the agreement and convergence of our shared team process, we have learned about the importance of validating the information our frameworks convey. In addition, we were able to unify our perspectives and ideas into a singular cohesive approach.

8. Updated Product Proposal

- When developing our product concepts, we began by exploring Honeywell's broad portfolio across industries. We realized that the Building Automation niche offered the most natural alignment with both our experiences and Honeywell's strengths.
- We approached ideation through the lens of customer empathy, drawing from our own familiarity with common homeowner frustrations like energy inefficiency, safety

concerns, and maintenance issues alongside broader security and sustainability challenges we've observed in real buildings.

- By combining these everyday insights with an understanding of how technology can enhance comfort, safety, and efficiency, we identified opportunities for smart, connected solutions that not only address individual needs but also contribute to larger goals of sustainable and secure building management.

Product Ideas for the Building Automation Sector of Honeywell			
Societal/ Customer Need	Product Name	Product Description	Rank
Intrusion detection and door locking mechanism	Smart Home Security System	This integrated door/window lock security system combines various sensors and smart locks into a single, cohesive security network. It operates using a central hub and mobile app, offering automated responses to potential security threats.	1
Buildings waste enormous energy on ventilating empty rooms because occupancy-based control isn't widespread.	Sensor-based AC System	A mesh of ceiling-mounted micro-sensors that detect motion, heat, and CO_2 levels to dynamically open/close vents and balance HVAC in real time. Integrates into Honeywell's BMS as a plug-in module for true "occupancy-responsive heating and cooling."	2
Building security often operates separately from environmental systems, leading to inefficient emergency response.	Security Management Hub	Integrates smoke, gas, and intrusion sensors into one AI event manager. Detects whether alarms are safety or security-related and activates appropriate ventilation, door locks, or evacuation lights automatically.	3
People forget to turn off appliances or lights when leaving a room, wasting power and increasing fire risk.	Bulb Management Hub	A low-cost sensor hub that detects both motion and body heat, auto-switching lights and outlets on/off accordingly. Integrates with Honeywell's lighting controllers for seamless retrofit into existing wiring.	4
Power outages and system downtime are affecting security	Low-Voltage UPS	A micro-backup system with smart switching to keep security cameras, sensors, and locks active during power failures. Syncs with building automation to prioritize critical circuits automatically.	5

9. Conclusion of Updated Product Proposal

Our approach leveraged an initial exploration of Honeywell's extensive portfolio. We focused our efforts on the Building Automation sector due to its alignment with our team's experience and Honeywell's established strengths. The brainstorming process was driven by customer empathy, which translated to common homeowner frustrations, such as energy inefficiency, security, and maintenance issues. This led to the development of five smart, connected solutions designed to improve comfort, safety, and efficiency. Our highest-ranked concept, the Smart Home Security System, directly addresses the core need for reliable intrusion detection. These product ideas successfully identified opportunities for Honeywell to offer solutions that meet customer needs while also contributing to broader goals of sustainability and secure building management.

As such, having determined in Phase I that Honeywell's strength lies in integrated ecosystem rather than low cost (competitive pricing), Phase 2 thus focuses on defining the technical metrics that will help support a high-quality, reliability-focused product.

Project Phase II: House of Quality

1. Overview of House of Quality

The following phase helped translate customer needs into definitive engineering requirements for our Honeywell Smart Home Security System. Through structured analysis, including competition reverse-engineering and House of Quality development, we were able to pinpoint the most technical drivers that would help build consumer confidence. From the HOQ, we were able to establish that sensor accuracy and wireless reliability are the most critical technical drivers for customer trust, directly impacting core needs like reliable security and minimal false alarms.

Key results include setting definitive performance targets: sensor actuation accuracy < 2 cm, wireless packet loss < 1%, false alarm rate < 5%, and notification latency ≤ 3 seconds. Further concluding the analysis, we identified the central design trade-off between polling frequency and battery life, driving our final design to be power-driven. Overall, these findings helped provide the quantitative foundation for the functional design and concept generation in the following phases, ensuring that our product is differentiated on reliability rather than the multitude of features it possesses.

2. Process Development

- To develop the conceptual design framework for our new product, we drew on the textbook Product Design & Development by Ulrich & Eppinger, focusing primarily on the chapters “Identifying Customer Needs” and “Product Specifications.” These chapters provided a structured approach for gathering customer information, establishing product requirements, and preparing for FAST and HOQ development.
- All members presented their individually prepared FAST diagrams, customer needs lists, technical requirements, and benchmarking findings. Through comparison and discussion, the team identified several overlapping themes and agreed to consolidate the individual inputs into a single unified process for completing the FAST and HOQ. The unified process established during this meeting consisted of the following steps:’
    1. **Finalize the set of related products to reverse engineer.**
       The team selected three related products that best reflected the core functionality and design features relevant to our proposed Smart Lock System. These products were chosen because they closely matched the intended purpose of our design, offered accessible technical and user information for analysis, and represented realistic expectations within the current competitive market. This ensured that our reverse engineering focused on making useful industry benchmarks.
    2. **Create a standardized FAST structure.**
       The group agreed on a consistent method for defining main functions, supporting

functions, and the logical Why and How relationships used in our function structure and FAST diagrams. We defined the main functions as the essential, outcome-oriented tasks that the Smart Lock System must accomplish to meet core customer needs, while supporting functions were identified as the sub-operations required to enable or enhance those main functions. To maintain clarity, we used Why and How logic to show the hierarchy between functions: a function placed above another answers “Why is this done?” and expresses its purpose, while a function placed below answers “How is this achieved?” and breaks the higher-level task into operational steps. This shared framework ensured that our system decomposition remained clear, logical, and consistent across the entire design process.

3. **Consolidate customer needs into a unified, structured list.**
   Members compared their individual customer need lists, merged duplicates, clarified ambiguous statements, and organized the final set into primary, secondary, and tertiary categories. The team also agreed on common justifications for importance rankings to ensure consistency and to bring more clarity in later stages of analysis.
4. **Establish a common set of technical requirements.**
   Technical metrics proposed by individual members were combined into a single agreed-upon list that aligned with the customer needs. The team clarified definitions, removed redundancies, and ensured that each technical requirement had a clear and measurable meaning.
5. **Decide on correlation rules for the HOQ.**
   The team established a unified set of rules for interpreting correlations within the House of Quality, including how to apply positive and negative relationships and how to differentiate between strong, medium, and weak correlations. This agreement ensured that each correlation judgment reflected consistent logic rather than individual interpretation.
6. **Set criteria for benchmarking and performance targets.**
   The team identified which competing products would be included in our benchmarking analysis and determined the specific data points to collect. We then aligned on how performance targets would be set by comparing our proposed design to market leaders and identifying areas where we could meet or exceed existing standards.
7. **Assign roles and responsibilities.**
   Each team member was assigned a clear role based on their strengths and areas of contribution, including tasks such as developing FAST diagrams, organizing customer needs, refining technical requirements, collecting benchmarking data, constructing the HOQ, and drafting portions of the final report. This ensured an efficient workflow and minimized overlap.

This agreed-upon unified process ensured that the entire team followed the same methodology, used consistent terminology, and worked towards a cohesive FAST analysis and HOQ.

3. Product Purpose

- The team identified the primary function of our proposed product. Establishing this function early allowed us to maintain a consistent direction throughout the reverse engineering, specification setting, and conceptual development stages.

Individual Preparation Prior To the First Team Meeting

- Before collaborating as a group, each team member independently identified existing products that served similar purposes to the one we aim to create. Members researched competitive offerings through online searches, product reviews, and available technical information to better understand the current market landscape.
- Each member also performed reverse engineering using a FAST diagram for one or more related products. This included identifying the main function, decomposing supporting functions, and exploring “Why?” and “How?” relationships to map each product’s underlying functional logic. In parallel, members gathered preliminary customer needs and corresponding technical requirements, assigning initial importance levels and collecting benchmarking information to prepare for HOQ construction.

4. FAST, Function Structure, and HOQ

With a unified approach established, the team completed the FAST diagrams for the selected related products and refined them to highlight key functional similarities and distinctions. These insights supported the development of a function structure for our new product, identifying its main function and required sub-functions and illustrating how each contributes to overall operation.

The team then developed the House of Quality by finalizing customer needs and technical requirements, assigning importance ratings, determining correlations, and incorporating benchmarking data. Performance targets were established based on competitive analysis and customer expectations, resulting in a completed HOQ that clearly connects user needs to technical specifications.

Review, Verification, and Concept Selection

After constructing the FAST diagrams, function structure, and HOQ, the team reviewed all components to ensure accuracy, consistency, and alignment with the agreed process. The results were then analyzed to draw meaningful conclusions. Selection criteria were developed,

weighted, and applied to evaluate the feasibility and performance of potential design directions. Through this evaluation, the team identified the strongest concept candidates for further development and summarized the insights gained during the analysis.

Communication Protocol

To ensure alignment, the coordinator posted summaries of each meeting within 24 hours. Team members participated in mid-week Slack check-ins for task updates. All FAST and HOQ drafts were submitted for review by Sunday before Meeting 2. Regular check-ins after each milestone enabled quick reflection on progress and recalibration of individual contributions.

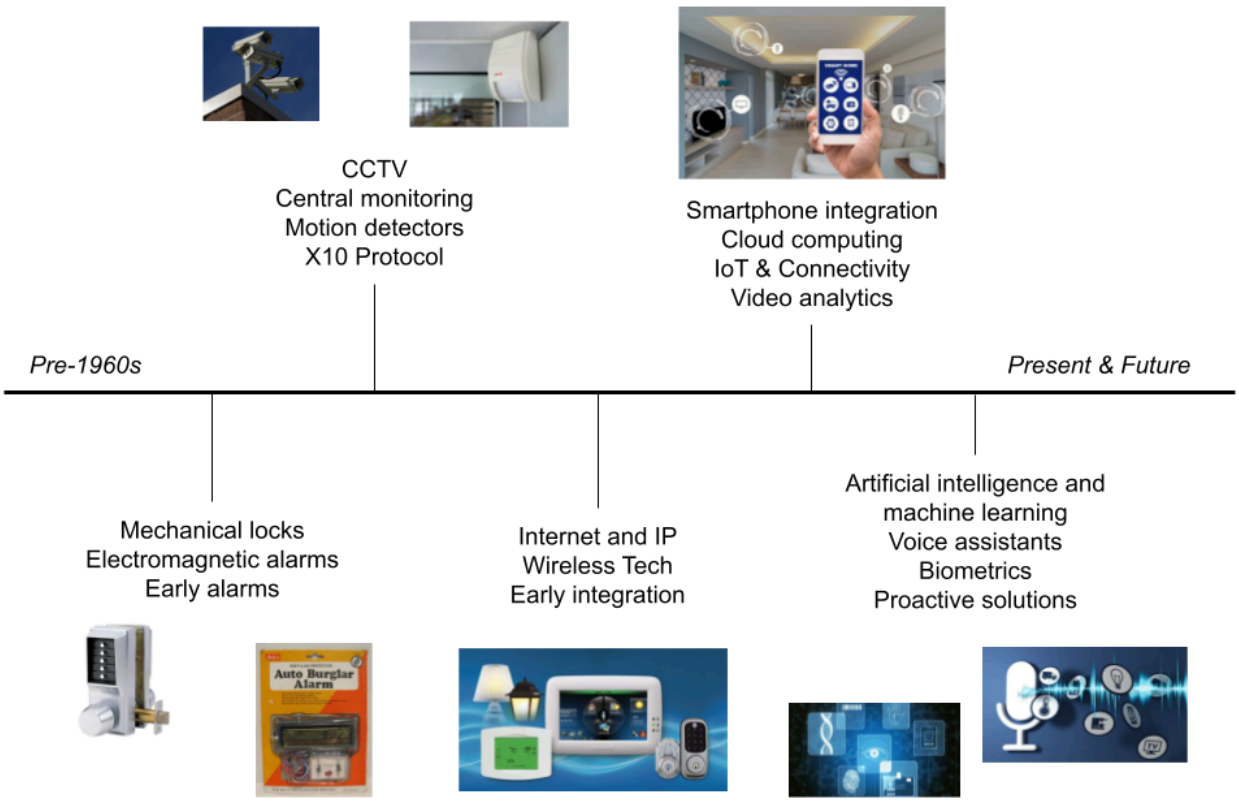
Tools and Resources

The team utilized a combination of digital tools and verified references to ensure efficiency, consistency, and data accuracy throughout the project:

- **Google Suite** - Centralized shared documents for real-time document editing, version control, and storage of all meeting notes, FAST drafts, and HOQ data.
- **Lucidchart** - Used to design and refine individual and team HOQ diagrams, allowing visual mapping of "Why-How" relationships and functional dependencies.
- **Honeywell Forge Documentation** - Consulted to understand Honeywell's IoT integration strategy, sensor management, and cloud-based analytics.
  Product Manuals and Technical Datasheets - Reviewed from Honeywell, SimpliSafe, Ring, and Ecobee to identify feature specifications and performance benchmarks.
- **Team Communication Platforms (Zoom/Discord)** - Supported coordination, quick updates, and screen-sharing during FAST and HOQ development sessions.

These tools were later assessed for their effectiveness in data consistency and collaborative iterations during our HOQ construction.

5. Evolution of Intersecting Fields and Smart Home Security Systems



The evolution of smart home security represents a merger of several technological fields, starting from simple physical locks to complex, connected ecosystems. Early security relied on mechanical locks and simple, localized electromagnetic alarms or early alarms to deter intruders. The mid-to-late 20th century incorporated CCTV, wired motion detectors, and service of central monitoring for professional response. The X10 protocol allowed simple communication between devices, followed by the emergence of the Internet and IP networks, and wireless technology. This enabled integration of security devices with digital systems. The modern era of IoT and Connectivity allowed thousands of devices to communicate through cloud computing, which drove the development of smartphone integration for managing security remotely. This network generates data that is used by video analytics and relies on sensors to detect logic operations, then process events, enabling the systems to become proactive solutions rather than just reactive alerts. The current and future state of smart home security systems consists of user-friendly interfaces like voice assistants and high-security measures like biometrics, which focus on seamless, intelligent, and highly personalized protection.

6. Functional Analysis

Each member created an individual FAST diagram for one benchmark product to reverse-engineer its core functions and identify innovation opportunities. We narrowed the FAST diagrams to 3 diagrams which illustrate their respective products' functionality, while eliminating the other proposed products as we did not feel they adhered too closely to our product focus.

SimpliSafe Smart Lock Product Dissection

Product Description

The SimpliSafe Smart Lock is a retrofit smart locking device designed to enhance security as a given entry point by adding digital control, monitoring, and automation to a standard deadbolt. This solution installs over an existing deadbolt, allowing it to maintain the traditional key-in-cylinder mechanism while also enabling extra security features. These features include app-based locking, PIN code access, automatic locking, tamper detection, and further integration with SimpliSafe's home security system. Priced at approximately $99 to $120, the SimpliSafe Smart Lock positions itself as a mid-range smart product that offers reliable security and access management without complex installation. Because this lock relies on the SimpliSafe Base Station and wireless keyboard, users can use remote features and system integration that they might find with other modern smart-home security products.

Main Function

The main function of the SimpliSafe Smart Lock is to "secure the primary entry point of a home by controlling and monitoring deadbolt access." This reflects how the lock ultimately aims to prevent unauthorized entries while also enabling users with convenient and authenticated access to remote controls and monitoring.

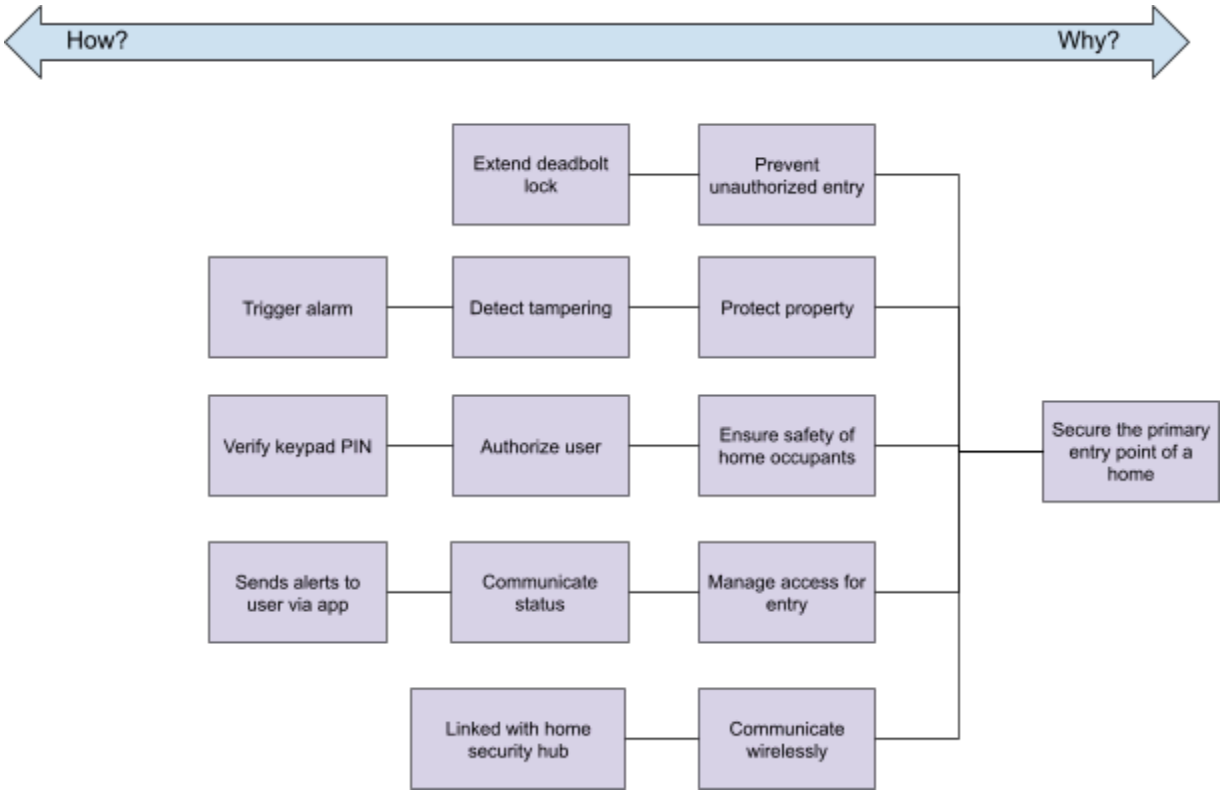
Sub-Function Breakdown

In order to perform its primary function, the SimpliSafe Smart Lock has multiple sub-functions. These sub-functions represent the operational logic required to manage access, verify users, and maintain the system's connectivity:

1. **Actuation of Deadbolt Lock:** The motorized lock mechanism physically engages or disengages the deadbolt based on app commands, keypad input, or configured automation (e.g., auto-lock timers).
2. **Authenticate User:** The keypad PIN or app credential validates a user's identity before granting entry, ensuring that only authorized individuals can unlock the door.
3. **Detect Tampering:** Sensors detect whether the lock is being forced, manually overridden, or manipulated, allowing the system to alert the user of potential security breaches.

4. **Trigger Alerts:** If unauthorized access or tampering is detected, the system sends notifications through the SimpliSafe app and may trigger additional alarm-system responses if paired with other SimpliSafe devices.
5. **Communicate System Status:** The lock reports lock/unlock status, battery level, and error conditions to the user and the base station, ensuring continuous awareness of entry-point security.
6. **Manage Access for Entry:** The system enables PIN creation, guest code scheduling, remote unlocking, and automatic locking, giving users flexible control over who can enter and when.
7. **Communicate Wirelessly with Base Station**
   The Smart Lock syncs with the SimpliSafe Base Station, enabling integration with the broader security ecosystem and allowing remote app control depending on the user's service plan.

FAST Diagram:



Comparison to a Basic Product

Compared to a traditional non-smart deadbolt, the SimpliSafe Smart Lock offers a deeper functional architecture. A basic lock, which only provides a mechanical locking function with a key, cannot authenticate users digitally, create/revoke guest access codes, trigger alerts, or integrate into a larger security system. In contrast, the SimpliSafe Smart Lock can perform multiple coordinated operations, including authorizing entry through PIN/app credentials, managing remote access/automation, detecting tampering, wirelessly communicating status, and integrating into a system-wide alarm network. These layered functions position the SimpliSafe Smart Lock as a more intelligent home-security solution while still relying on the physical deadbolt for mechanical security.

Ring Alarm Pro Product Dissection

Product Description

The Ring Alarm Pro is a smart home security hub designed to provide comprehensive intrusion monitoring and reliable real-time alerts. Its primary goal is to enhance home safety by combining traditional security functions such as door, window, and motion sensing with modern connected capabilities, including app-based notifications, smart device integration, and backup internet through an embedded Eero router. By pairing local processing with cloud-based communication, the Ring Alarm Pro ensures continuous operation even during outages while offering users a seamless, connected security experience. The system typically retails for approximately $249.99, making it a competitive option within the consumer smart home security market.

Main Function

The main function identified for the Ring Alarm Pro is to “provide reliable home security monitoring and notification”. This function captures the essential purpose of the product: detecting intrusion-related events and ensuring the user is promptly informed so they can take action. All supporting features of the Ring Alarm Pro exist to enhance or enable this primary function.

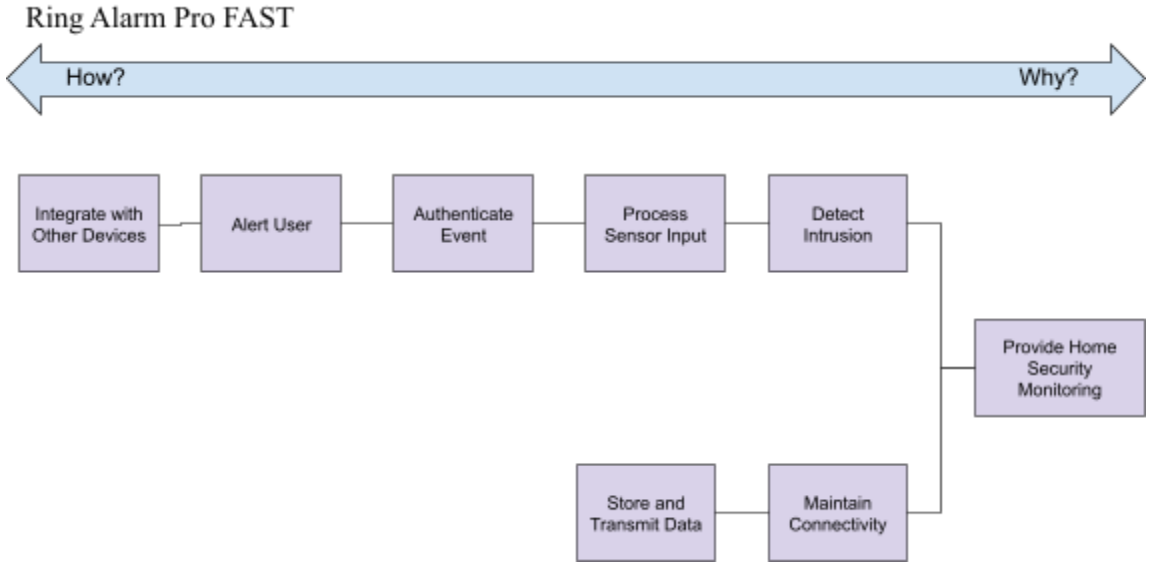
Sub Function Breakdown

To carry out its main function, the Ring Alarm Pro performs several key sub-functions. These represent the operational steps required for the system to detect, interpret, and communicate security events:

1. **Detect Intrusion:** The system uses door and window contact sensors, motion detectors, and compatible Ring devices to identify potential intrusion events and monitor activity throughout the home.
2. **Process Sensor Input:** Incoming signals are analyzed to determine whether they represent meaningful activity. The system filters minor disturbances and uses built-in logic to reduce false alarms before escalating an event.
3. **Authenticate Event:** The device evaluates system status, such as whether it is armed or disarmed, and classifies the detected event to determine whether it requires notification, escalation, or automated action.
4. **Alert User:** Notifications are delivered through the Ring mobile app, and the system can trigger an audible siren or display visual alerts to ensure the user becomes aware of a potential security issue in real time.
5. **Maintain Connectivity:** The Ring Alarm Pro relies on Wi-Fi and cloud services, with optional backup internet provided by its integrated Eero router, ensuring that alerts and system communications remain available even during outages.
6. **Store and Transmit Data:** Event logs, sensor activity, and system updates are stored locally or synchronized with the Ring cloud service, providing users with a historical timeline of security-related events.

7. **Integrate with Other Devices:** The system connects seamlessly with Ring cameras, smart lighting, and Alexa-enabled devices, enabling coordinated home automation routines and enhanced security responses such as activating cameras or lights during an event.

FAST Diagram



Comparison to a Basic Product

A comparison to a simple, non-smart home alarm system highlights the significantly greater functional depth of the Ring Alarm Pro. A basic alarm system typically detects a door or window opening, triggers a siren, and offers no remote monitoring, data storage, or automation capabilities. In contrast, the Ring Alarm Pro detects intrusion through multiple sensor types, processes and validates events to reduce false alarms, authenticates the system state, and determines the appropriate response. It also sends alerts through cloud-connected channels, maintains internet connectivity even during outages, logs data for later review, and integrates with a broader smart home ecosystem. This contrast clearly illustrates the increased complexity of modern smart products and underscores how much more robust their operational structures must be. Such functional insights directly support subsequent steps in identifying customer needs, defining technical specifications, and constructing the House of Quality.

Ecobee Smart Security System Product Dissection

Product Description

The Ecobee Smart Security System is a home monitoring platform built around Ecobee's ecosystem of smart thermostats, smart sensors, cameras, and door/window entry sensors. Designed as a subscription-optional, hub-free security solution, Ecobee leverages its existing home comfort hardware, especially occupancy and temperature sensors, to provide whole-home awareness beyond traditional intrusion detection. Its goals include detecting unexpected activity, verifying events using multi-sensor input and video, and notifying the user through the Ecobee mobile app. The system connects directly over Wi-Fi and integrates seamlessly with Ecobee's smart thermostat for coordinated home/away modes and automation routines. Depending on the selected bundle, Ecobee Smart Security packages typically range from $99 to $249, positioning the system competitively within the consumer smart home security market.

Main Function

The main function identified for the Ecobee Smart Security System is to "monitor home activity and notify the user of unexpected events." This description reflects the system's purpose of detecting intrusion-related or unexpected in-home events and ensuring that the user is promptly informed through notifications, automated responses, or connected devices. All supporting features from video capture to environmental sensing exist to enable or strengthen this core function.

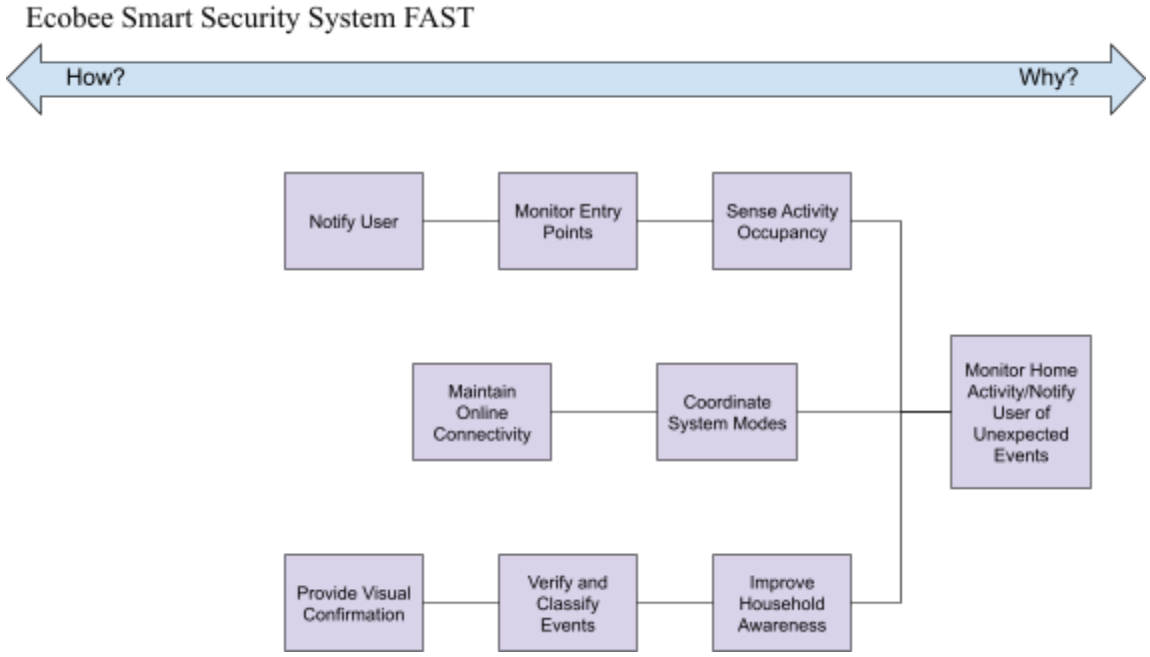
Sub-Function Breakdown

To achieve its primary monitoring and notification role, the Ecobee Smart Security System performs several sub-functions that differ slightly from traditional intrusion-only systems:

1. **Sense Activity and Occupancy:** Ecobee's multifunction sensors measure motion, presence, and in some cases temperature to determine whether activity is expected or unusual.
2. **Monitor Entry Points:** Door and window sensors detect opening or closing events that may signal an intrusion.
3. **Verify and Classify Events:** The system cross-checks sensor data with occupancy patterns, thermostat home/away settings, and camera feeds to differentiate normal activity from potential security issues.
4. **Capture Visual Evidence:** Integration with the Ecobee SmartCamera or Doorbell Camera allows video confirmation of events.
5. **Notify the User:** Push notifications, video clip alerts, and in-app timelines keep the user informed in real time.
6. **Coordinate Home Modes:** The system updates or responds to thermostat modes (Home/Away/Sleep) and triggers automation routines for lighting or climate control.

7. **Maintain Cloud-Based Operation:** All data and notifications rely on secure Wi-Fi communication with Ecobee's servers.

FAST Diagram



Comparison to Basic Product

Compared to a traditional, non-smart home alarm system, which typically detects only door/window openings and activates a siren, the Ecobee Smart Security System offers a broader and more integrated approach to home monitoring. It uses motion and occupancy sensing to detect unexpected activity, classifies events by comparing them to normal behavior patterns, and provides visual confirmation through connected cameras. The system notifies users through dynamic app alerts rather than relying solely on loud alarms and logs events for later review. Additionally, Ecobee coordinates security with comfort settings, adjusting thermostat modes and triggering automation routines based on detected activity. This level of integration illustrates how modern smart security systems expand well beyond basic intrusion detection, informing the development of customer needs, system requirements, and the House of Quality for new product concepts.

8. House of Quality

The House of Quality was developed by translating customer requirements (“Whats”) into engineering characteristics (“Hows”) derived from the FAST diagram.

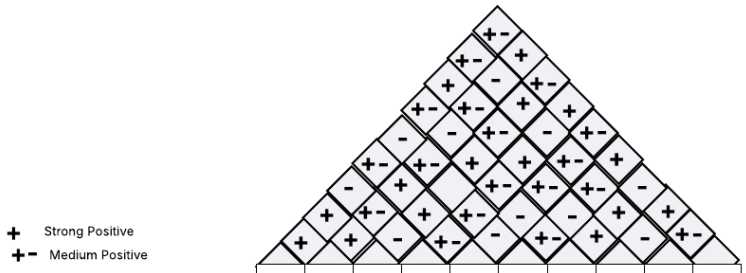
Customer Requirements

Customer Needs	Importance (1-5)
Reliable door security	5
Minimal False Alerts	5
Instant mobile notifications	4
Smart home integration	4
Remote lock control	4
Accurate door position detection	5
Accurate window position detection	5
Accurate door lock status detection	5
Loud, audible alarm	5
Easy installation	3
Tamper-resistant hardware	4
Affordable price point	4
Simple mobile app	3

Engineering Characteristics

Technical Metrics	Importance (1-5)
Door Sensor Accuracy (%)	5
Lock Detection Time (seconds)	5
Notification Delivery Time (seconds)	5
False Alert Rate (%)	5
Bluetooth Communication Range (meters)	3
Battery Life (months)	4
Installation Time (minutes)	3
IOS/Android app Compatibility	5

House of Quality Diagram



+ Strong Positive
+- Medium Positive
- Medium Negative
● Strong Correlation
○ Moderate Correlation
△ Weak/No Correlation

		Door sensor actuation	Window sensor actuation	Polling/Sensing refresh frequency	Notification Latency	Wireless Communication Frequency	Wireless Signal Reliability	Siren Volume	False trigger rate	System uptime/reliability	Battery Life	Our Product	Google Nest Protect	Ring Alarm Pro	Ecobee Smart Security
		4	4	4	5	4	5	5	4	4	3				
Reliable door security	5	●	●	○	○	○	●	△	△	●	○	4	3	3	3
Minimal false alerts	5	○	○	○	△	△	○	△	●	△	△	5	4	4	2
Instant mobile notifications	5	△	△	●	●	●	●	△	○	●	○	5	5	3	4
Smart-home integration	3	△	△	△	△	●	●	△	△	○	○	5	5	4	5
Remote lock control	3	○	○	●	●	●	●	△	○	●	○	5	4	4	3
Accurate door position detection	5	●	△	●	△	○	●	△	△	●	●	5	5	4	3
Accurate window position detection	5	△	●	●	△	○	●	△	△	●	●	4	4	5	4
Accurate door lock status detection	4	○	△	●	●	●	●	△	△	●	○	4	3	3	3
Loud, audible alarm	4	△	△	○	●	△	○	●	○	●	○	4	3	4	4
Easy installation	3	○	○	△	△	△	○	△	△	○	○	5	3	3	2
Tamper-resistant hardware	4	●	●	○	△	○	●	○	○	●	●	4	3	4	3

Measurement Units	**Engineering Characteristics**									
	cm	cm	Hz	seconds	Ghz	PL%	Db	False %	Uptime	Battery Life
Our product	2	2	500	3	2.4+5	1	105	5	99.9	48
SimpliSafe	5.08	5.08	433	5	2.4	Not disclosed	105	Not Disclosed		36
Ring Alarm Pro	2.54	2.54	908	3	2.4+5		104			36
Ecobee Smart Security	3.2	3.2	915	5	2.4+5		75			36
Technical Difficulty	3	3	4	3	1	4	1	3	2	5

Targets for Customer Needs & Technical Metrics

Customer Needs

Customer Needs	Target Scale (1/5)
Reliable door security	5/5
Minimal False Alerts	5/5
Instant mobile notifications	3/5
Smart home integration	4/5
Remote lock control	4/5
Accurate door position detection	5/5
Accurate window position detection	3/5
Accurate door lock status detection	5/5
Loud, audible alarm	4/5
Easy installation	4/5
Tamper-resistant hardware	4/5

Analysis of Targets Set for Customer Needs

After assessing the importance of each customer need, we believe that we must prioritize the needs that are the most critical to the success of our product and represent our competitive advantage. Some of the main ones include:

- **Reliable Door Security:** Reliable door security is central to the system's ability to protect the home. Ensuring consistent and dependable sensing at primary entry points inspires user confidence and supports the overall purpose of maintaining a secure home environment.
- **Minimal False Alerts:** Reducing false alerts is vital for maintaining user trust and preventing fatigue from notifications. Frequent false alarms could cause users to dismiss genuine warnings and lose confidence in the system. Prioritizing the minimization of false alerts ensures the meaningfulness and credibility of the system's alerts.
- **Accurate door position detection:** Precise sensing to verify accurate door position detection enables the system to properly respond to unauthorized access as well as avoid false alerts. This directly affects the security performance of the system as well as the user experience.

- **Accurate Door Lock Status Detection:** Users will rely on the system to verify that their home remains protected. Correctly identifying whether a door is locked or unlocked is a crucial part of maintaining a secure entry point.

Technical Metrics

Technical Metrics	Target Scale (1/5)
Door Sensor Actuation	5/5
Window sensor actuation	5/5
Polling/sensing refresh frequency	4/5
Notification Latency	4/5
Wireless Communication Frequency	3/5
Wireless Signal Reliability	5/5
Siren Volume	3/5
False alarm trigger rate	5/5
System uptime/reliability	4/5
Battery Life	3/5

Analysis of Targets Set for Technical Metrics

After assessing the importance of each technical metric, we believe we must prioritize the metrics most critical to our product's success and that represent the core functionalities that let our product remain with a competitive advantage over others in their niche. Some of the main ones include:

- **Door Sensor Actuation:** High accuracy is essential for correctly detecting door open/close events, which directly affects user trust in the lock's security. Incorrect readings can lead to safety risks and undermine the core function of the product.
- **Window Sensor Actuation:** High actuation accuracy ensures the system reliably detects when a window is opened or closed, which is critical for full-home intrusion coverage. Inaccurate sensing can leave key entry points unmonitored and weaken overall security.
- **Wireless Signal Reliability:** Strong wireless reliability ensures consistent communication between sensors, the smart lock, and the user's mobile app. Unstable signals can lead to missed events, delayed alerts, or system downtime, compromising user trust and safety.

- **False Alarm Trigger Rate:** A low false alarm rate is essential to maintaining user confidence and reducing notification fatigue. Frequent false triggers undermine the system's credibility and may cause users to ignore or disable important alerts.

HOQ Insights and Recommendations

The House of Quality analysis helped translate customers' needs into actionable engineering targets and showed major issues in the system that needed to be prioritized. The key results from this analysis allowed for the following design targets to be moved as a top priority:

1) Primary Engineering Targets
    - Sensor Actuation Accuracy: Target < 2 cm.
        - The following metric displayed the strongest correlation with top consumer needs (reliable security, minimal false alerts). It is the main decider of the system's reliability.
    - Wireless Signal Reliability (Packet Loss): Target < 1%.
        - The following metric is the most mandated requirement; this metric alone impacts the notification speed, false alarm rate, and user trust. This metric must be a core design focus.
    - False Alarm Rate: Target < 5%.
        - This metric was noted as a key competitive weakness (competitors within this sector range from 8-36%); this target is a must-have to convince users to make the switch.
    - Notification Latency: Target ≤ 3 seconds.
        - Required to meet the "Instant Mobile Notifications" need, benchmarked against leading competitors.

2) Critical Design Imperatives from Results
    - Competitive Differentiation: The largest issue within the market is not revolving around which product has the most features; it's about which product is the most reliable. Our system must ensure that accuracy and robustness are a large priority to separate ourselves from our competitors, not just our connectivity.
    - Main Trade-Off: The HOQ assessed the Polling Frequency (connectivity) vs. Battery Life trade-off. This forces an event-driven, smart power design as a non-negotiable system requirement.

3) Actionable Recommendations to Reach Targets

- Prioritize Sensor Actuation Accuracy
    - Invest in high-consistency magnetic reed switches or even Hall-effect sensors with stable mounting to consistently hit the <2 cm actuation target. This would directly address one of the larger consistent consumer issues.

- Optimize Wireless Reliability
    - We'd implement a dual-protocol strategy (a primary 2.4 GHz network with a Bluetooth fall-back system in place). To meet the <1% packet loss target, the key design elements would be antenna placement and radio frequency shielding, while still supporting all other system functions.
- Implement Smart Power Management
    - The goal here would be to develop an event-driven sensor firmware that would use low-power sleep modes, utilizing high-frequency polling upon an initial trigger activation. This would be key to resolving the identified trade-off and achieving both the 3-second notification and 48-month battery life targets.

9. Analysis and Interpretation of Findings

Our completed analysis of the HOQ highlights the engineering characteristics that most directly influence both the highest priority customer needs and the technical specifications, providing a clear roadmap for optimizing our smart-lock system. The strongest correlations appear between customer expectations of reliable security, minimal false alarms, and accurate position detection, and the technical metrics of door sensor actuation, wireless signal reliability, and window and door position detection accuracy. These findings reinforce that the foundation of a customer's trust is built on consistent and accurate sensing along with dependable cross-system communication. The HOQ also reveals areas where our product demonstrates competitive advantages, particularly in battery life, detection time, and lock status accuracy, which perform at or above the level of leading competitors. However, the competitive benchmarking also highlights clear gaps in false-alarm mitigation and smart-home integration. These gaps present opportunities for expansion into different market segments and for better alignment with evolving customer expectations.

A key insight from the HOQ is the strong influence of wireless communication on almost every major customer requirement. When wireless signal reliability is weak, the system experiences slower notifications, inaccurate detections, and a higher rate of false alarms. Improving this single factor has the potential to elevate the performance of the overall system. After completing the House of Quality, the team carried out a post-processing analysis to translate the matrix results into an actionable design and implementation strategy. This ensured that the relationships between customer needs and engineering characteristics directly informed how the Smart Lock System would be refined beyond the conceptual stage. The HOQ identified detection accuracy, minimizing false alarms, and rapid notification delivery as the most influential customer needs. These insights guided targeted design refinements through improved sensors and more vigilant door and window tampering event validation, strengthened system reliability, and reduced unnecessary alarms.

To translate this into a product-to-market scenario, our developmental strategy should prioritize high-accuracy sensing modules, improvements in wireless reliability, and reductions in false-alarm frequency through better filtering and validation. Expanding compatibility with major smart-home ecosystems and maintaining strengths in battery life and detection time will also be essential for customer adoption and long-term competitiveness.

By grounding our product decisions in the insights from the HOQ analysis, we ensure that we are focusing on the technical factors with the largest impact on overall customer satisfaction. This approach not only improves our product's competitive position but also establishes a clear strategy for future expansion and continued iterations of our Smart Lock System.

10. Conclusions of Phase II

The team’s structured and iterative approach throughout phase 2 proved essential for connecting customer needs with engineering design choices. By starting with individual research and then consolidating insights as a group, we ensured that our understanding of smart home security systems was both broad and detailed. This process helped us clearly identify the functional drivers that matter most in the market: minimizing false alarm rates, reliable sensor uptime, and communication interoperability, all of which were validated through our FAST diagram and the correlations revealed in the HOQ.

One of the major takeaways from this phase was learning how systematically the FAST diagram and HOQ complement each other. The fast diagrams enable us to break down each benchmark product into functional pathways, making its design needs transparent. When we later built the HOQ, these functional insights converted directly into engineering characteristics, ensuring that every “How” in the HOQ was backed by real product logic rather than assumptions. This alignment strengthened the validity of our final HOQ rankings and ensured that our chosen priorities reflected genuine customer value rather than arbitrary scoring.

A key reflection is the importance of false alarm minimization as the dominant customer priority. Through competitive analysis, we recognized that even strong competitors like Google Nest, Ring, and Ecobee struggle with false positives, which is often due to the products' sensing features or limited cross verification. Our HOQ validates that user trust and satisfaction depend heavily on accuracy, reinforcing our direction to incorporate high-accuracy sensors and granting customers the ability to closely monitor the ecosystem, to detect irregularities using their own judgment. This insight laid the foundation for Honeywell to reclaim differentiation in the smart home market by improving the accuracy of our sensors and providing a robust, yet simple ecosystem, rather than just hardware capability.

Another important learning outcome was the value of structured peer review and cross-functional collaboration. Reviewing each other's FAST diagrams and HOQ contributions helped maintain consistency across sections and ensured each technical relationship was justified. This process highlighted how each team member's perspective uncovered blind spots, whether related to sensor limitations, integration requirements, or product specifications that meet customer needs. The collaboration ultimately elevated the clarity and professionalism of the final project.

Looking forward, the next logical steps include product development, usability testing with real users, and iterative refinement of the HOQ once user testing is conducted. Customer feedback will be especially important, as the accuracy and usefulness of the sensors can only be validated with real-world edge cases. Additionally, future iterations should assess long-term reliability factors such as battery performance, interoperability, and fail-safe behaviors during outages.

Overall, Phase 2 strengthened our understanding of the engineering trade-off between smart home systems and demonstrated the importance of pairing user-centric thinking with technical structure. This phase not only clarifies the technical direction of our Smart Home Security System but also reinforces the value of methodical design tools like FAST and HOQ in supporting rigorous product development aligned with Honeywell's strategic strengths. In Phase 3, we will begin to introduce conceptual designs that prioritize solution principles (like vibration sensors) that will be specifically selected to help mitigate false positives.

Project Phase III: Conceptual Design

1. Conceptual Design Overview

About Honeywell

Honeywell is known for consumer appliances, including thermostats, power generators, air filters, air conditioners, refrigerators, smart-door locks, and other products. Our products are available in a variety of retailers, including Home Depot, Costco, Walmart, Amazon, and Lowe's. This showcases a robust line of consumer products built with Honeywell technologies. Many of our products require sensors to work properly, such as temperature sensors, power gauges, and carbon monoxide sensors. The combination of these sensors will allow us to introduce a high-tech Smart Home Security System to the market.

Objectives of the Project

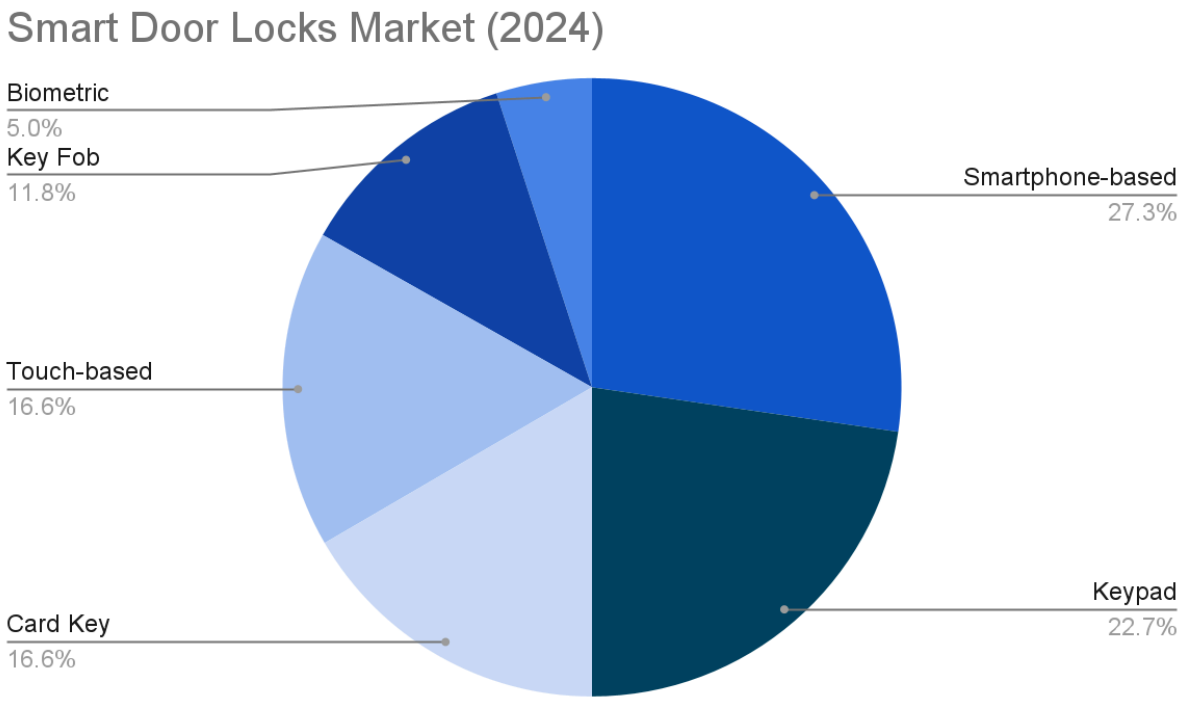
The objective of the project is to develop a Smart Home Security System. This phase of the project aims to establish a framework for conceptual design, implement an initial conceptual design, implement a final conceptual design, and explicitly demonstrate the improvements between our initial and final conceptual design implementations.

Evolution of Home Security Systems

Function	Gen 0: Standalone Contact Alarms	Gen 1: Panel-Based Security Systems (Wired or Monitored)	Gen 2: Early Smart Systems (App-Connected)	Gen 3: Advanced Smart Systems (IoT)
Detect	Single magnetic contact sensor triggers a loud local buzzer when a door/window opens. Very sensitive to misalignment and false alarms	Multiple wired door/window contact sensors connected to a central alarm panel. More reliable but still basic open/close detection.	Wireless battery-powered contact sensors for doors/windows. Sensors self-report battery status, signal strength, and tamper events. Can differentiate between door vs. window zones.	Integrated door + window sensor nodes with status tracking, tamper detection, smart-lock integration, and context-aware detection (e.g., distinguishing "unlocked but closed" vs. "opened after unlock").
Verify	No verification, any	The monitoring	Homeowner verifies	The system

	separation of the magnetic contacts triggers the alarm.	center verifies entry events by calling the customer (human verification).	through app notifications, event history, and status dashboards (open/closed/unlocked).	cross-checks events with smart lock activity, known user codes, time-of-day patterns, and can detect unusual unlock behavior. AI-based filtering reduces false triggers due to vibration, weather, or misalignment.
Notify	Only a local buzzer; the homeowner must be nearby to hear it.	Alarm siren + phone call from monitoring service. Very little user visibility of entry logs.	Push notifications, text alerts, and in-app event logs showing real-time door/window status and unlock method.	Multi-channel alerts: push, SMS, smart speakers, wearables, shared family access. Includes detailed context (who unlocked, how, when, door vs. window, status changes). Supports escalation paths (neighbors, family, emergency services).
Respond	User must manually investigate and contact authorities. The system cannot interact with other devices.	The monitoring service can dispatch police/fire, but the home takes no automatic action.	The app allows remote arming/disarming, smart lock control, and a manual emergency button to call law enforcement. Some automation is possible (e.g., sound alarm when opened).	Coordinated smart-home responses: auto-trigger siren, lock or unlock depending on emergency scenario, turn on lights, activate cameras, share data with emergency services. The app allows emergency override. Can integrate with whole-home routines for evacuation or lockdown.

Smart Door Locks Market



The smart door lock market has evolved from traditional mechanical locks to electronic keypad systems and, more recently, to internet-connected devices that integrate with broader smart home ecosystems via Wi-Fi, Bluetooth, and other wireless protocols. This progression has been driven by growing consumer demand for enhanced security, convenience, and remote access capabilities. As a result, modern smart locks now offer features such as smartphone-controlled entry, voice assistant integration, digital keys, and biometric authentication. The market continues to expand as advancements in AI, next-generation biometrics, and deeper interoperability with other smart home devices further shape user expectations and system capabilities.

2. Framework Development Process

Developing the Conceptual Design Process Framework

The primary resource for developing the framework for conceptual design is the Product Development & Design by U&E. The group specifically focused on chapters titled “Concept Generation” and “Concept Selection” to review examples of conceptual design. This resource was selected because it provides a clear framework that we can modify to fit our needs, provides real case studies of existing products, and will assist in our product development objective.

The framework follows:

1. Identify the primary function of the product
2. Establish customer & technical metrics for the new product using a House of Quality.
    a. HOQ
        i. Structured and prioritized list of customer needs organized by primary, secondary, and tertiary importance. Include a reasoning for the ranking of each item included.
        ii. Structured and prioritized list of technical requirements. Include a reasoning for the ranking of each item included.
        iii. Correlate customer needs and technical metrics. If they are correlated, include a reason for positive or negative correlation.
        iv. Establish dependencies within technical metrics.
        v. Benchmarking competing products by utilizing online customer reviews, sensor compatibility, and judgment.
        vi. Set targets for customer needs and technical specifications.
3. Identify two related products for reverse engineering using FAST.
    a. FAST
        i. Understand how a product works.
        ii. Identify the main function of the product.
        iii. Systematically ask “Why?” and “How?” questions.
        iv. Represent the found results in a diagram, where “Why?” questions are closer to the right and “How?” questions are closer to the left.
4. Create a Function Structure, which is an abstract functional representation of the new intended product.
    a. Function Structure
        i. Identify the primary function of the product and sub-functions to realize how a product works. The goal is to understand “Why?” of a product.
        ii. Represent the found results in a diagram.
        iii. Provide an analysis of the found results.

5. For each sub-function in the function structure, generate solution principles for realizing that sub-function, then organize the solution principles into a morphological matrix.
    a. For each conceptual design, represent the product within a morphological matrix.
    b. For each solution principle, identify existing patents for possible conflicts.
6. Generate at least 6 alternative concepts by combining multiple solution principles.
    a. Establish a drawing template for a room that has a door, a window, and a pet.
    b. Using the drawing template, create a simple sketch for each concept design.
    c. Provide an analysis of the found results.
7. Develop a selection criterion to generate a utility function.
8. Utilize a utility function to select 2 feasible concepts for further development.
    a. The utility function will provide us with a detailed method of identifying which concepts are feasible based on the weight of importance for sub-functions realized.
    b. Provide an analysis of the found results.

Summary of the Conceptual Design Process Framework

After individually reviewing the HOQ, FAST diagram, and conceptual design steps, our team met to compare findings and quickly saw common gaps in sub-functions and correlations. This led us to refine both diagrams together to establish a clearer functional foundation. With an improved understanding of the product's required functions, we generated multiple solution principles for each sub-function and combined them into several concept variations. We then applied importance weights to evaluate feasibility, allowing us to collectively determine which conceptual designs were the strongest candidates for our final implementation.

Project Toolkit

The team will work collaboratively, utilizing digital tools to ensure efficiency:

- Google Docs - Centralized shared documents for real-time document editing, version control, and storage of all meeting notes. This is where we will collaboratively present our conceptual design report.
- Google Drawings - Centralized drawing tool with real-time editing for creating FAST and functional structure diagrams.
- Lucidchart - Centralized higher-level diagram tool to design and refine HOQ diagrams.
- Product Manuals and Technical Datasheets - Reviewed from Honeywell, Google Nest, Ecobee, and SimpliSafe
- Digital Team Communication Platforms (Zoom/Slack) - Supported coordination, quick updates, and screen-sharing during FAST and HOQ development sessions.

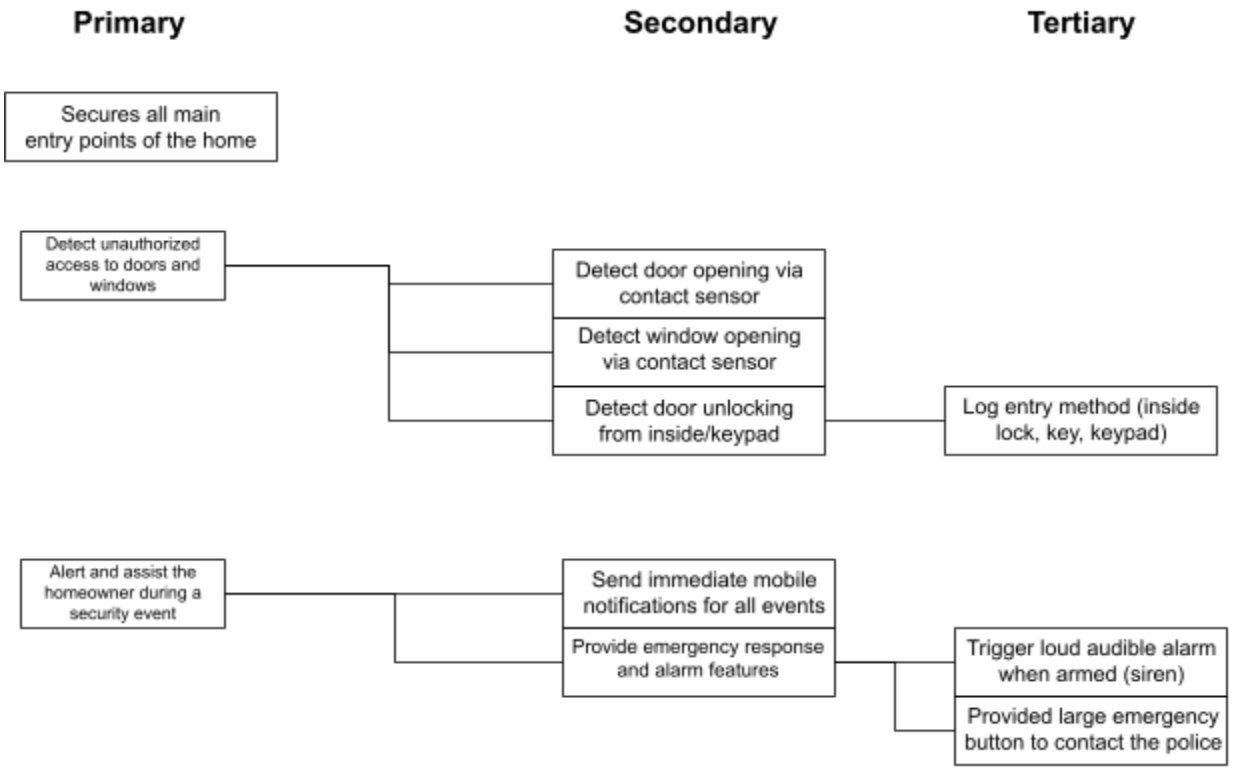
3. Initial Implementation of the Conceptual Design Process

Primary Function of the Product

The primary function of the Smart Home Security System is that it must reliably secure all household entry points by detecting unauthorized access and alerting the homeowner for further action.

House of Quality

Hierarchical Representation of Customer Needs



Prioritized List of Customer Needs:

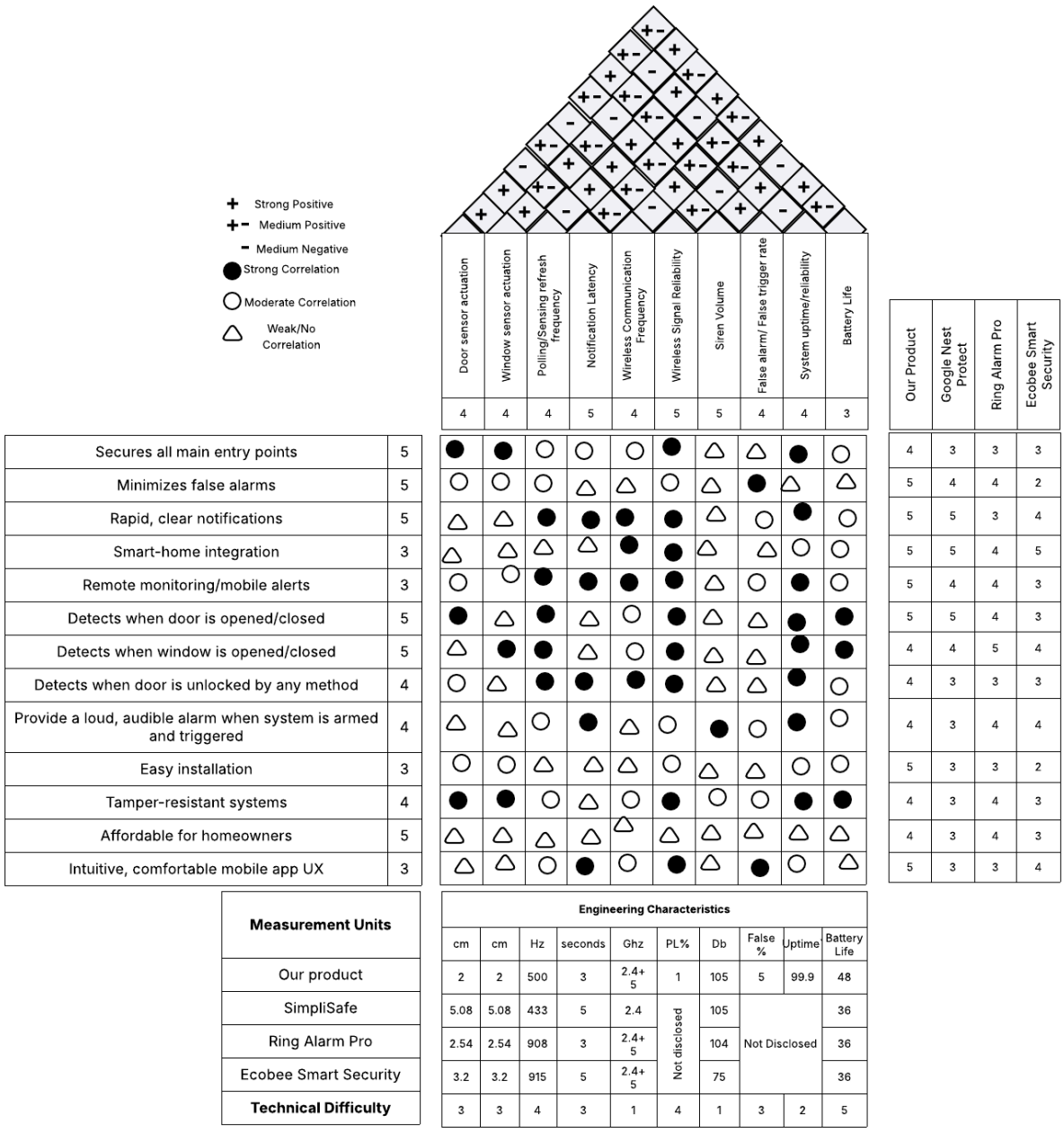
Customer Needs	Target Scale (1/5)
Secures all main entry points	5
Minimizes false alarms	5
Rapid, clear notifications	5
Smart-home integration	3
Remote monitoring/mobile alerts	3
Detect when the door is opened/closed	5
Detect when the window is opened/closed	5
Detect when the door is unlocked by any method	4
Provide a loud, audible alarm when the system is armed and triggered	4
Easy installation	3
Tamper-resistant systems	4
Affordable for homeowners	5
Intuitive, comfortable mobile app UX	3

Prioritized List of Technical Requirements:

Technical Metrics	Target Rating(1/5)	Target Metric
Door sensor actuation (cm)	4	2 cm
Window sensor actuation (cm)	4	2 cm
Polling/Sensing refresh frequency (Hertz/Hz)	4	500 Hz
Notification Latency (seconds)	5	3000 seconds
Wireless Communication Frequency (GHz)	4	2.4 Ghz and 5 Ghz
Wireless Signal Reliability (Packet Loss %)	5	1%

Siren Volume (decibels)	5	105 Db
False alarm/ False trigger rate (%)	4	5%
System uptime/reliability (%)	4	99.9%
Battery Life (Months)	3	48 months

House of Quality Diagram



Function Analysis System Technique (FAST)

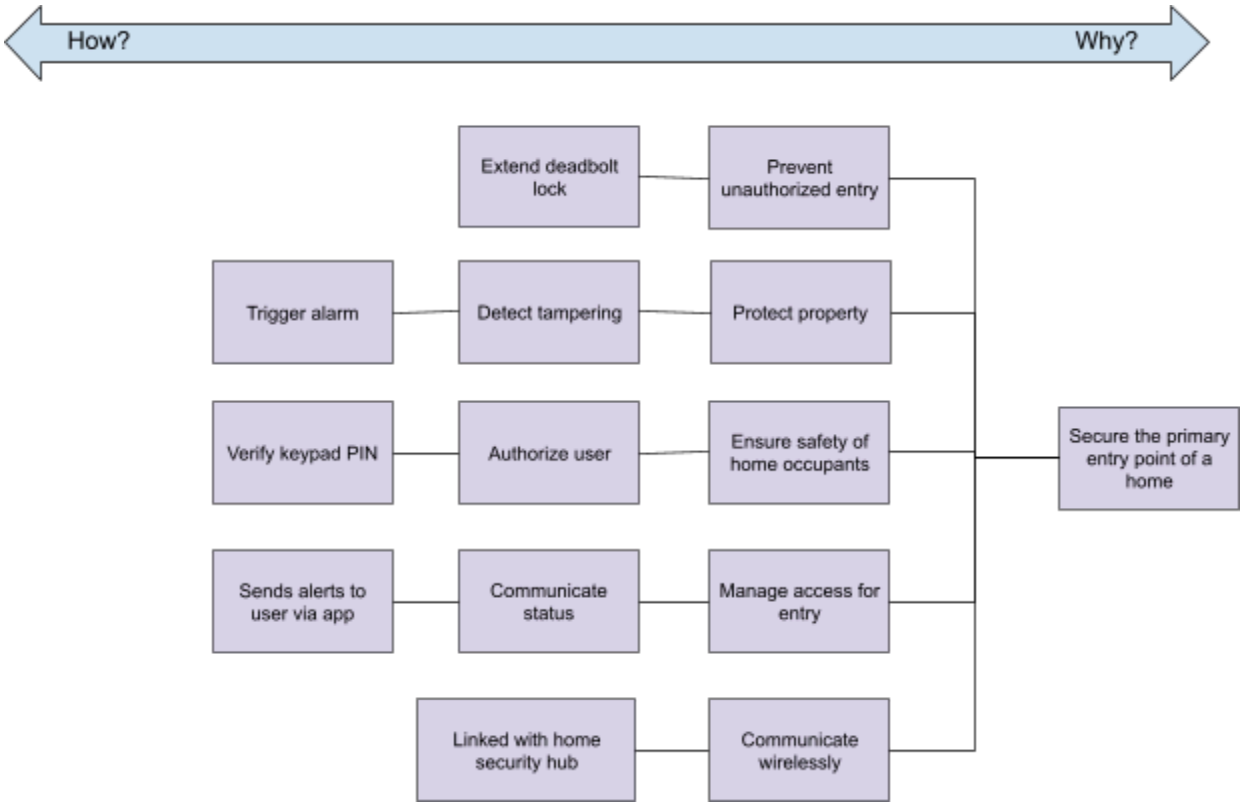
SimpliSafe Smart Lock FAST

The SimpliSafe Smart Lock is an electronic deadbolt attachment that integrates with the SimpliSafe home security system/hub. It replaces the interior part of the deadbolt and is controlled through the system's keypad, mobile app, or automatically with the system's arm/disarm status. It can lock and unlock the door via the app, be programmed with unique PINs for different people, or automatically lock when the system is armed. A key feature is that when the system is armed, the lock engages automatically, and when the system is disarmed, the lock unlocks to prevent false alarms.

It costs approximately $99 to $120 for the lock itself. It also requires the SimpliSafe Base station and a keypad to function, and may need a monthly monitoring plan to access the app and other features remotely.



SimpliSafe Smart Lock FAST diagram:

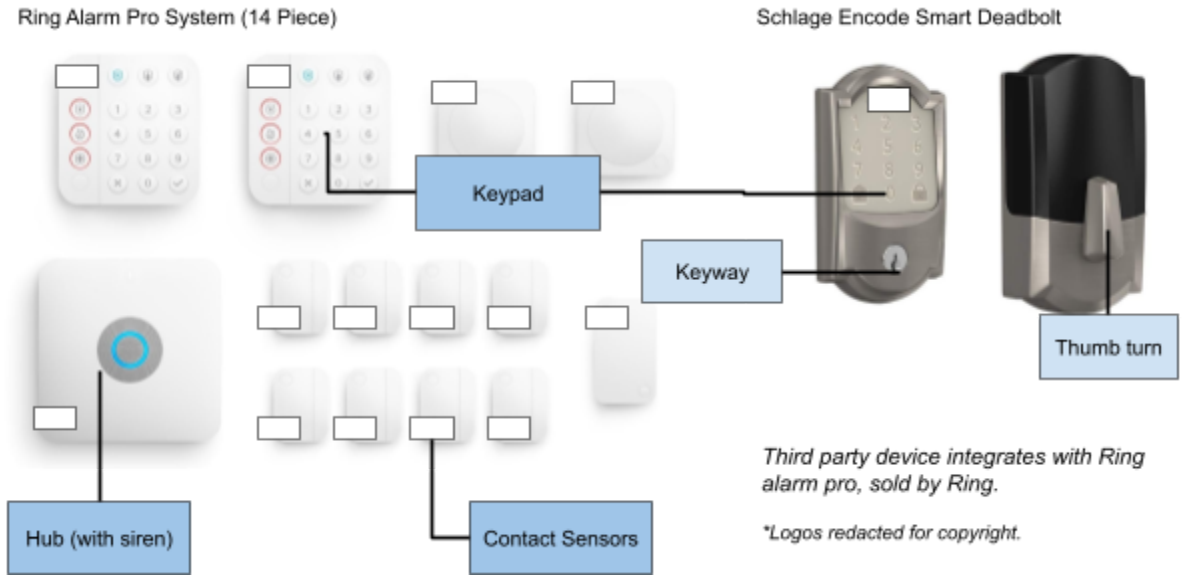


Ring Alarm Pro FAST

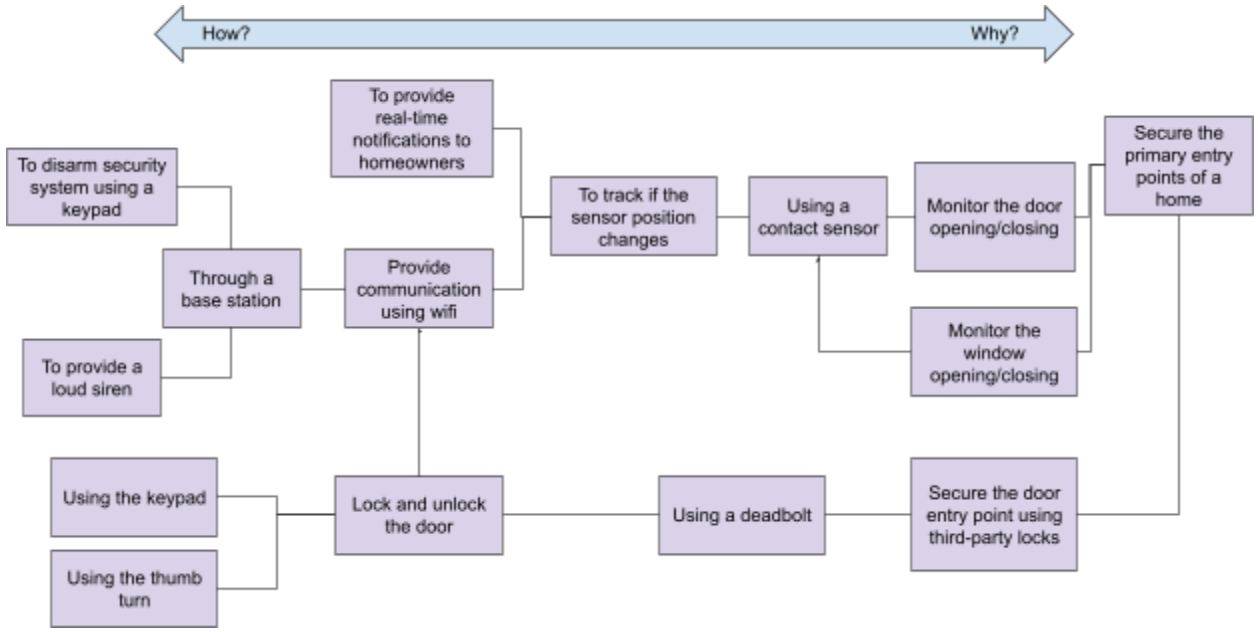
The Ring Alarm Pro is an advanced security system designed to protect homes or businesses. The product is sold by "Ring," which is owned by Amazon. The company sells a 14-piece alarm system with an MSRP of $499.95 and can integrate with the Schlage Encode Smart Deadbolt, with an MSRP of $299.99, bringing the cost of the security system to $799.94.

The primary purpose of Ring Alarm Pro is to secure and monitor the primary entry points of a home, providing homeowners with a sense of security. The product works by ringing a powerful siren and calling law enforcement when the system detects a security event. Security events can only be triggered when the security system is armed (enabled) and can be triggered by opening a door or a window. To minimize false alarms, the security system allows a short amount of time to disarm the system when entering the home. This allows homeowners a brief period to enter their home to disarm the system without calling law enforcement.

Ring Alarm Pro System with Schlage Encode Smart Deadbolt



Ring Alarm Pro with Schlage Encode Smart Deadbolt FAST:



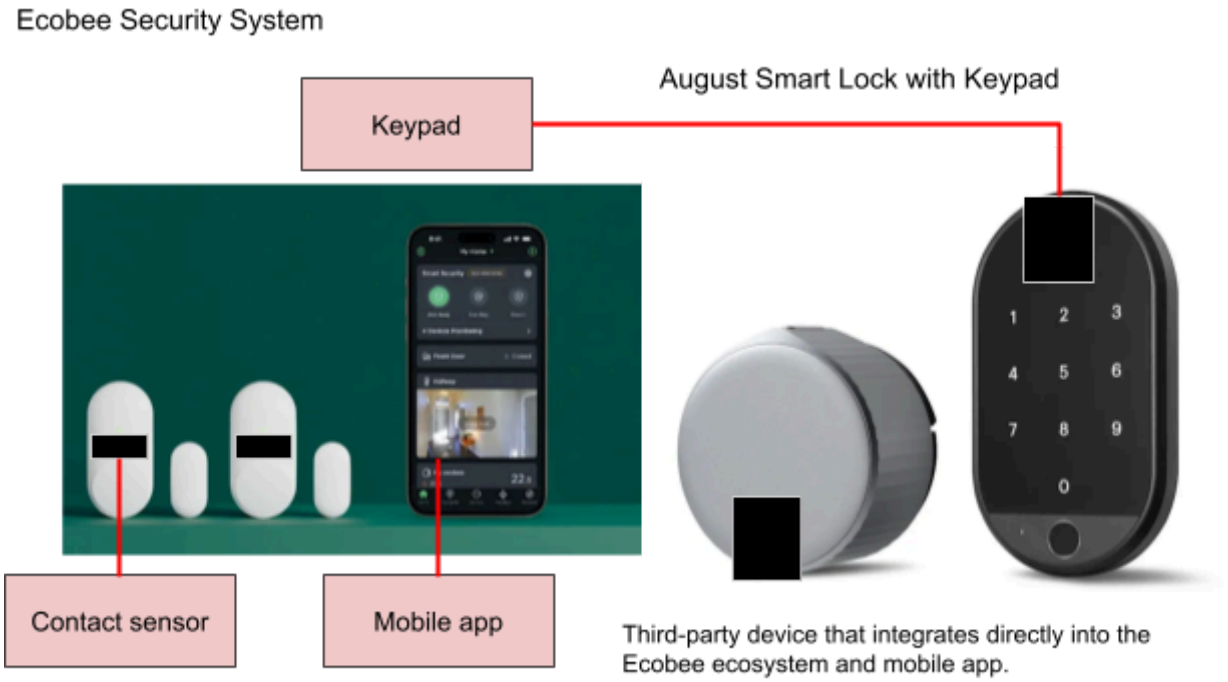
Ecobee Smart Security System FAST

The Ecobee Smart Security System is a series of products that secure the most vulnerable parts of your home, taken to the next level with their smart technology. The products are sold by “Ecobee”, a Canada-based home automation company that makes a variety of home-security technologies. Through the integration of their SmartSensors into Ecobee’s mobile app, the main function of the product is the ability to distinguish between standard (normal) household behavior and potential threats to not only reduce false alarms but also respond appropriately when needed. The sensors are designed, under normal circumstances, to alert the customer as soon as windows and doors open or close. When the sensors detect unusual/unexpected activity when the system is armed, a notification is sent directly to the Ecobee mobile app, notifying you of potentially dangerous activity.

Although Ecobee itself does not produce a smart lock, its systems directly support third-party companies that do produce a smart lock to integrate themselves within the Ecobee ecosystem. Ecobee offers a range of bundles for their security systems, ranging from $200 - $600 MSRP. The smart locks are a separate product offered by companies like August, with their deadbolt smart locks being offered at $249.99 MSRP.

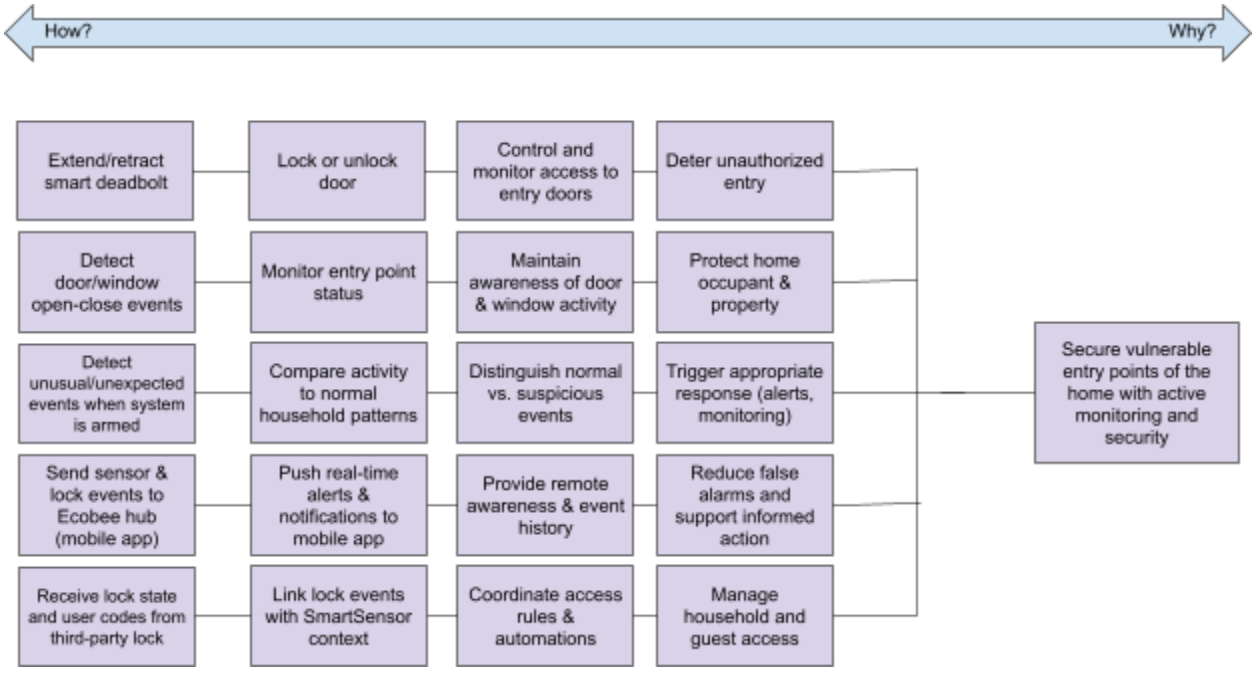
Ecobee’s technology directly integrates into their mobile app, where users are offered varying levels of subscription-based plans that determine the level of security they are offered. The standard plan, currently priced at $5.00/month, provides some self-monitoring features such as alerts, basic automation, and cloud video storage for a single camera, focusing on homeowners who prioritize awareness and need remote access to their home without heavy monitoring. The complete plan, on the other hand, provides 24/7 monitoring, unlimited storage for footage, and prioritizes emergency response capabilities. This plan is priced at $10.00/month and is intended for users who prioritize fast escalation and responses to home security threats.

Ecobee Security System and August Smart Lock



**Logos redacted for copyright.*

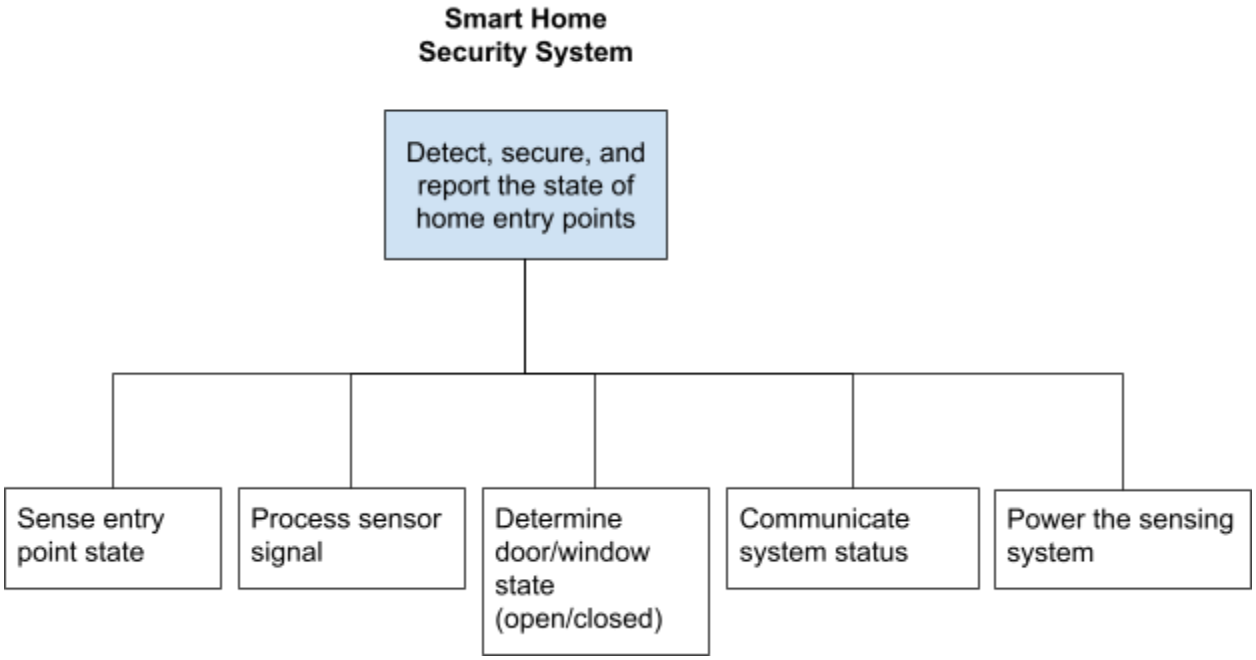
Ecobee Smart Security System FAST



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Function Structure

The function structure was developed by first defining the system's core purpose of detecting, securing, and reporting the state of home entry points and then breaking that objective into the essential actions required to achieve it. Before constructing the diagram, we completed a session of reviewing user needs, clarifying system boundaries, and analyzing current smart lock technologies to determine the necessary sensing, processing, communication, and power requirements. This groundwork allowed us to logically decompose the system into its key sub-functions, such as sensing the entry state, processing sensor signals, determining whether the door or window is open or closed, communicating system status, and powering the sensing components. The resulting structure provides a clear, organized foundation for subsequent concept generation and system design.



Morphological Matrix

The purpose of the initial morphological analysis is to systematically explore the solution space for each major sub-function identified in the functional structure. By creating a structured option set and combining solution principles analytically, the team develops early system-level concepts grounded in functional reasoning rather than arbitrary brainstorming.

The morphological matrix below represents the first iteration of mapping solution principles (SPs) to core sub-functions. At this stage, the matrix emphasizes breadth over selectivity to ensure that solution pathways are not prematurely constrained. The focus is on capturing technically plausible variations, even if some may later be eliminated due to feasibility constraints.

Sub Function	SP1	SP2	SP3	SP4
Sense entry point state	Magnetic reed switch (contact sensor)	Hall-effect sensor	Optical break-beam sensor	Mechanical microswitch
Process Sensor Signal	Low-pass digital filtering	Debounce algorithm	Threshold comparison logic	Stateful logic + error checking
Determine door/window state	Binary open/closed evaluation	Compare against calibrated reference	Integrate lock status (inside/keypad)	Time-based pattern recognition
Communicate system status	Bluetooth Low Energy	Wi-Fi to cloud/app	Zigbee to Hub	Local siren + LED indicator output
Power the sensing system	Replace the alkaline battery	Rechargeable Li-ion battery	Solar/Light Powered	Low-voltage wired power

Alternative Product Concepts

Using the morphological matrix, the team generated six early system-level concepts. These are not final designs, but structured prototypes that represent coherent combinations of solution principles. Refined designs will be presented in the "Final Implementation of the Conceptual Design Process" section of this document.

Concept 1: Optical Detection LE

Sub Function	SP1	SP2	SP3	SP4
Sense entry point state	Magnetic reed switch (contact sensor)	Hall-effect sensor	Optical break-beam sensor	Mechanical microswitch
Process Sensor Signal	Low-pass digital filtering	Debounce algorithm	Threshold comparison logic	Stateful logic + error checking
Determine door/window state	Binary open/closed evaluation	Compare against calibrated reference	Integrate lock status (inside/keypad)	Time-based pattern recognition
Communicate system status	Bluetooth Low Energy	Wi-Fi to cloud/app	Zigbee to Hub	Local siren + LED indicator output
Power the sensing system	Replace the alkaline battery	Rechargeable Li-ion battery	Solar/Light Powered	Low-voltage wired power

Selected Solution Principles:

- Sense entry point state: Optical break-beam sensor
- Process sensor signal: stateful logic + error checking
- Determine state: Compare against calibrated reference
- Communicate system status: Bluetooth Low Energy
- Power the sensing system: Rechargeable Li-ion battery

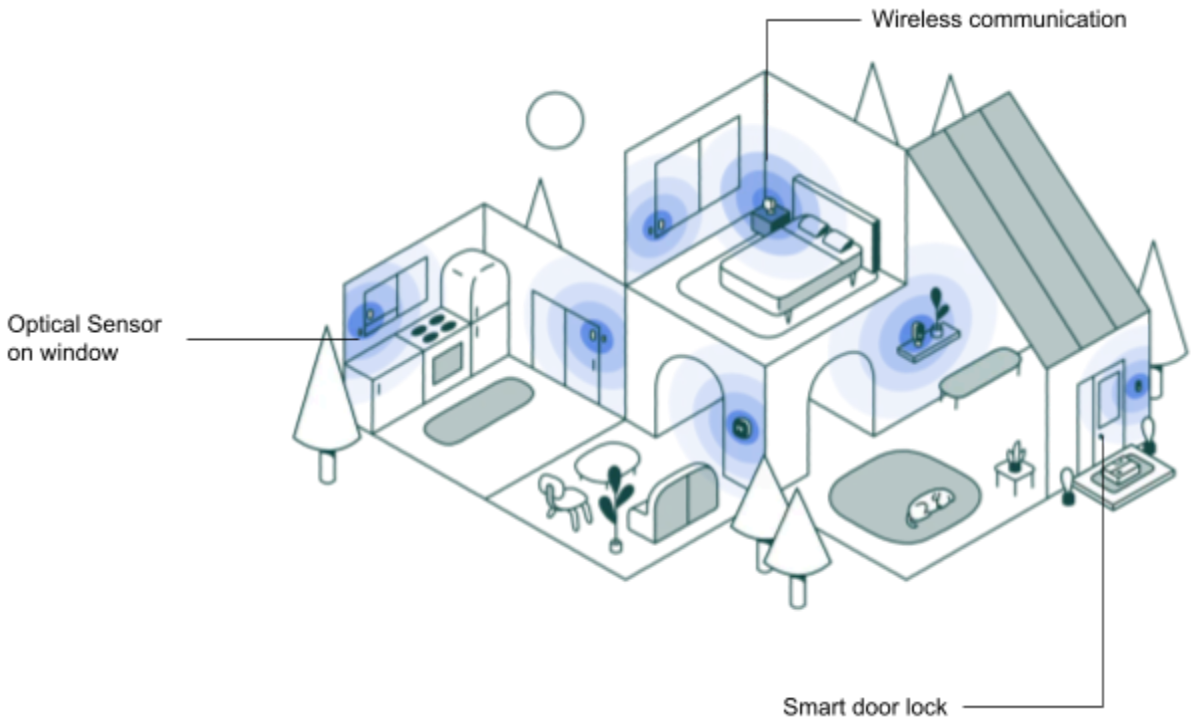
Analytical Strengths:

- Low power consumption and efficient battery usage
- Simple and reliable sensing
- Minimizes false readings and signal noise
- High initial accuracy and consistent measure quality after setup
- Easy, direct communication to consumer devices

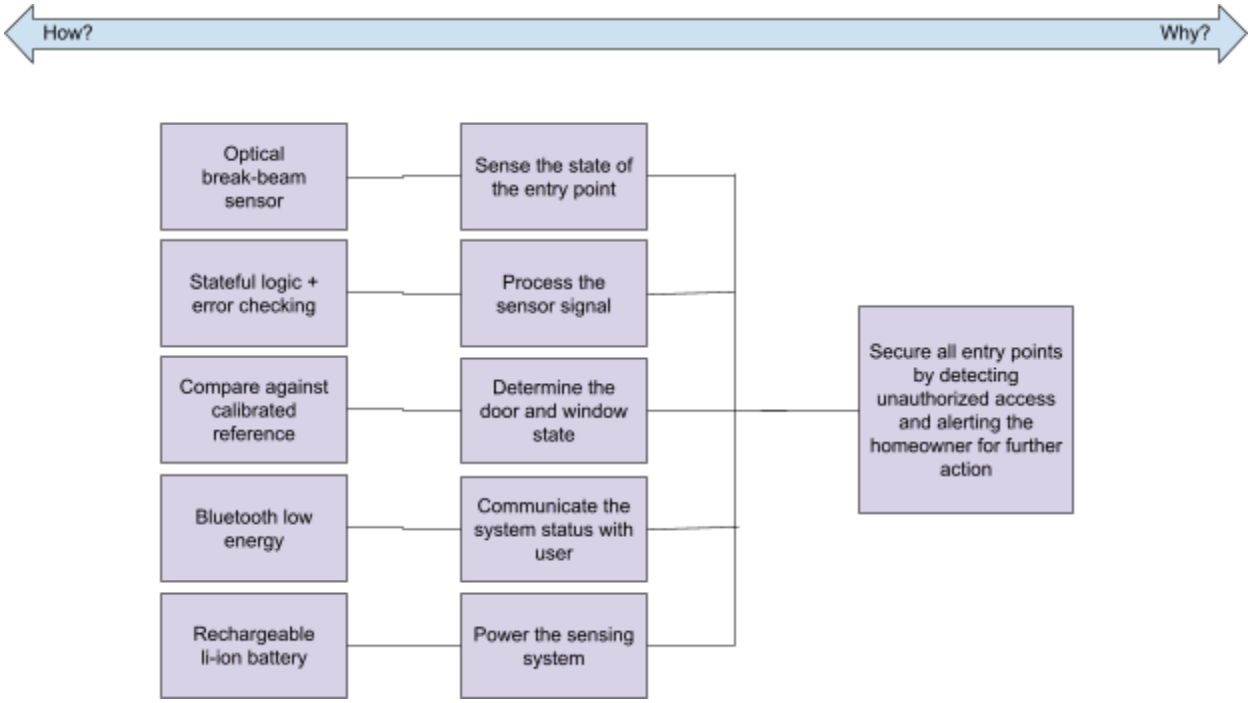
- A rechargeable battery allows for portability and reduces costs and waste

Analytical Weaknesses:

- Optical break-beam sensors are highly susceptible to dust, dirt, and condensation.
- Sensor drift or environmental changes over time may require frequent recalibration.
- BLE has a short effective range
- Implementing and debugging stateful logic can be complex
- Battery management requires additional hardware



FAST diagram:



Concept 2: Magnetic Detection LE

Sub Function	SP1	SP2	SP3	SP4
Sense entry point state	Magnetic reed switch (contact sensor)	Hall-effect sensor	Optical break-beam sensor	Mechanical microswitch
Process Sensor Signal	Low-pass digital filtering	Debounce algorithm	Threshold comparison logic	Stateful logic + error checking
Determine door/window state	Binary open/closed evaluation	Compare against calibrated reference	Integrate lock status (inside/keypad)	Time-based pattern recognition
Communicate system status	Bluetooth Low Energy	Wi-Fi to cloud/app	Zigbee to Hub	Local siren + LED indicator output
Power the sensing system	Replace the alkaline battery	Rechargeable Li-ion battery	Solar/Light Powered	Low-voltage wired power

Selected Solution Principles:

- Sense entry point state: Magnetic reed switch
- Process sensor signal: Low-pass digital filtering
- Determine state: Binary open/closed evaluation
- Communicate system status: Bluetooth Low Energy
- Power the sensing system: Replaceable alkaline battery

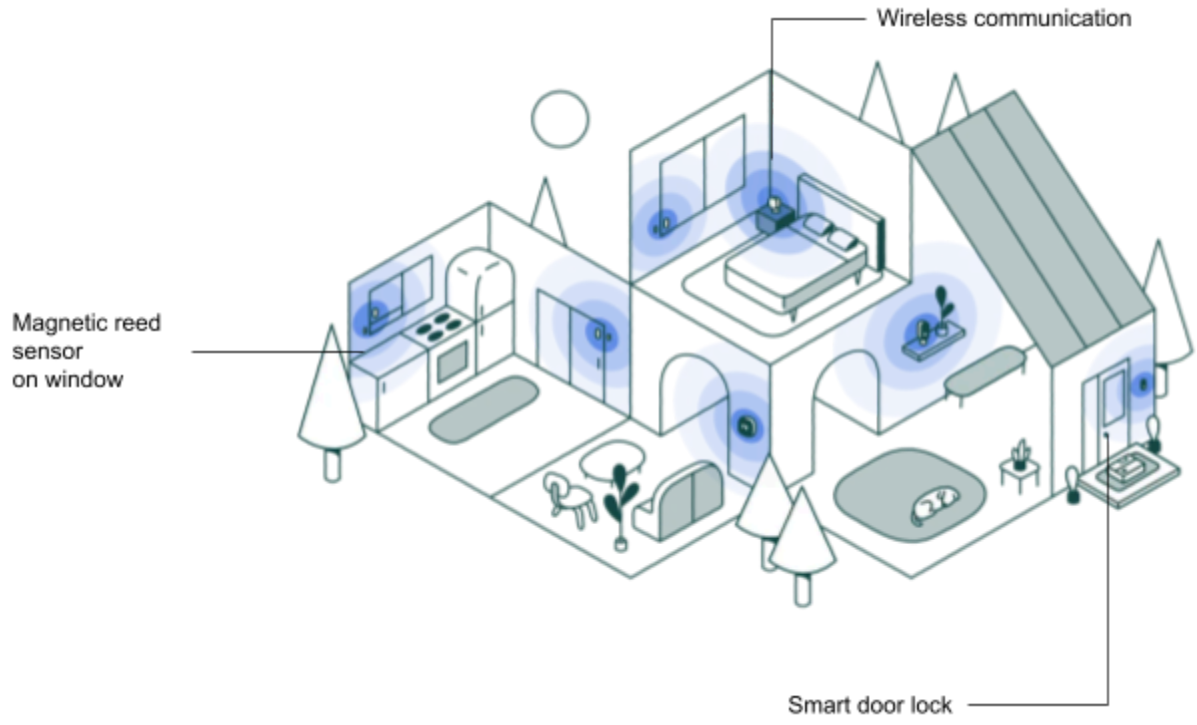
Analytical Strengths:

- Highly reliable and well-established sensing method
- Lowest power consumption means a longer battery life
- Minimal false positives due to stable filtering
- Very low cost and easy to manufacture
- BLE is simple, secure, and ideal for short-range home communication

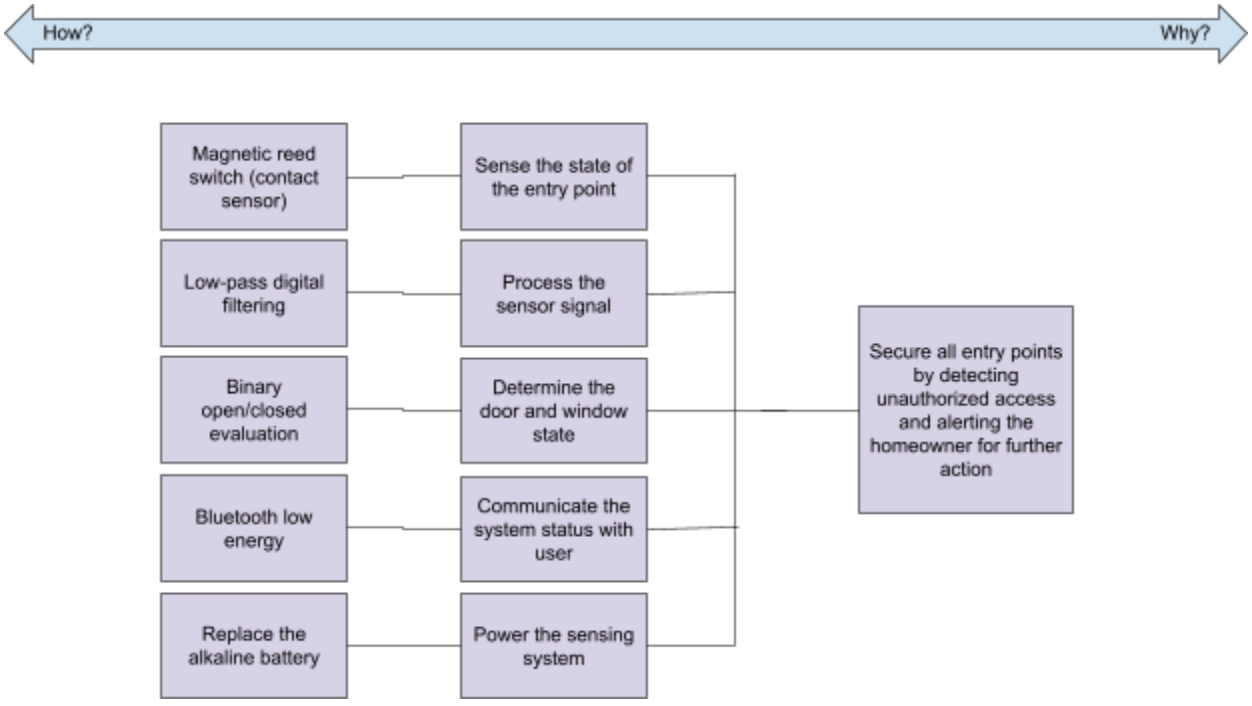
Analytical Weaknesses:

- Lacks door-unlock context (no smart-lock integration)

- Limited range compared to Wi-Fi or Zigbee
- No advanced signal interpretation beyond binary logic
- Not optimal for larger homes or multi-floor deployments



FAST diagram:



Concept 3: Hall Effect Detection

Sub Function	SP1	SP2	SP3	SP4
Sense entry point state	Magnetic reed switch (contact sensor)	Hall-effect sensor	Optical break-beam sensor	Mechanical microswitch
Process Sensor Signal	Low-pass digital filtering	Debounce algorithm	Threshold comparison logic	Stateful logic + error checking
Determine door/window state	Binary open/closed evaluation	Compare against calibrated reference	Integrate lock status (inside/keypad)	Time-based pattern recognition
Communicate system status	Bluetooth Low Energy	Wi-Fi to cloud/app	Zigbee to Hub	Local siren + LED indicator output
Power the sensing system	Replace the alkaline battery	Rechargeable Li-ion battery	Solar/Light Powered	Low-voltage wired power

Selected Solution Principles:

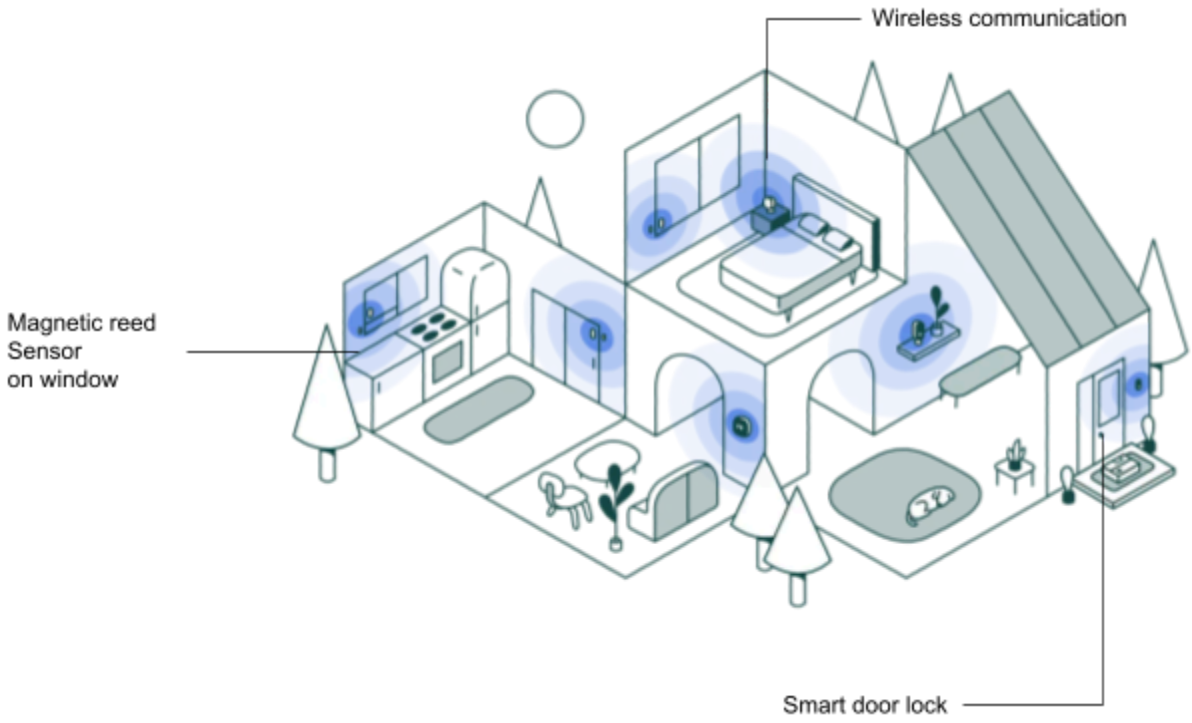
- Sense entry point state: Hall-effect sensor
- Process sensor signal: Debounce algorithm
- Determine state: Integrate lock status
- Communicate system status: Zigbee to Hub
- Power the sensing system: Solar/Light powered

Analytical Strengths:

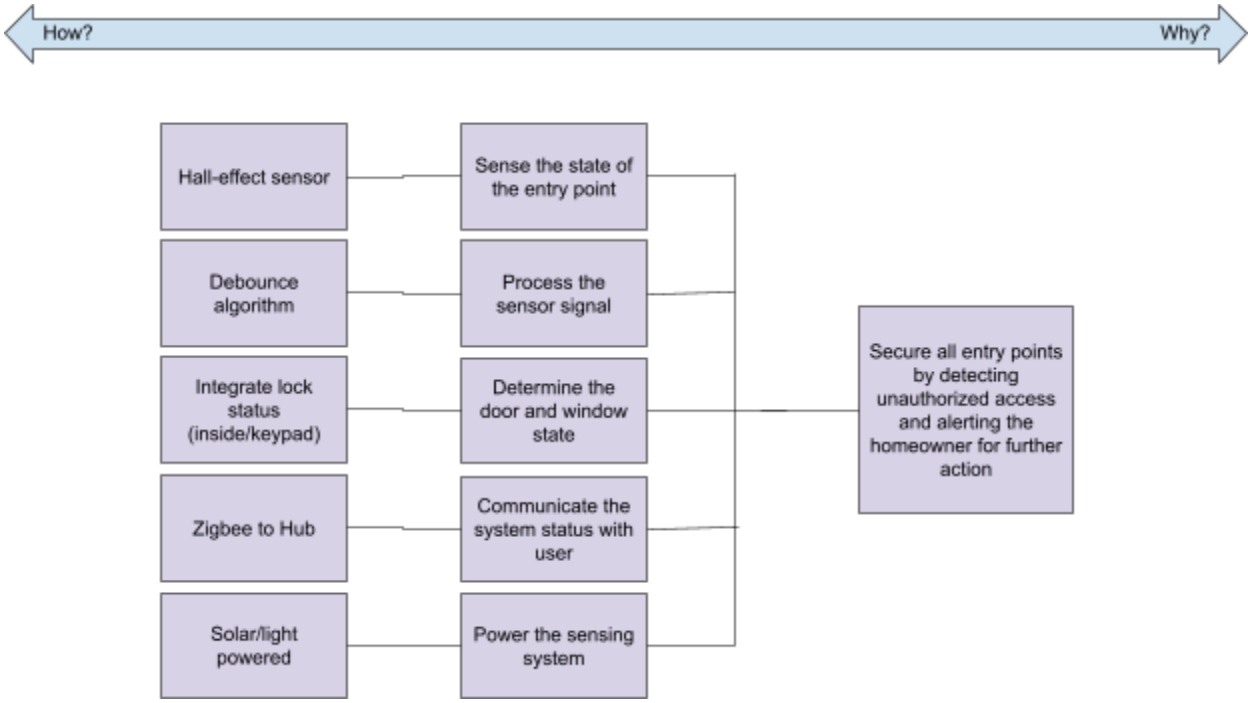
- Contactless sensing helps to improve stability and durability
- Less sensitive to alignment than reed switches
- Zigbee provides a strong range with mesh support
- Debounce algorithm logic reduces false triggers
- Reduced battery maintenance through solar power
- Lock integration provides proper feedback to the user

Analytical Weaknesses:

- Solar power becomes less reliable in low-light conditions and areas
- The Zigbee hub contributes to cost and setup complexity
- Debounce algorithm logic can lead to a slight detection delay
- Lock integration increases firmware complexity



FAST diagram:



Concept 4: Cloud-Based

Sub Function	SP1	SP2	SP3	SP4
Sense entry point state	Magnetic reed switch (contact sensor)	Hall-effect sensor	Optical break-beam sensor	Mechanical microswitch
Process Sensor Signal	Low-pass digital filtering	Debounce algorithm	Threshold comparison logic	Stateful logic + error checking
Determine door/window state	Binary open/closed evaluation	Compare against calibrated reference	Integrate lock status (inside/keypad)	Time-based pattern recognition
Communicate system status	Bluetooth Low Energy	Wi-Fi to cloud/app	Zigbee to Hub	Local siren + LED indicator output
Power the sensing system	Replace the alkaline battery	Rechargeable Li-ion battery	Solar/Light Powered	Low-voltage wired power

Selected Solution Principles:

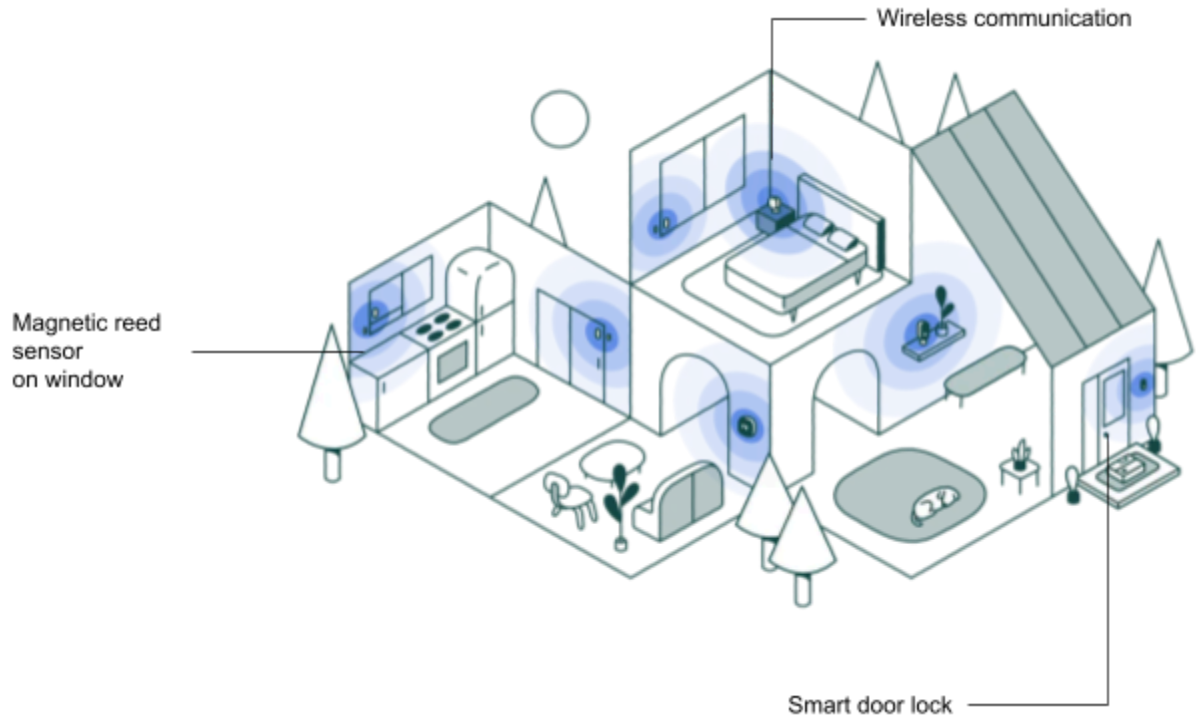
- Sense entry point state: Magnetic reed switch
- Process sensor signal: Debounce Algorithm
- Determine state: Integrate lock status
- Communicate system status: Wifi to cloud/app
- Power the sensing system: Rechargeable Li-ion battery

Analytical Strengths:

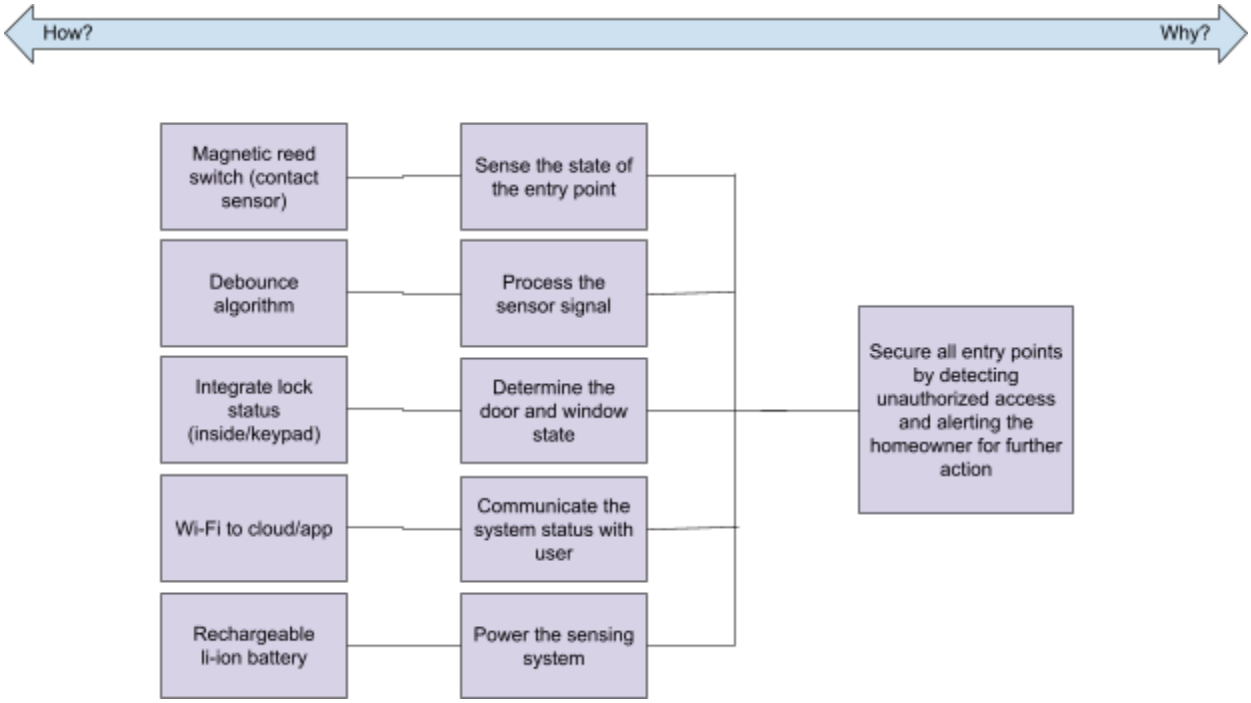
- Reed switch offers reliable and low-cost sensing
- Debounce Algorithm logic minimizes small false alarms
- Wifi connectivity allows for long-range and hub-free connectivity
- Rechargeable battery reduces long-term cost
- Provides lock-status data

Analytical Weaknesses:

- High power usage due to Wifi connectivity
- Vulnerable to network outages and congestion
- Reed switches require precise alignment
- Cloud-based alerts may be susceptible to latency
- Lock interaction complicates firmware



FAST diagram:



Concept 5: Microswitch Siren

Sub Function	SP1	SP2	SP3	SP4
Sense entry point state	Magnetic reed switch (contact sensor)	Hall-effect sensor	Optical break-beam sensor	Mechanical microswitch
Process Sensor Signal	Low-pass digital filtering	Debounce algorithm	Threshold comparison logic	Stateful logic + error checking
Determine door/window state	Binary open/closed evaluation	Compare against calibrated reference	Integrate lock status (inside/keypad)	Time-based pattern recognition
Communicate system status	Bluetooth Low Energy	Wi-Fi to cloud/app	Zigbee to Hub	Local siren + LED indicator output
Power the sensing system	Replace the alkaline battery	Rechargeable Li-ion battery	Solar/Light Powered	Low-voltage wired power

Selected Solution Principles:

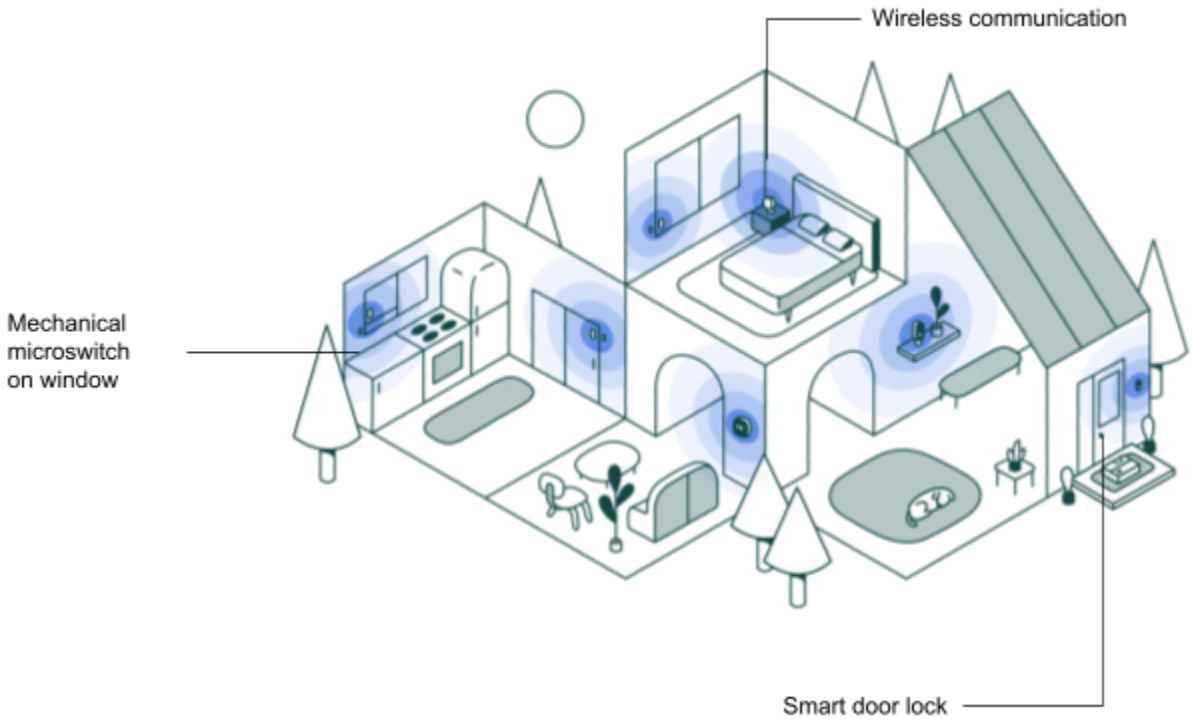
- Sense entry point state: Mechanical microswitch
- Process sensor signal: Threshold comparison logic
- Determine state: Binary open/closed evaluation
- Communicate system status: Local siren + LED indicator output
- Power the sensing system: Low-voltage wired power

Analytical Strengths:

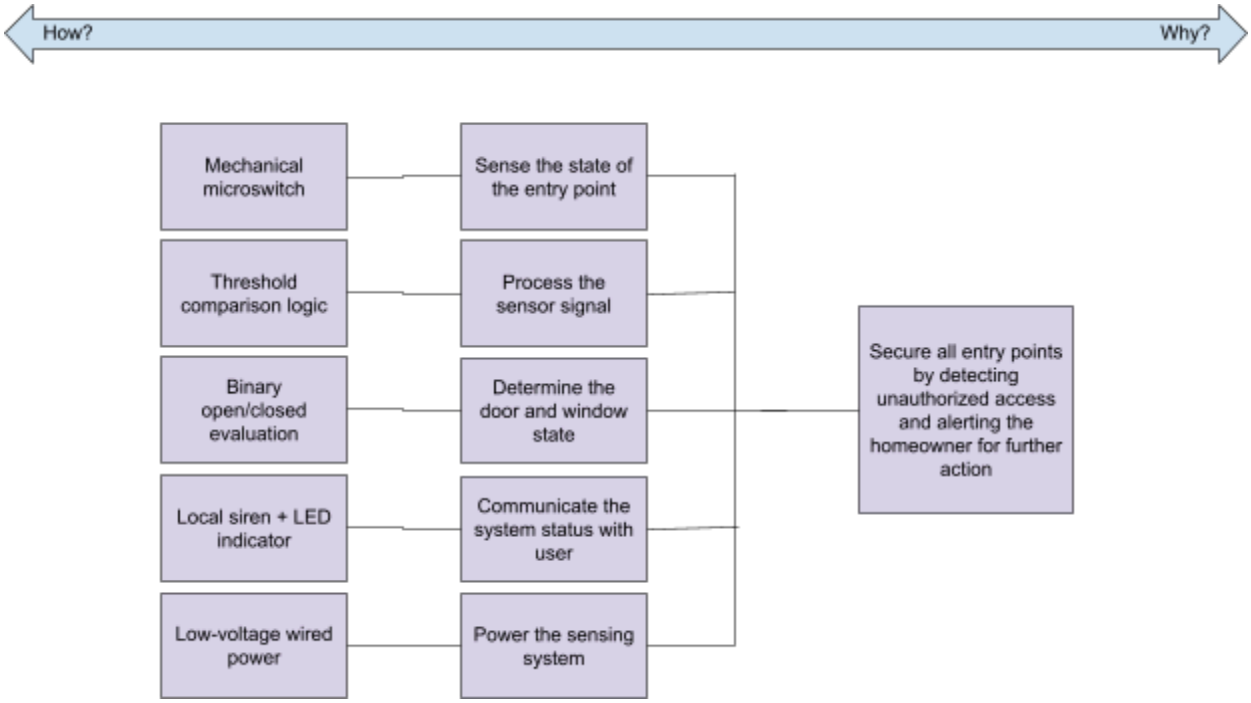
- Physical actuation of lock reduces false positives
- Simple and reliable logic
- Low cost and robust
- Local siren and LED indicator offers instant feedback
- Continuous uptime with wired power

Analytical Weaknesses:

- No mobile alerts or remote notifications
- Harder to scale in comparison to wireless systems
- Limited feedback (only open/closed data)
- Requires precise alignment of lock actuation



FAST diagram:



Concept 6: High Coverage System

Sub Function	SP1	SP2	SP3	SP4
Sense entry point state	Magnetic reed switch (contact sensor)	Hall-effect sensor	Optical break-beam sensor	Mechanical microswitch
Process Sensor Signal	Low-pass digital filtering	Debounce algorithm	Threshold comparison logic	Stateful logic + error checking
Determine door/window state	Binary open/closed evaluation	Compare against calibrated reference	Integrate lock status (inside/keypad)	Time-based pattern recognition
Communicate system status	Bluetooth Low Energy	Wi-Fi to cloud/app	Zigbee to Hub	Local siren + LED indicator output
Power the sensing system	Replace the alkaline battery	Rechargeable Li-ion battery	Solar/Light Powered	Low-voltage wired power

Selected Solution Principles:

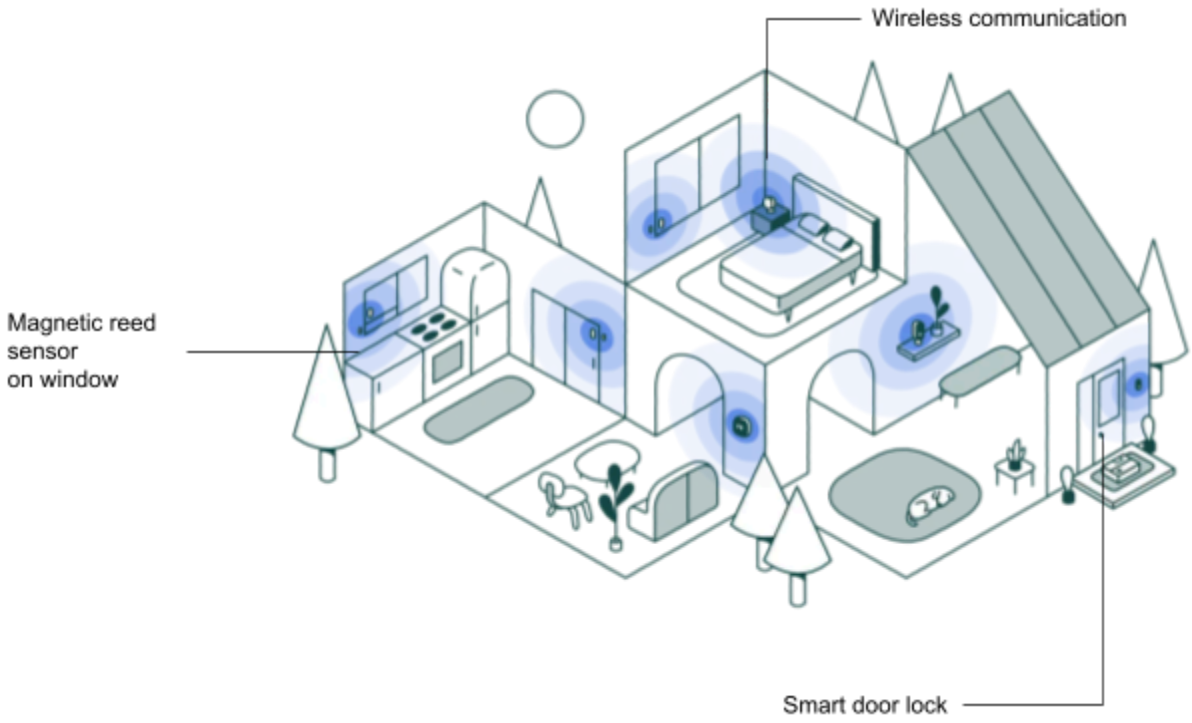
- Sense entry point state: Magnetic reed switch
- Process sensor signal: Low-pass digital filtering
- Determine state: Binary open/close evaluation
- Communicate system status: Zigbee to hub
- Power the sensing system: Replace the alkaline battery

Analytical Strengths:

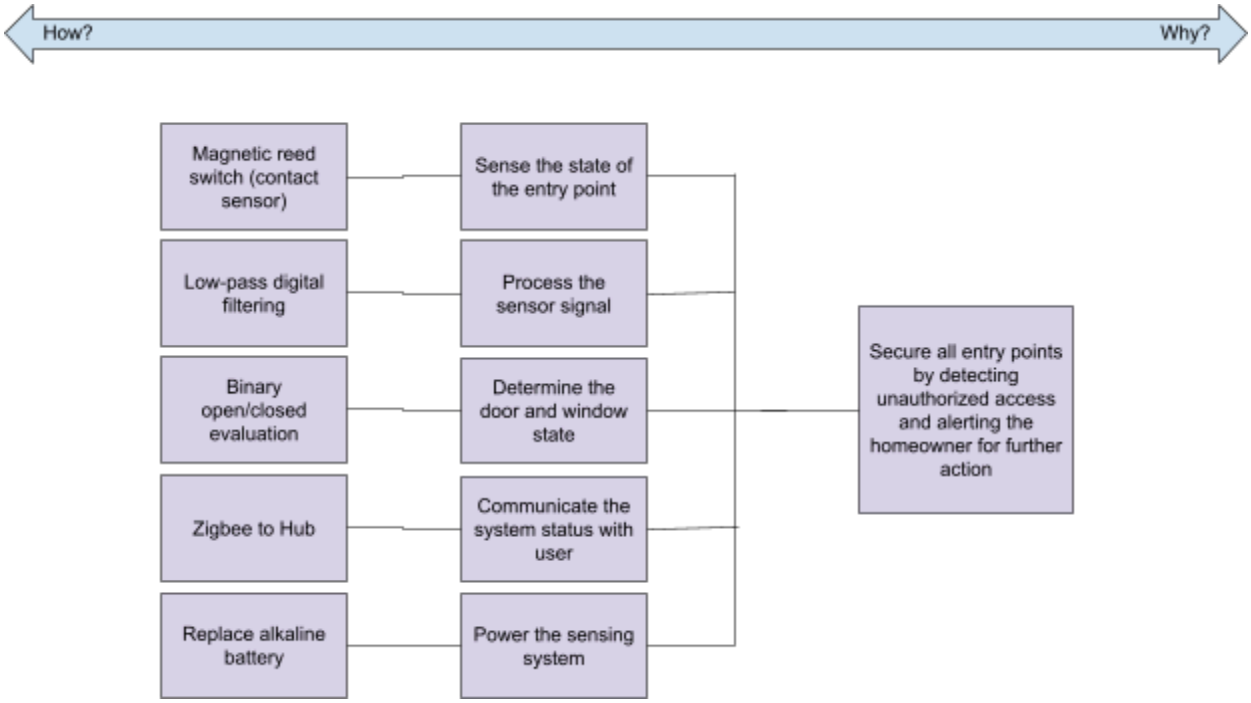
- Reed switch offers reliable and low-cost sensing
- Low-pass digital filtering reduces noise and false alarms
- Zigbee mesh contributes to whole-home coverage
- Lower power consumption extends battery life
- Easy installation with simple batteries

Analytical Weaknesses:

- Alkaline batteries require replacement periodically.
- Lack of context for the user with binary feedback
- Requires Zigbee hub software
- Redd sensors can drift out of alignment over time



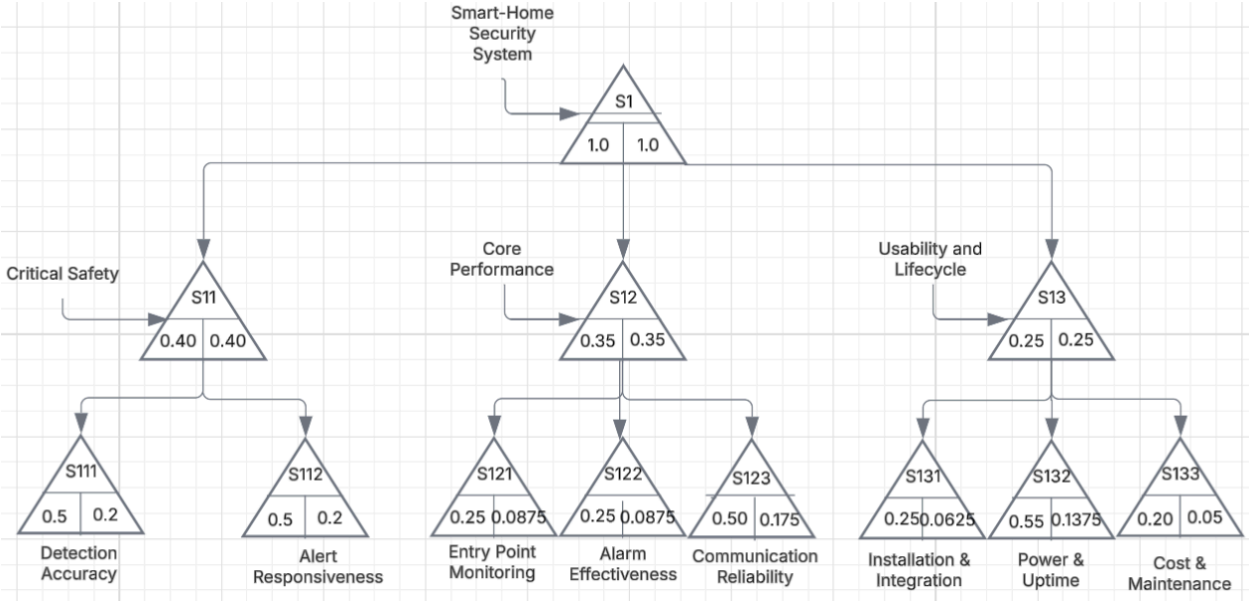
FAST diagram:



Utility Function

The purpose of the Utility Function is to help provide us with a structured, quantitative, and unbiased framework for evaluating and selecting the most viable product concept from the set of alternative concepts we created. It helps translate customer needs into more measurable criteria, allowing for a data-driven decision-making process that isn't just subjective.

- **Objective Comparison:** Allows for a direct comparison of fundamentally different design concepts against a common criterion.
- **Traceability for Customer Needs:** Each criterion and its weight are based on the HOQ, allowing for a selection that is traceable to the customer's needs.
- **Identification of Trade-offs:** The scoring process defines the strengths and weaknesses of each concept, helping identify the critical design compromises related to cost, performance, and feasibility.



	Selection Criteria	Absolute Weight	C1		C2		C3		C4		C5		C6	
			Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility
Detection Accuracy	S111	0.2	3	0.6	4	0.8	3	0.6	3	0.6	3	0.6	3	0.6
Alert Responsiveness	S112	0.2	2	0.4	3	0.6	2	0.4	3	0.6	4	0.8	3	0.6
Entry Point Monitoring	S121	0.08	3	0.24	3	0.24	3	0.24	3	0.24	2	0.16	4	0.32
Alarm Effectiveness	S122	0.08	3	0.24	2	0.16	2	0.16	2	0.16	5	0.4	2	0.16
Communication Reliability	S123	0.1	2	0.2	2	0.2	3	0.3	2	0.2	2	0.2	4	0.4
Installation & Integration	S131	0.06	4	0.24	3	0.18	2	0.12	2	0.12	2	0.12	3	0.18
Power & Uptime	S132	0.1	3	0.3	2	0.2	4	0.4	2	0.2	4	0.4	2	0.2
Cost & Maintenance	S133	0.05	5	0.25	3	0.15	2	0.1	4	0.2	3	0.15	3	0.15
Total Utility				2.23		2.53		2.32		2.32		2.83		2.61

Feasible Product Concepts

Based on the utility function, Concepts 5 and 6 are feasible to pursue development.
Concept 5 prioritizes alert responsiveness while optimizing the usage of power. This concept has the highest feasibility with a high rating of 2.83 out of 5.
Concept 6 is more oriented to detection accuracy, ensuring that there is less likelihood for false responses based on our forms of sensor usage. This concept has the second-highest feasibility, with a high rating of 2.61 out of 5.

4. Critical Review of Initial Implementation

Related Product Use Case Analysis

Product Dissected: Ecobee Smart Security System
Scenario: A homeowner is at work. An intruder attempts to force open the back door, which is equipped with a SmartSensor but no smart lock. The system detects the vibration/open event, verifies that the system is "armed," and sends an alert to the user's phone.

Sub Function + Subsystem Table

Purpose	Sub Functions	Sub Systems	Represented in FAST diagram?
Intruder forces door/window	Detects door/window open-close events.	Contact Sensor	Yes
System checks mode	Detects unusual events when armed.	Micro controller	Yes
System ignores cat	Differentiate between normal vs. suspicious events.	Motion Processing Algorithm	Yes
User alerted	Push real-time alerts to the mobile app.	Wi-Fi Module/Cloud Server	Yes

Correlation and Resolution of Differences

Match Quality	Relatively high, the core detection loop in the use case aligns with the manage household/guest access branch of the FAST diagram
Identified Difference	In the use case, the homeowner attempts to view a video feed to verify the intrusion

Resolution	The current FAST diagram focuses on sensors and locks but lacks a specific function box for "Capture/Stream Video," though it implies "Remote Awareness." To align these, we would need to add "Capture visual evidence" to the How direction of the FAST diagram or acknowledge that this specific diagram excludes the camera subsystem.

New Product Use Case

Our Intended Product: Smart Home Security System
Scenario: Unexpected Entry
The user is not currently at home and does not expect any visitors. While they are away from their home, they receive an urgent notification that the system's sensors detected that a window was opened unexpectedly, with a prompt that allows them to verify a break-in and quickly contact law enforcement.

Key Sub-Functions	Actions Performed
Monitor armed entry sensors	Continuously watch window/door sensors while the user is away and/or the system is armed.
Detect window-open event	Detect when a closed window has been opened.
Check the event against the expected behavior.	Compare the event to previous occurrences, occupancy status, and arming state to see if it's unexpected.
Classify the event as a potential intrusion.	Decide whether this window event is suspicious enough to trigger an alert state.
Generate an urgent alert message	Compose a notification with sensor ID, location, time, and risk level.
Deliver an alert to the user's mobile app	Push urgent notifications to the user's phone in real time.
Present verification options in the app	Show context (sensor ID, recent activity) and let the user decide between "false alarm" or "possible break-in."
Initiate protective response	When break-in is confirmed (or no response within a set time), trigger the siren and lock

	available smart locks.
Provide quick contact to law enforcement/monitoring service	Offer a one-tap solution for the user (or monitoring center) to be able to contact police/emergency services.
Log and update incident status.	Record the event and user/monitoring actions in the system history for future reference and update the system state after the incident is closed.

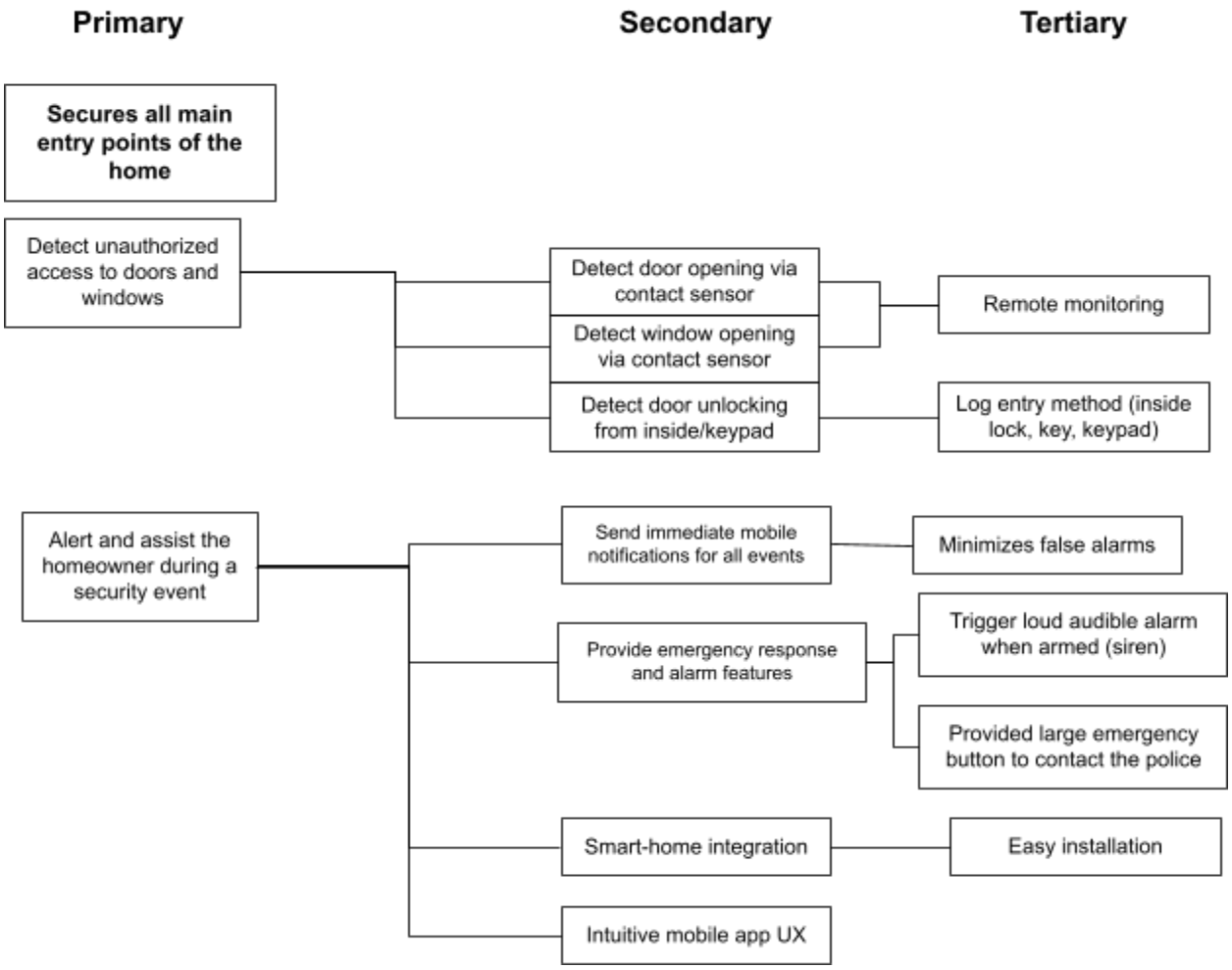
5. Final Implementation of the Conceptual Design Process

Primary Function of the Product

The primary function of the Smart Home Security System is that it must reliably secure all household entry points by detecting unauthorized access and alerting the homeowner for further action.

House of Quality

Hierarchical Representation of Customer Needs



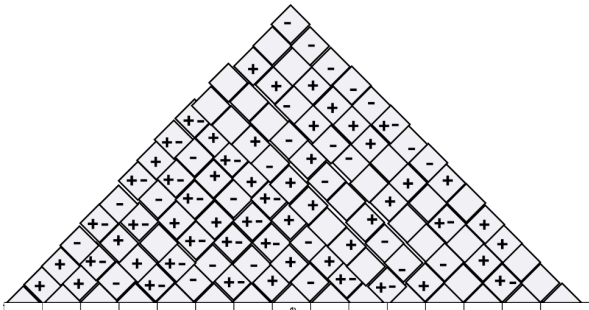
Prioritized List of Customer Needs:

Customer Needs	Target Scale (1/5)
Secures all main entry points	5
Minimizes false alarms	5
Rapid, clear notifications	5
Smart-home integration	3
Remote monitoring/mobile alerts	3
Detect when the door is opened/closed	5
Detect when the window is opened/closed	5
Detect when the door is unlocked by any method	4
Provide a loud, audible alarm when the system is armed and triggered	4
Easy installation	3
Tamper-resistant systems	4
Affordable for homeowners	5
Intuitive, comfortable mobile app UX	3

Prioritized List of Technical Requirements

Technical Metrics	Target Rating(1/5)	Target Metric
Door sensor actuation (cm)	4	2 cm
Window sensor actuation (cm)	4	2 cm
Polling/Sensing refresh frequency (Hertz/Hz)	4	500 Hz
Notification Latency (seconds)	5	3000 seconds
Wireless Communication Frequency (GHz)	4	2.4 Ghz and 5 Ghz
Wireless Signal Reliability (Packet Loss %)	5	1%
Siren Volume for the Hub (decibels)	5	105 Db
False alarm/ False trigger rate (%)	4	5%
System uptime/reliability (%)	4	99.9%
Battery Life (Months)	3	48 months
Bandwidth Consumption	3	8 Mbps
Expandability	2	60 - 70
Tamper Detection Response Time	3	5 seconds
Wireless Operating Range	4	80 meters
End-to-End Data Security Level	4	AES-256 + TLS 1.3

House of Quality Diagram



+ Strong Positive
+- Medium Positive
- Medium Negative
● Strong Correlation
○ Moderate Correlation
△ Weak/No Correlation

		Door sensor actuation	Window sensor actuation	Polling/Sensing refresh frequency	Notification Latency	Wireless Communication Frequency	Wireless Signal Reliability	Siren Volume (Hub)	False alarm/ False trigger rate	System uptime/reliability	Battery Life	Bandwidth Consumption	Expandability	Tamper Detection Response Time	Wireless Operating Range	End-to-End Data Security Level	Our Product	SimpliSafe	Ring Alarm Pro	Ecobee Smart Security
		4	4	4	5	4	5	5	4	4	3	3	2	3	4	4				
Secures all main entry points	5	●	●	○	○	○	●	△	△	●	○	△	●	○	●	○	4	3	3	3
Minimizes false alarms	5	○	○	○	△	△	○	△	●	△	△	△	△	○	○	△	5	4	4	2
Rapid, clear notifications	5	△	△	●	●	●	●	△	○	●	○	△	△	●	○	△	5	5	3	4
Smart-home integration	3	△	△	△	△	●	●	△	△	○	○	○	○	△	○	○	5	5	4	5
Remote monitoring/mobile alerts	3	○	○	●	●	●	●	△	○	●	○	○	○	●	○	●	5	4	4	3
Detects when door is opened/closed	5	●	△	●	△	○	●	△	△	●	●	△	○	△	●	○	5	5	4	3
Detects when window is opened/closed	5	△	●	●	△	○	●	△	△	●	●	△	○	△	●	○	4	4	5	4
Detects when door is unlocked by any method	4	○	△	●	●	●	●	△	△	●	○	△	○	△	●	●	4	3	3	3
Provide a loud, audible alarm when system is armed and triggered	4	△	△	○	●	△	○	●	○	●	○	△	△	○	△	△	4	3	4	4
Easy installation	3	○	○	△	△	△	○	△	△	○	○	○	○	△	●	△	5	3	3	2
Tamper-resistant systems	4	●	●	○	△	○	●	○	○	●	●	△	△	●	△	●	4	3	4	3
Affordable for homeowners	5	△	△	△	△	△	△	△	△	△	△	△	△	△	△	△	4	3	4	3
Intuitive, comfortable mobile app UX	3	△	△	○	●	○	●	△	●	○	△	○	○	○	△	○	5	3	3	4

Measurement Units	Engineering Characteristics														
	cm	cm	Hz	seconds	Ghz	PL%	Db	False %	Uptime	Battery Life	Mbps	# of units	sec	meters	Encryption Standard
Our product	2	2	500	3	2.4+ 5	1	105	5	99.9	48	8	70	5	80	AES-256 + TLS 1.3
SimpliSafe	5.08	5.08	433	5	2.4	Not disclosed	105	Not Disclosed		36	6	18	30 - 60	76	AES-128 + TLS/SSL
Ring Alarm Pro	2.54	2.54	908	3	2.4+ 5		104			36	5	232	30	6	AES-128 + AES-256
Ecobee Smart Security	3.2	3.2	915	5	2.4+ 5		75			36	6	32	N/A	20	TLS 1.2 + AES-128
Technical Difficulty	3	3	4	3	1	4	1	3	2	5	3	3	2	3	3

Function Analysis System Technique (FAST)

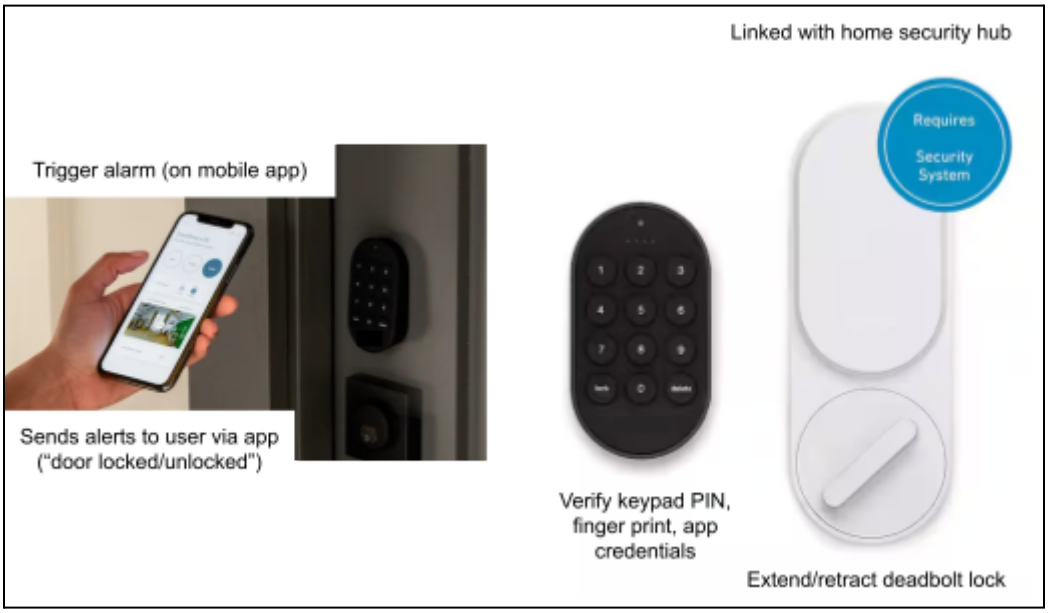
FAST is a systematic, structured technique to represent relationships between Function (WHY) and Form (HOW). The main goal of a FAST diagram is to define the relationship between the functions through HOW and WHY logic. We will be dissecting three existing products to understand their functionality, which will help us generate sub-functions for our product.

SimpliSafe Smart Lock FAST

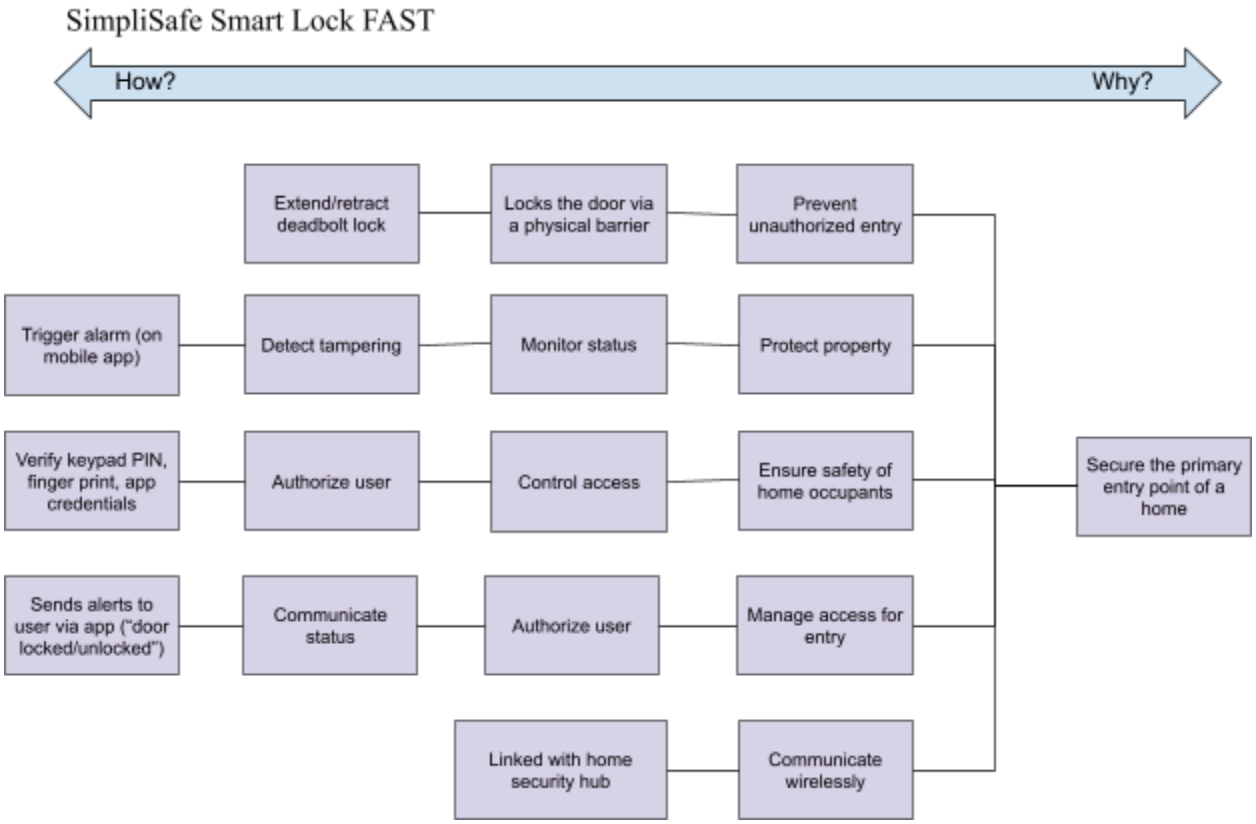
The SimpliSafe Smart Lock is an electronic deadbolt attachment that integrates with the SimpliSafe home security system/hub. It replaces the interior part of the deadbolt and is controlled through the system's keypad, mobile app, or automatically with the system's arm/disarm status. It can lock and unlock the door via the app, be programmed with unique PINs for different people, or automatically lock when the system is armed. A key feature is that when the system is armed, the lock engages automatically, and when the system is disarmed, the lock unlocks to prevent false alarms.

It costs approximately $99 to $120 for the lock itself. It also requires the SimpliSafe Base station and a keypad to function, and may need a monthly monitoring plan to access the app and other features remotely.

SimpliSafe Smart Lock



SimpliSafe Smart Lock FAST diagram:

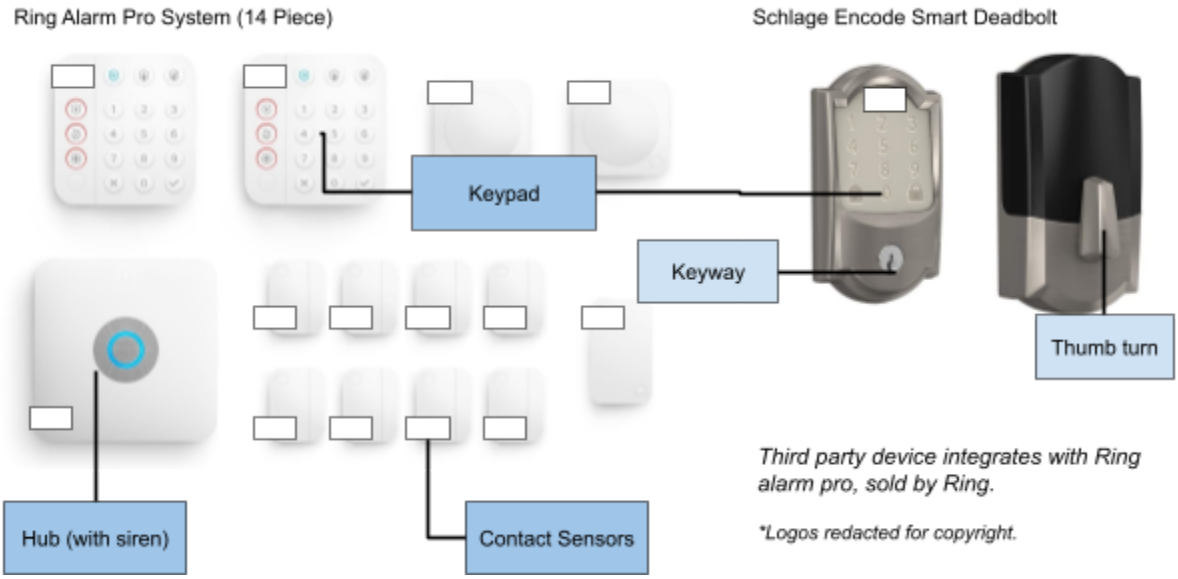


Ring Alarm Pro FAST

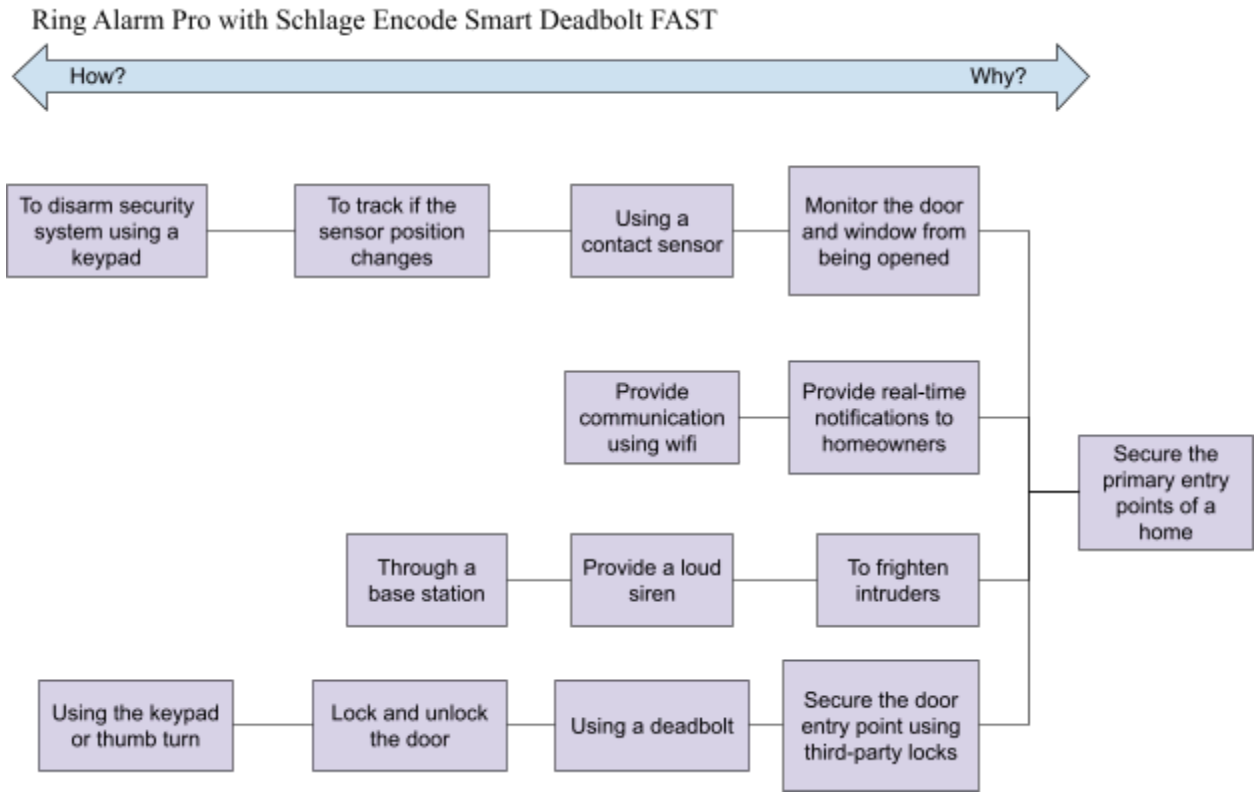
The Ring Alarm Pro is an advanced security system designed to protect homes or businesses. The product is sold by "Ring," which is owned by Amazon. The company sells a 14-piece alarm system with an MSRP of $499.95 and can integrate with the Schlage Encode Smart Deadbolt, with an MSRP of $299.99, bringing the cost of the security system to $799.94.

The primary purpose of Ring Alarm Pro is to secure and monitor the primary entry points of a home, providing homeowners with a sense of security. The product works by ringing a powerful siren and calling law enforcement when the system detects a security event. Security events can only be triggered when the security system is armed (enabled) and can be triggered by opening a door or a window. To minimize false alarms, the security system allows a short amount of time to disarm the system when entering the home. This allows homeowners a brief period to enter their home to disarm the system without calling law enforcement.

Ring Alarm Pro System with Schlage Encode Smart Deadbolt



Ring Alarm Pro with Schlage Encode Smart Deadbolt FAST:

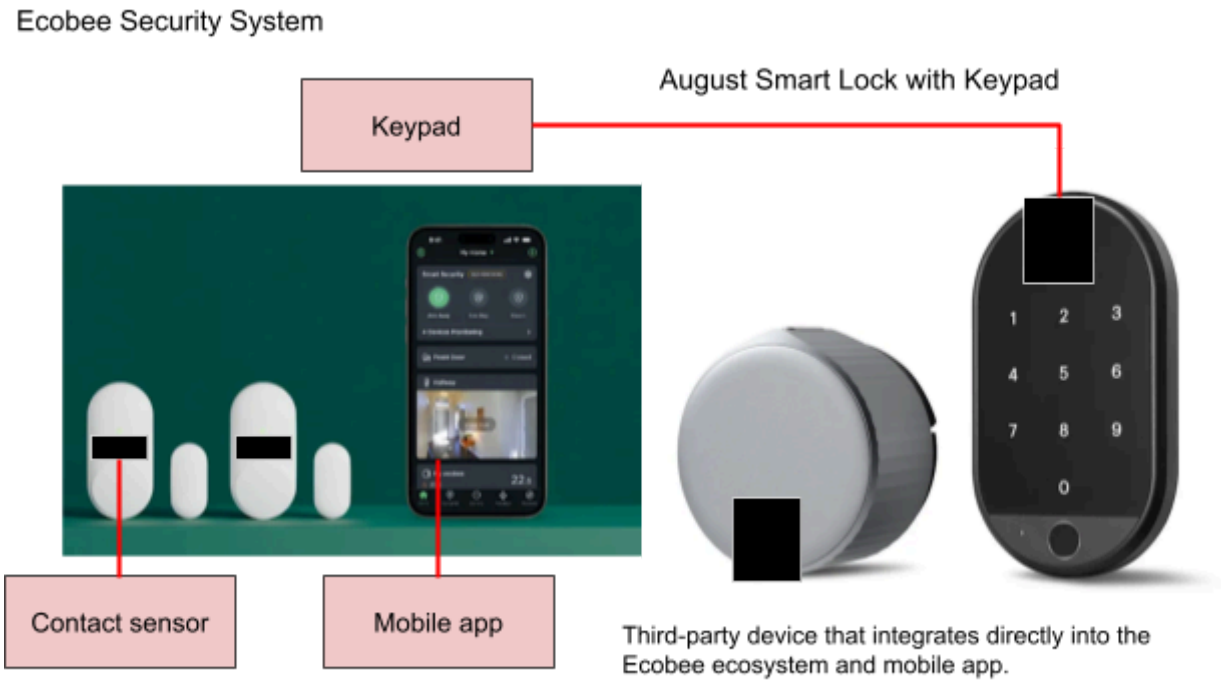


Ecobee Smart Security System FAST

The Ecobee Smart Security System is a series of products that secure the most vulnerable parts of your home, taken to the next level with their smart technology. The products are sold by "Ecobee", a Canada-based home automation company that makes a variety of home-security technologies. Through the integration of their SmartSensors into Ecobee's mobile app, the main function of the product is the ability to distinguish between standard (normal) household behavior and potential threats to not only reduce false alarms but also respond appropriately when needed. The sensors are designed, under normal circumstances, to alert the customer as soon as windows and doors open or close. When the sensors detect unusual/unexpected activity when the system is armed, a notification is sent directly to the Ecobee mobile app, notifying you of potentially dangerous activity.

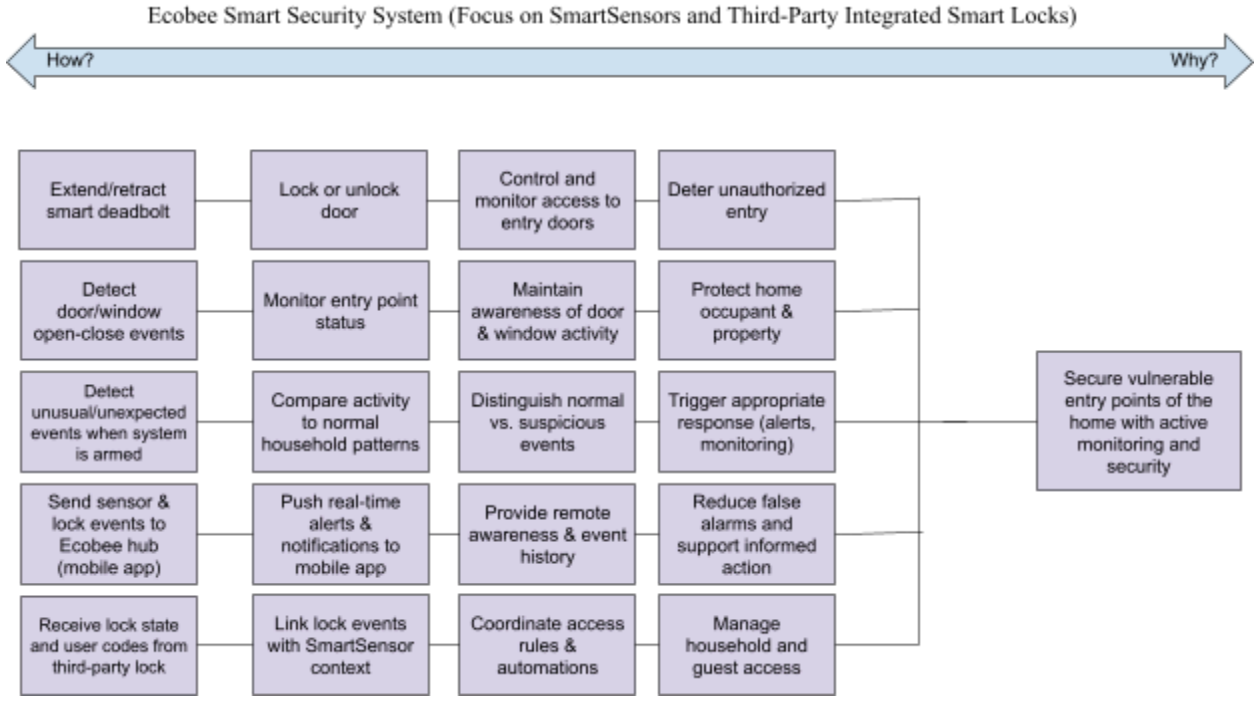
Although Ecobee itself does not produce a smart lock, its systems directly support third-party companies that do produce a smart lock to integrate themselves within the Ecobee ecosystem. Ecobee offers a range of bundles for their security systems, ranging from $200 - $600 MSRP. The smart locks are a separate product offered by companies like August, with their deadbolt smart locks being offered at $249.99 MSRP.

Ecobee Security System and August Smart Lock



**Logos redacted for copyright.*

Ecobee Smart Security System FAST

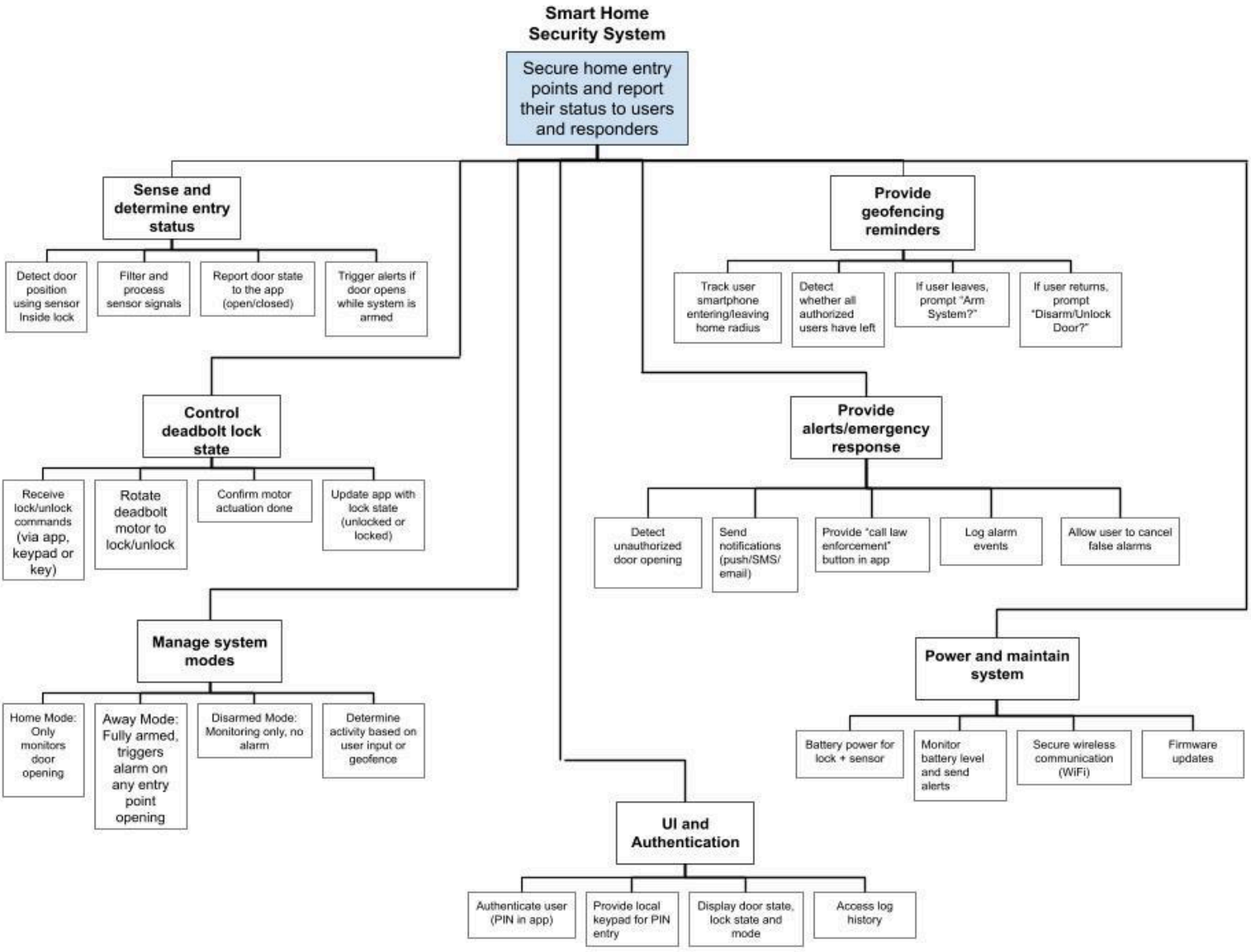


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Function Structure:

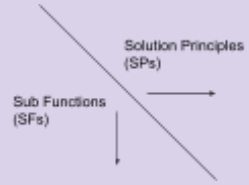
The function structure shown below represents a solution-neutral breakdown of the Smart Home Security System. At the highest level, the system must secure home entry points and report their status to users and responders.

This primary function branches into major secondary functions, such as sensing and determining entry status, controlling the deadbolt lock, managing system modes, authenticating users, providing geofencing reminders, issuing alerts or emergency responses, and maintaining power to the system. Each of these secondary functions is further decomposed into supporting functions that ultimately express the requirements. By organizing the system in this way, the function structure reveals and clarifies how each supporting function contributes to the overall purpose, ultimately ensuring complete coverage of user needs.



Morphological Matrix

The row headers for the Morphological Matrix are taken directly from the Function Structure that we developed in the previous section. We then brainstormed multiple "Solution Principles" columns" to physically realize each of these abstract functions.

	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sensor	MEMS contact vibration sensor (detects opening as a contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actuator	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based on multi-user geofencing	Mode selection via lock keypad
UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on the lock face	App-based authentication (PIN/biometric)	Voice-assistant integration (Alexa/Google/ Siri)
Provide geofencing reminders	Basic GPS geofence on a single user	Multi-user presence detection (last user leaving triggers a reminder)	Wi-Fi/Bluetooth presence override (detect if someone is still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed)	SMS or email alerts	"Call law enforcement" button in app	Audible lock alarm (local siren on tamper/forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optional low-voltage power	Solar trickle charger (for exterior units)

Internet connectivity	Wi-Fi	Mobile Data	/	/
Deadbolt + Lock material	Stainless Steel	Hardened Steel	Bronze	Brass
Micro Processor	SnapDragon	Samsung Exynos Chip	Custom ARM Chip	/
Emergency Power	Power Terminals	Lithium-ion Rechargeable Port	Solar Power	/

Alternative Product Concepts

Each of the following concepts represents a distinct “path” through the Morphological Matrix. For example, Concept 1 combines the “Reed switch sensor" with a physical keypad, whereas Concept 6 combines “MEMS vibration sensors” with app-based authentication instead.

Concept 1: Minimalist Smart Deadbolt

	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sensor	MEMS contact vibration sensor (detects opening as a contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actuator	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based on multi-user geofencing	Mode selection via lock keypad
UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on the lock face	App-based authentication (PIN/biometric)	Voice-assistant integration (Alexa/Google/Siri)
Provide geofencing reminders	Basic GPS geofence on a single user	Multi-user presence detection (last user leaving triggers a reminder)	Wi-Fi/Bluetooth presence override (detect if someone is still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed)	SMS or email alerts	"Call law enforcement" button in app	Audible lock alarm (local siren on tamper/forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optional low-voltage power	Solar trickle charger (for exterior units)
Internet connectivity	Wi-Fi	Mobile Data	/	/
Deadbolt + Lock material	Stainless Steel	Hardened Steel	Bronze	Brass
Micro Processor	SnapDragon	Samsung Exynos Chip	Custom ARM Chip	/
Emergency Power	Power Terminals	Lithium-ion Rechargeable Port	Solar Power	/

Design Logic: A simple, cost-effective smart lock optimized for reliability and essential security, with minimal electronics.

Selected Solution Principles

- Sense entry status: Reed switch

- Control lock: DC motor + gear train
- Modes: App toggles Home/Away/Disarmed
- UI/Auth: Physical keypad
- Geofencing: Basic single-user GPS reminder
- Alerts: Push notifications
- Power: Replaceable AA battery pack with backup emergency power terminals
- Connectivity: Wi-Fi
- Processor: Snap Dragon
- Material: Stainless Steel

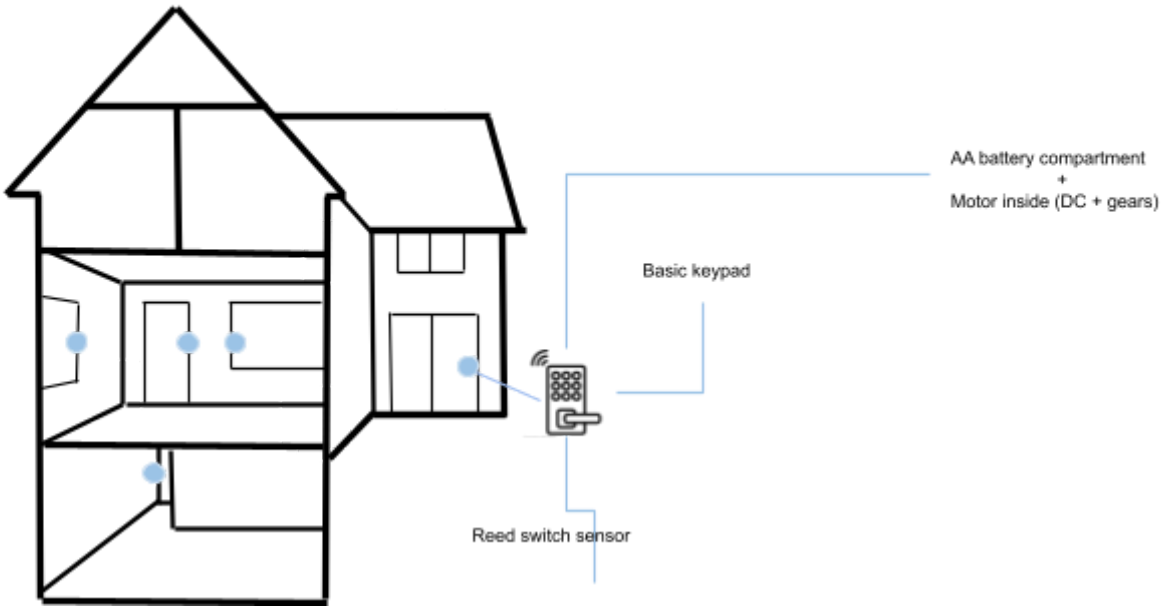
Strengths

- Lowest manufacturing cost
- Extremely power-efficient
- Very intuitive physical keypad

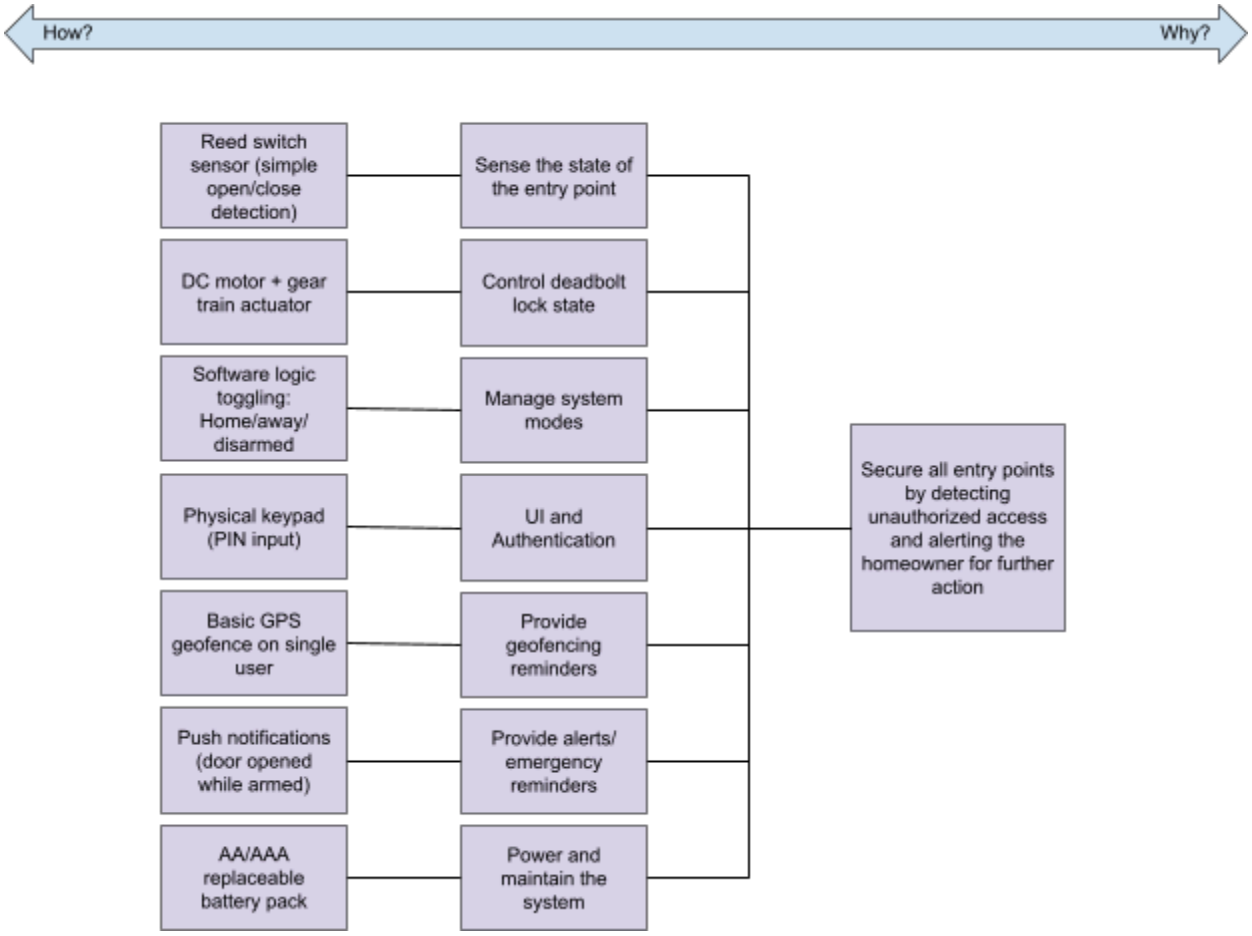
Weaknesses

- No multi-user geofencing
- No advanced connectivity options
- Batteries changed more often

Concept Sketch:



FAST diagram:



Concept 2: Touch Pad Lock

	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sensor	MEMS contact vibration sensor (detects opening as a contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actuator	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based on multi-user geofencing	Mode selection via lock keypad
UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on the lock face	App-based authentication (PIN/biometric)	Voice-assistant integration (Alexa/Google/Siri)
Provide geofencing reminders	Basic GPS geofence on a single user	Multi-user presence detection (last user leaving triggers a reminder)	Wi-Fi/Bluetooth presence override (detect if someone is still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed)	SMS or email alerts	"Call law enforcement" button in app	Audible lock alarm (local siren on tamper/forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optional low-voltage power	Solar trickle charger (for exterior units)
Internet connectivity	Wi-Fi	Mobile Data	/	/
Deadbolt + Lock material	Stainless Steel	Hardened Steel	Bronze	Brass
Micro Processor	SnapDragon	Samsung Exynos Chip	Custom ARM Chip	/
Emergency Power	Power Terminals	Lithium-ion Rechargeable Port	Solar Power	/

Design Logic: A sleek high-end product emphasizing user experience through touch controls and high-precision actuation.

Selected Solution Principles

- Sense entry status: Hall-effect sensor
- Control lock: Stepper motor
- Modes: App mode selection (with status animations)
- UI/Auth: Capacitive touchpad
- Geofencing: Multi-user presence detection
- Alerts: Push + SMS
- Power: Rechargeable Li-ion pack with a backup solar power option
- Connectivity: Mobile Data
- Processor: Custom Arm Chip
- Material: Bronze

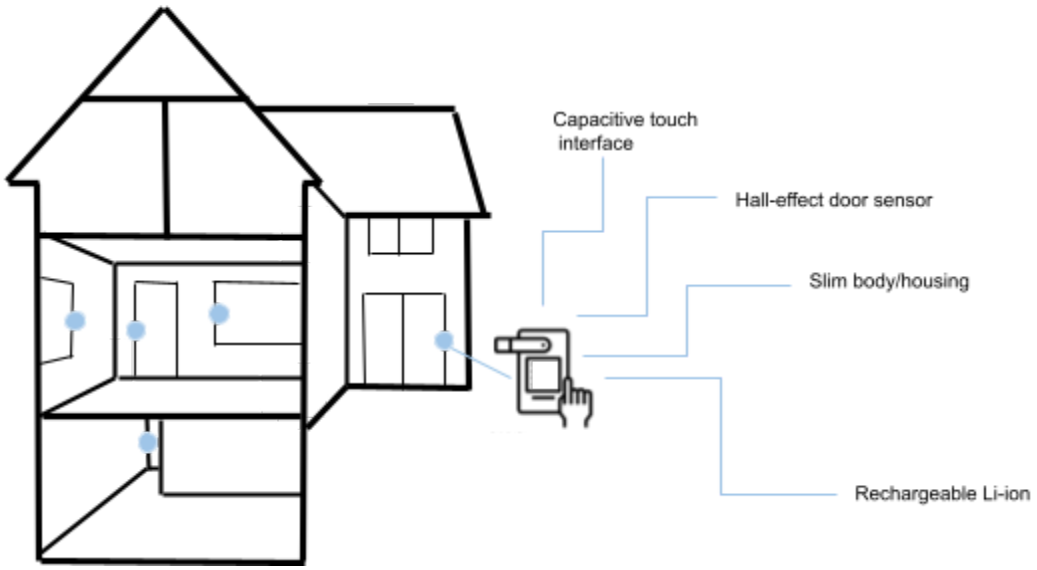
Strengths

- Premium look and feel
- Quiet and smooth motor actuation
- Accurate door-state detection

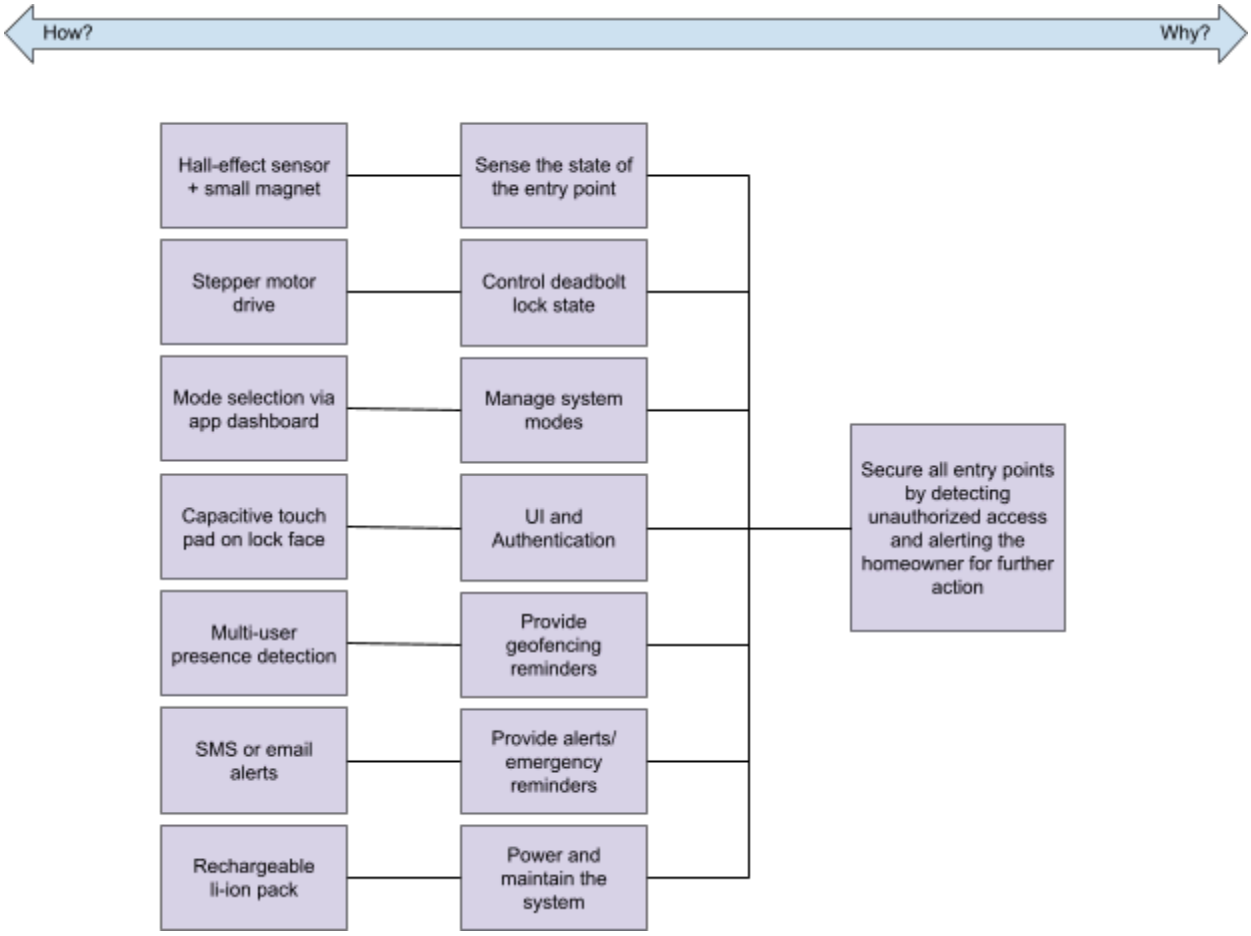
Weaknesses

- More expensive to produce
- Requires periodic recharging

Concept Sketch:



FAST diagram:



Concept 3: Fully Connected Smart Lock

Sub-Function	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sens	MEMS contact vibration sensor (detects opening as contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actu	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based multi- er geofencing	Mode selection via lock keypad
UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on lock face	App-b ed authe ation (PIN/biometric)	Voice-assistant integration (Alexa / Google / Siri)
Providing geofencing	Basic GPS geofence on single user	Multi-user presence detection (last user leaving triggers reminder)	Wi-Fi / Bluetooth prese override (detect if someone still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed)	SMS or email alerts	"Call law enfor ent" button in app	Audible lock alarm (local siren on tamper / forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optio low-voltage power	Solar trickle charger (for exterior units)
Internet Connectivity	Wi-Fi	Mobile Data	Ethernet (wired LAN)	Bluetooth-only (local-only control)
Microprocessor	SnapDragon	Samsung Exynos Chip	Custom ARM Chip	Low-power Microcontroller (MCU)
Deadbolt and Lock material	Stainless Steel	Harde d Steel	Bronze	Brass
Emergency Power	Power Terminals	Lithium-ion Rechargeable Port	Solar Power	Replaceable AA/AA Battery Backup

Design Logic: A connected lock that integrates deeper automation and multi-user presence detection for maximum convenience.

Selected Solution Principles

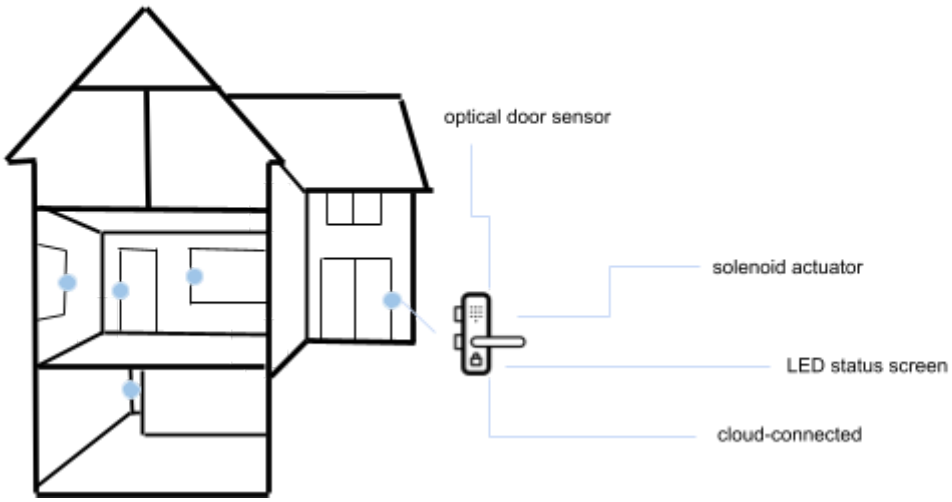
- Sense entry status: Optical interrupter
- Control lock: Solenoid bolt actuator
- Modes: Auto-arming via geofencing (multi-user)
- UI/Auth: App-based biometric authentication
- Geofencing: Multi-user + Wi-Fi presence detection
- Alerts: Push + SMS + email
- Power: Hard-wired low voltage (optional battery backup)
- Internet connectivity: Wi-Fi
- Lock material: Hardened steel
- Processor: Custom ARM chip
- Emergency power: External power terminals

Strengths

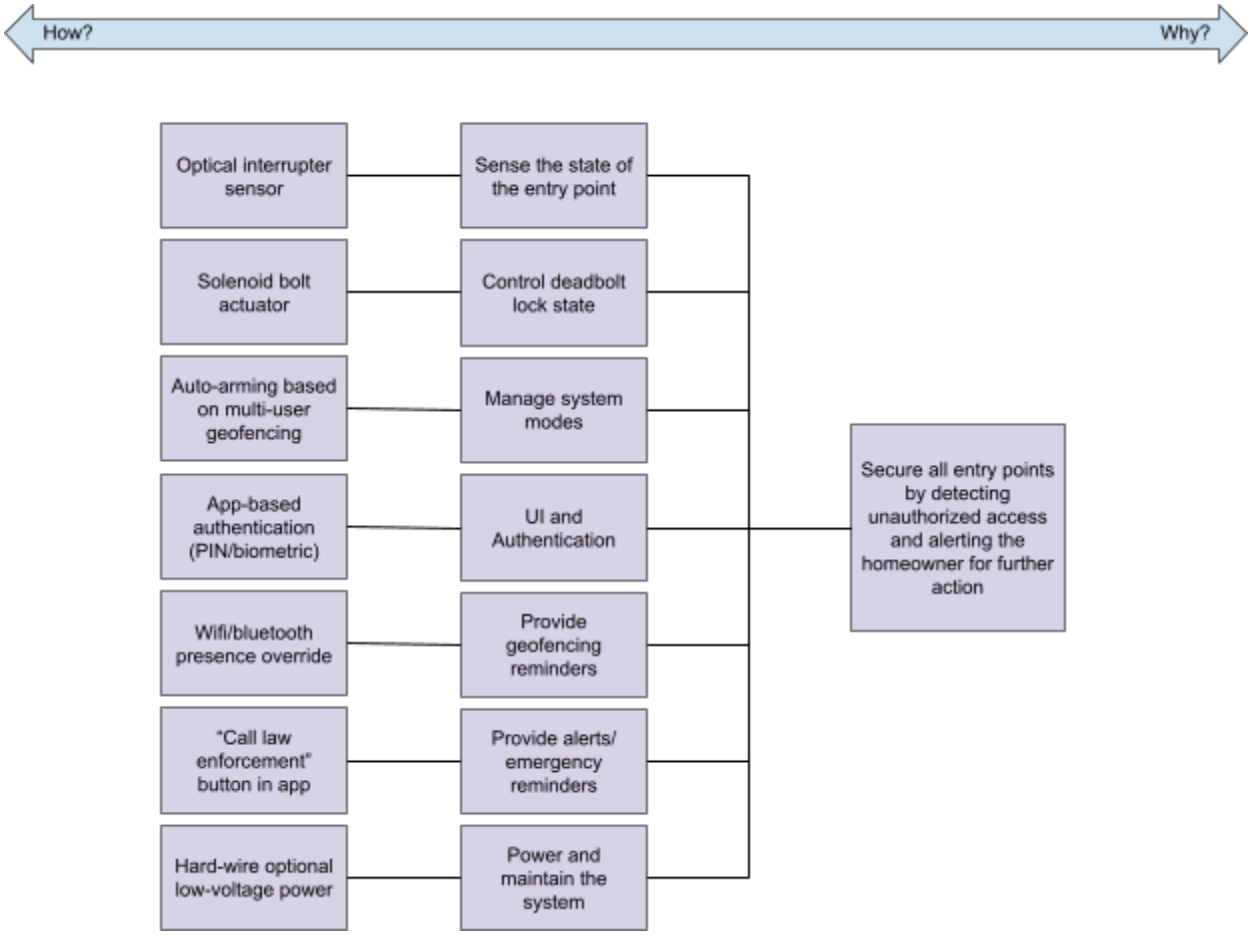
- Most accurate presence detection
- Fast lock actuation
- Very low maintenance due to wired power

Weaknesses

- Install complexity (needs wiring)
- A solenoid can be noisy compared to motors



FAST diagram:



Concept 4: Accessibility Voice First Lock

Sub-Function	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sensor	MEMS contact vibration sensor (detects opening as contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actuator	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based on multi-user geofencing	Mode selection via lock keypad
UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on lock face	App-based authentication (PIN/biometric)	Voice-assistant integration (Alexa / Google / Siri)
Providing geofencing	Basic GPS geofence on single user	Multi-user presence detection (last user leaving triggers reminder)	Wi-Fi / Bluetooth presence override (detect if someone still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed)	SMS or email alerts	"Call law enforcement" button in app	Audible lock alarm (local siren on tamper / forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optional low-voltage power	Solar trickle charger (for exterior units)
Internet Connectivity	Wi-Fi	Mobile Data	Ethernet (wired LAN)	Bluetooth-only (local-only control)
Microprocessor	SnapDragon	Samsung Exynos Chip	Custom ARM Chip	Low-power Microcontroller (MCU)
Deadbolt and Lock material	Stainless Steel	Hardened Steel	Bronze	Brass
Emergency Power	Power Terminals	Lithium-ion Rechargable Port	Solar Power	Replaceable AA/AAA Battery Backup

Design Logic: Optimized for accessibility and households wanting voice-controlled arming/disarming.

Selected Solution Principles

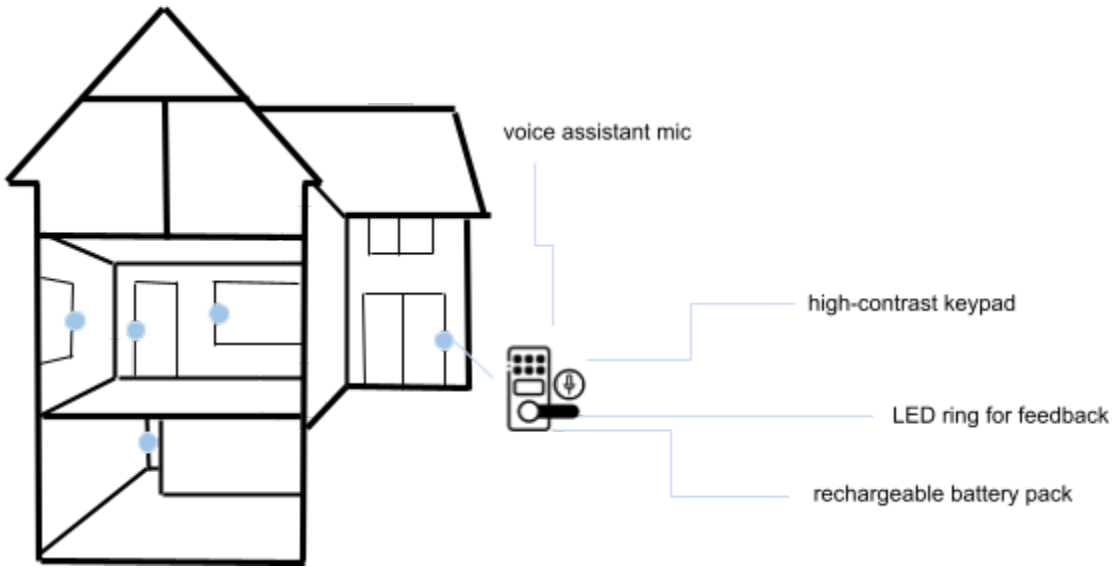
- Sense entry status: Reed switch
- Control lock: DC motor
- Modes: Voice-controlled mode selection (Home/Away)
- UI/Auth: Voice assistant + simple keypad backup
- Geofencing: GPS-based + arrival reminders
- Alerts: Push + in-home audible warnings
- Power: Li-ion rechargeable pack
- Internet connectivity: Wi-Fi
- Lock material: Brass
- Processor: SnapDragon
- Emergency power: Lithium-ion rechargeable port

Strengths

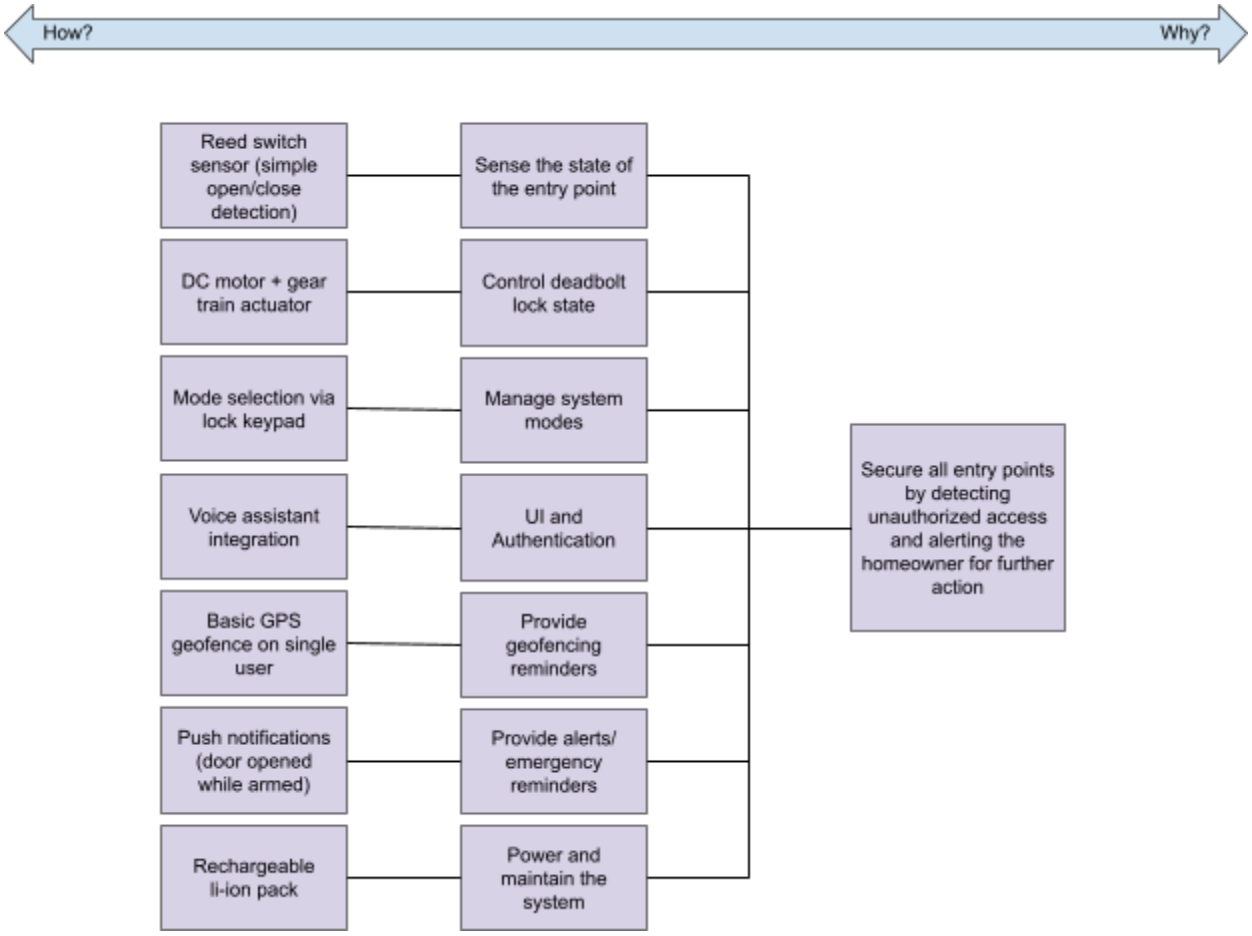
- Helps seniors, kids, or accessibility users
- Voice assistant reduces app dependency
- Reliable simple sensor

Weaknesses

- Voice assistants can be inconsistent offline
- Privacy/security concerns around voice control



FAST diagram:



Concept 5: Outdoor Optimized Solar Lock

	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sensor	MEMS contact vibration sensor (detects opening as a contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actuator	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based on multi-user geofencing	Mode selection via lock keypad
UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on the lock face	App-based authentication (PIN/biometric)	Voice-assistant integration (Alexa/Google/Siri)
Provide geofencing reminders	Basic GPS geofence on a single user	Multi-user presence detection (last user leaving triggers a reminder)	Wi-Fi/Bluetooth presence override (detect if someone is still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed)	SMS or email alerts	"Call law enforcement" button in app	Audible lock alarm (local siren on tamper/forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optional low-voltage power	Solar trickle charger (for exterior units)
Internet connectivity	Wi-Fi	Mobile Data	/	/
Deadbolt + Lock material	Stainless Steel	Hardened Steel	Bronze	Brass
Micro Processor	SnapDragon	Samsung Exynos Chip	Custom ARM Chip	/
Emergency Power	Power Terminals	Lithium-ion Rechargeable Port	Solar Power	/

Design Logic: A self-sustaining lock system for environments where battery replacement is difficult.

Selected Solution Principles

- Sense entry status: Hall-effect sensor
- Control lock: Stepper motor
- Modes: Mode selection via keypad
- UI/Auth: Capacitive touchpad
- Geofencing: Multi-user presence + Wi-Fi fallback
- Alerts: Push + local siren alarm
- Power: Solar trickle charger + Li-ion
- Internet connectivity: Wi-Fi
- Deadbolt + lock material: Stainless steel
- Microprocessor: Samsung Exynos chip
- Emergency power: Lithium-ion rechargeable port

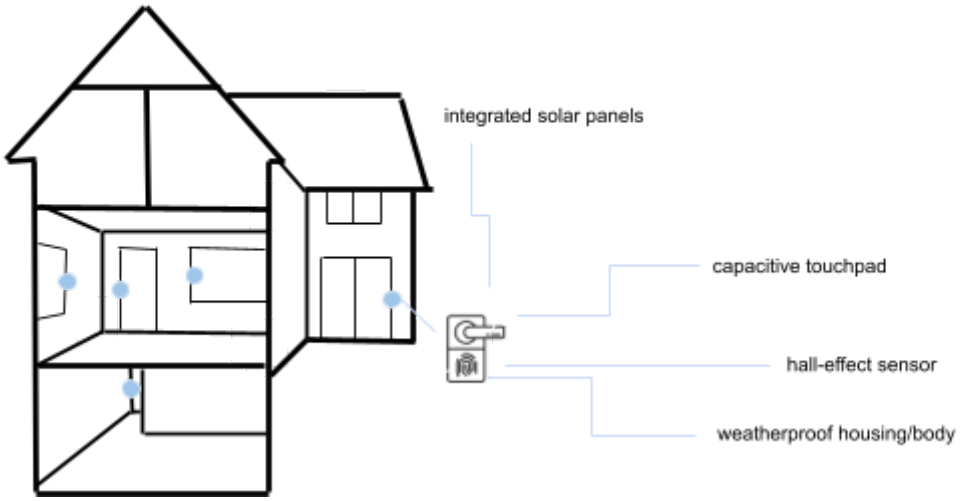
Strengths

- Nearly zero maintenance
- Ideal for rental properties, remote houses
- Strong presence detection
- Consists of a backup power supply

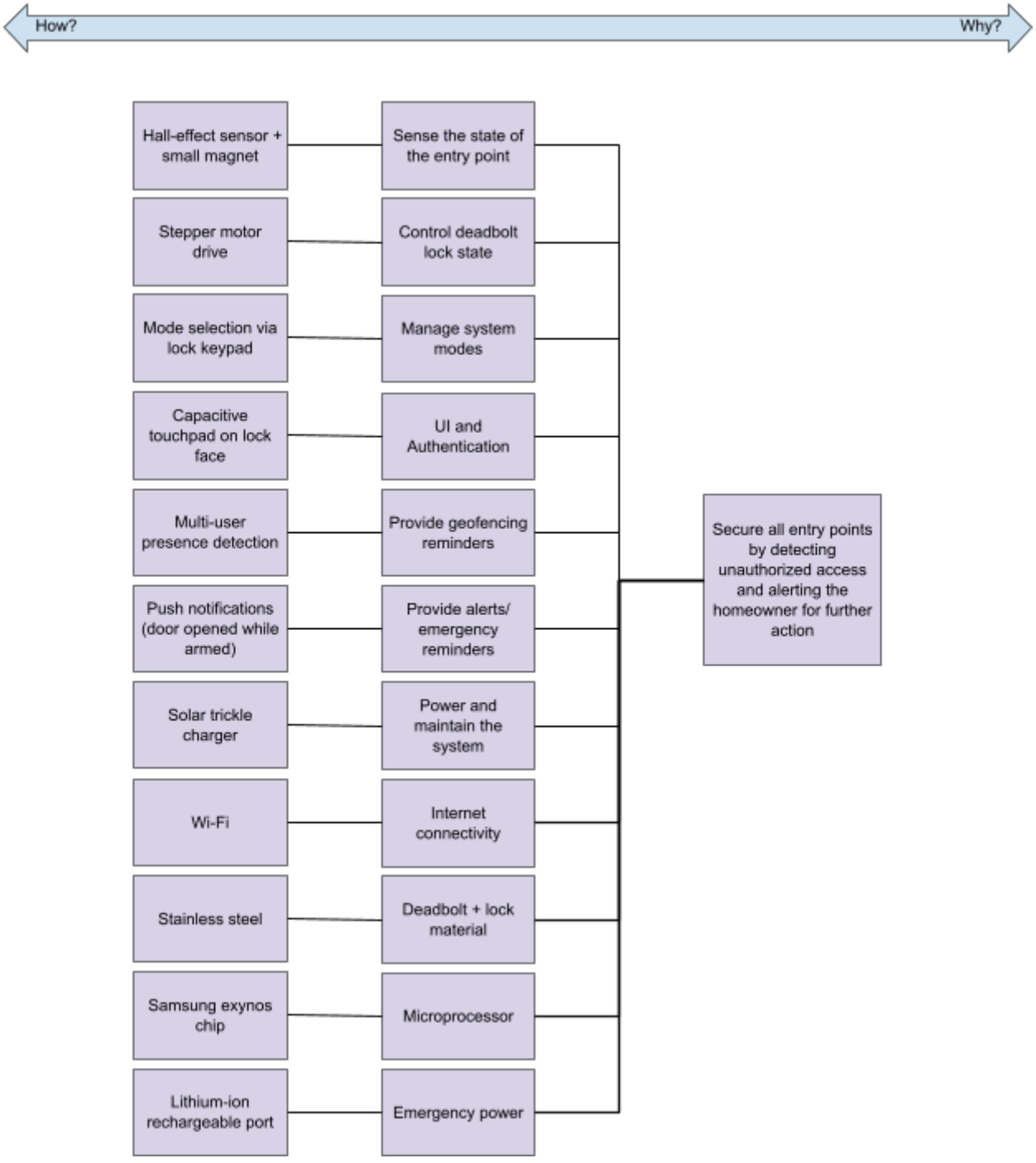
Weaknesses

- Needs sunlight exposure
- Slightly bulkier exterior housing

Concept sketch:



FAST diagram:



Concept 6: Emergency Ready Lock

	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sensor	MEMS contact vibration sensor (detects opening as a contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actuator	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based on multi-user geofencing	Mode selection via lock keypad
UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on the lock face	App-based authentication (PIN/biometric)	Voice-assistant integration (Alexa/Google/Siri)
Provide geofencing reminders	Basic GPS geofence on a single user	Multi-user presence detection (last user leaving triggers a reminder)	Wi-Fi/Bluetooth presence override (detect if someone is still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed)	SMS or email alerts	"Call law enforcement" button in app	Audible lock alarm (local siren on tamper/forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optional low-voltage power	Solar trickle charger (for exterior units)
Internet connectivity	Wi-Fi	Mobile Data	/	/
Deadbolt + Lock material	Stainless Steel	Hardened Steel	Bronze	Brass
Micro Processor	SnapDragon	Samsung Exynos Chip	Custom ARM Chip	/
Emergency Power	Power Terminals	Lithium-ion Rechargeable Port	Solar Power	/

Design Logic: Emphasizes rapid alerts, tamper detection, and fast emergency escalation.

Selected Solution Principles

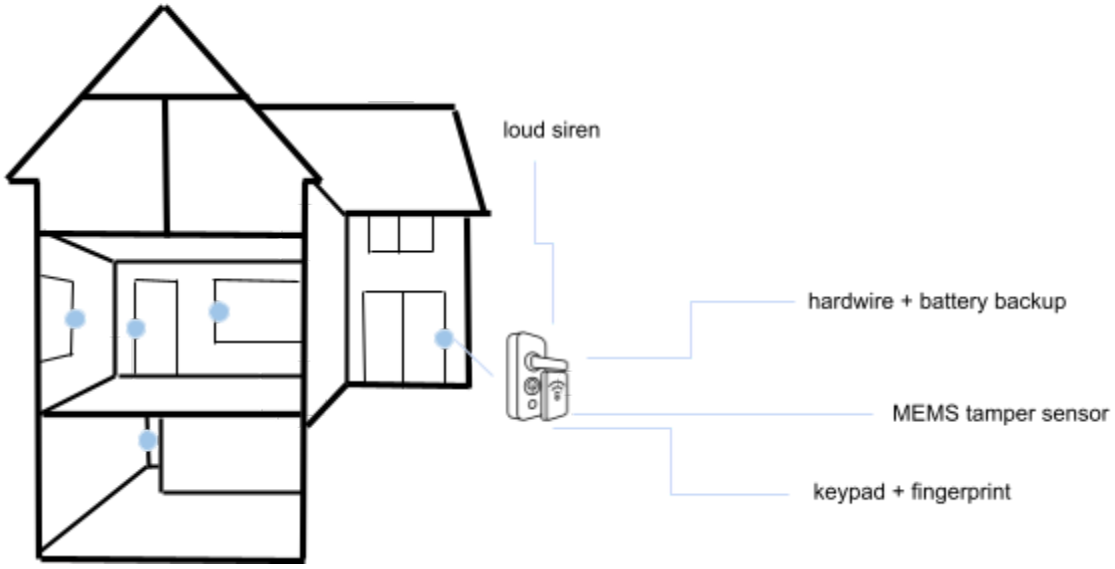
- Sense entry status: MEMS vibration/contact sensor (detects forced entry attempts)
- Control lock: Stepper motor
- Modes: App-based modes with emergency override
- UI/Auth: Physical keypad + app biometric
- Geofencing: Multi-user presence detection
- Alerts: Push + SMS + “Call law enforcement” button + loud siren
- Power: Hard-wire option + battery backup
- Connectivity: Mobile Data
- Processor:Samsung Exynos Chip
- Emergency Power: Lithium-ion rechargeable Port

Strengths

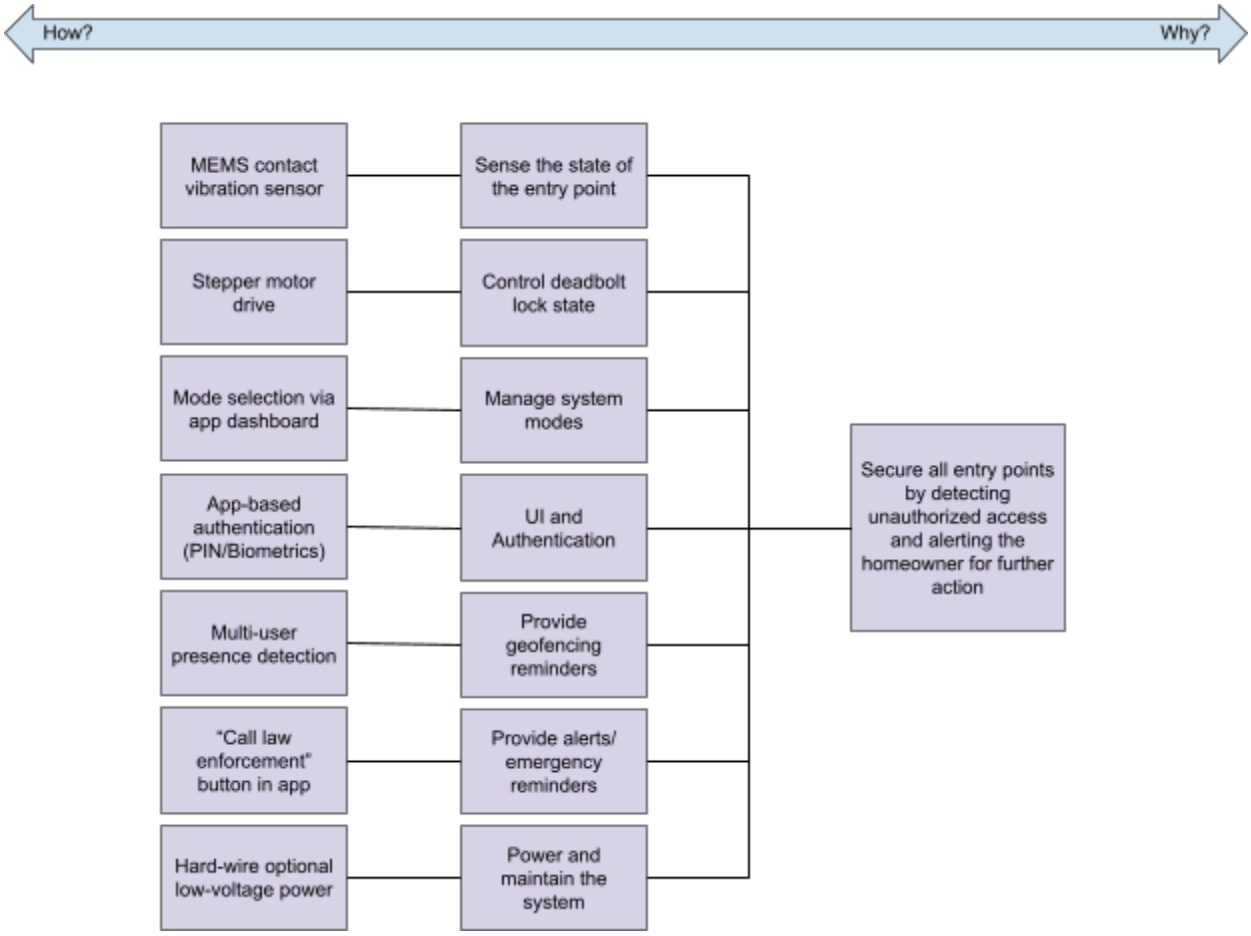
- Best anti-tamper capabilities
- Strongest emergency features
- Redundant power options

Weaknesses

- Higher cost
- MEMS sensors require calibration



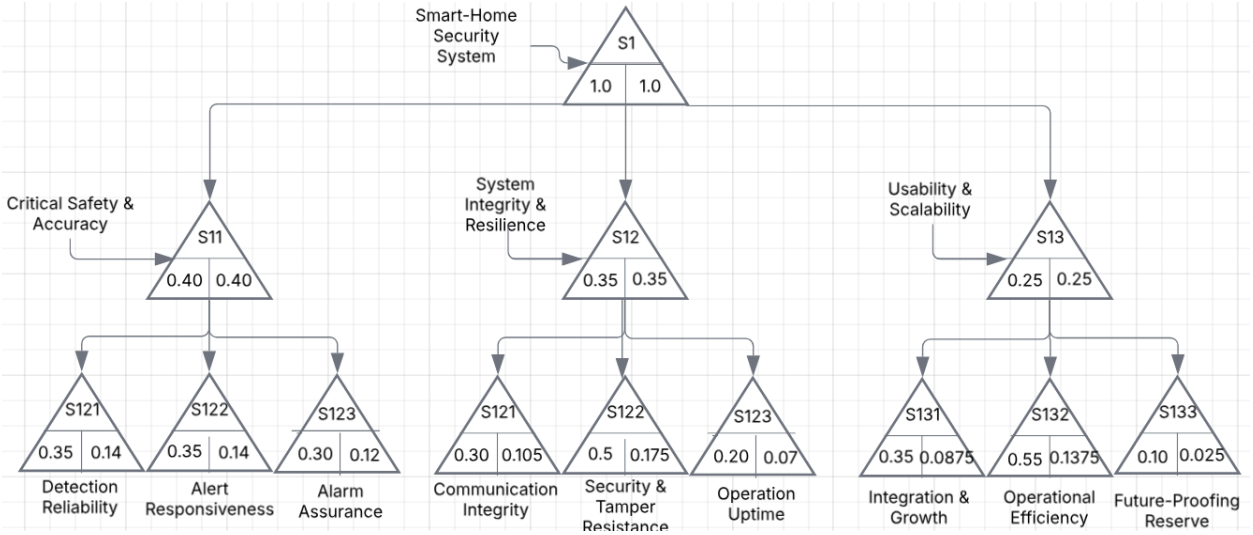
FAST diagram:



Utility Function

The purpose of the Utility Function is to help provide us with a structured, quantitative, and unbiased framework for evaluating and selecting the most viable product concept from the set of alternative concepts we created. It helps translate customer needs into more measurable criteria, allowing for a data-driven decision-making process that isn't just subjective.

- **Objective Comparison:** Allows for a direct comparison of fundamentally different design concepts against a common criterion.
- **Traceability for Customer Needs:** Each criterion and its weight are based on the HOQ, allowing for a selection that is traceable to the customer's needs.
- **Identification of Trade-offs:** The scoring process defines the strengths and weaknesses of each concept, helping identify the critical design compromises related to cost, performance, and feasibility.



	Selection Criteria	Absolute Weight	Concept 1		Concept 2		Concept 3		Concept 4		Concept 5		Concept 6	
			Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility	Concept Rating	Utility
Detection Reliability	S111	0.14	3	0.42	3	0.42	4	0.56	3	0.42	3	0.42	4	0.56
Alert Responsiveness	S112	0.14	1	0.14	2	0.28	2	0.28	2	0.28	3	0.42	3	0.42
Alarm Assurance	S113	0.12	1	0.12	2	0.24	3	0.36	2	0.24	3	0.36	3	0.36
Communication Integrity	S121	0.105	1	0.105	3	0.315	1	0.105	2	0.21	1	0.105	2	0.21
Security & Tamper Resistance	S122	0.175	2	0.35	2	0.35	3	0.525	4	0.7	3	0.525	3	0.525
Operation Uptime	S123	0.07	2	0.14	2	0.14	3	0.21	2	0.14	4	0.28	3	0.21
Integration & Growth	S131	0.0875	2	0.175	2	0.175	3	0.2625	3	0.2625	2	0.175	3	0.2625
Operational Efficiency	S132	0.1375	2	0.275	2	0.275	3	0.4125	2	0.275	3	0.4125	3	0.4125
Future-Proofing Reserve	S133	0.025	2	0.05	2	0.05	3	0.075	4	0.1	2	0.05	3	0.075
Total Utility				1.775		2.245		2.79		2.6275		2.7475		3.035

Based on the utility function, Concepts 3 and 6 are feasible to pursue development. Concept 6 prioritizes the detection system and alert response while optimizing the usage of power. This concept has the highest feasibility with a high rating of 3.035 out of 5.

Concept 3 is more well structured and rounded, with its only main weak point being its method of communication. This concept has the second-highest feasibility, with a high rating of 2.79 out of 5.

Emergency Ready Lock Concept

The Emergency Ready Lock (Concept 6) is a security system designed to prioritize rapid alerts, tamper detection, and fast emergency escalation. Unlike standard smart locks, it utilizes vibration sensors to detect forced entry attempts, such as hammering or drilling. This allows the product to detect an early intrusion, before a bad actor enters a home. The system is built for reliability and redundancy, featuring a stepper motor for lock control, a loud local siren to deter intruders. It also mitigates power failure risks by offering a hard-wired connection with a battery backup. Its superior anti-tamper capabilities and strong emergency features made it the most well-rounded and viable candidate for final implementation. It is complemented by the mobile application integration, found in the next section.

Mobile Application Integration

Application name: “Home Security By Honeywell.”

The purpose of a mobile application is to provide customers with basic functionality and information about their smart home security system from their smartphone. Competitors all provide a mobile application with similar functionality.

Application Logo



The logo is a lock icon that is blue with the application name. This combination of elements is associated with trust, loyalty, and security.

Home Screen

In the home screen, users can enable the security system and disable the security system by tapping the lock icon (see illustrations below) in the home screen menu and see the status of the home security system. This screen also features a button to call local law enforcement to your home, available regardless of whether the security system is enabled or disabled.

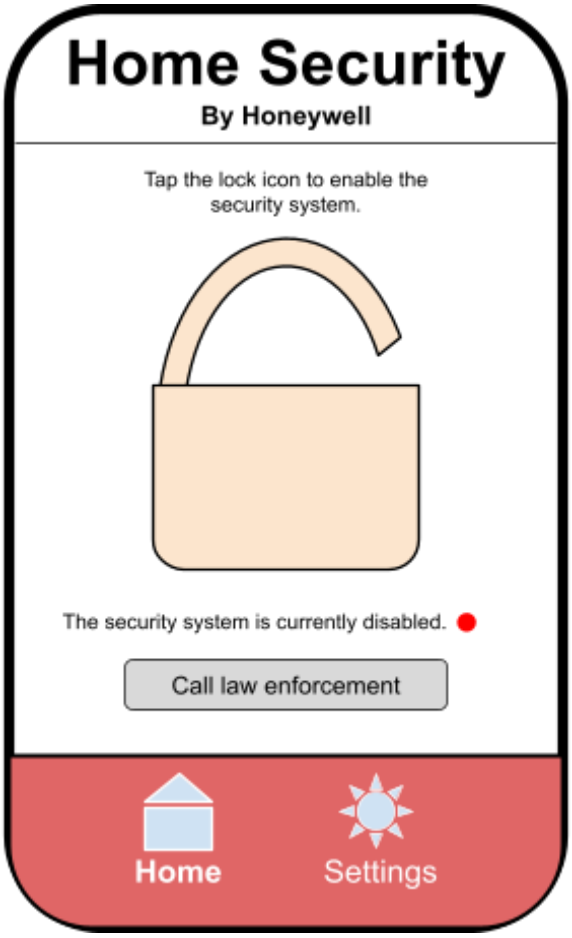
The application color theme will correspond accordingly:
→ If the system is enabled, the app theme will be blue, signalling trust and security.
→ If the system is disabled, the app theme will be red, signalling distrust and vulnerability.

Home screen when the security system is enabled:



The navigation bar will be "blue" when the system is enabled.

Home screen when the security system is disabled:

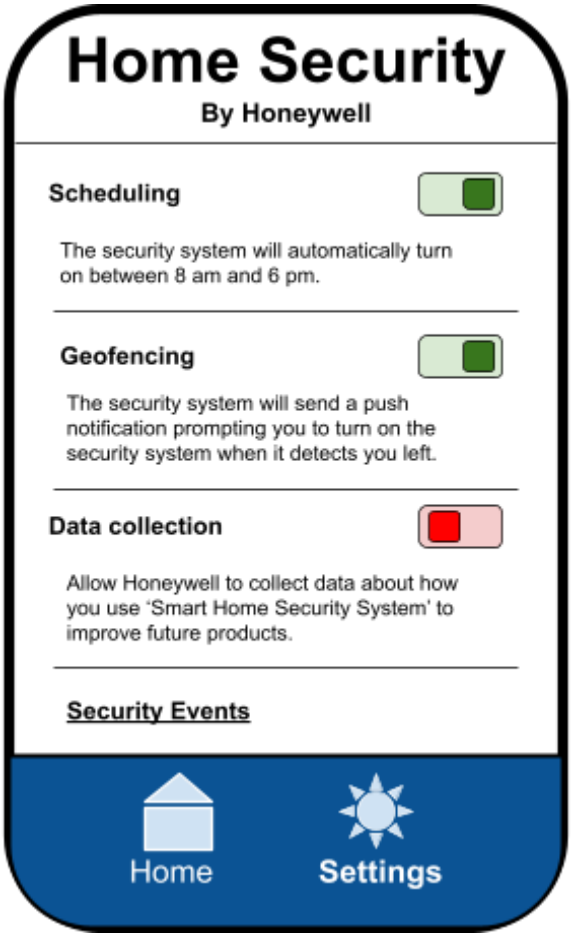


The navigation bar will be "red" when the system is disabled.

Settings Screen

In the settings menu, users can automatically enable the home security system through scheduling and geofencing to increase the amount of time the home is secured. By default, scheduling will enable the security system between 8 am and 6 pm, which is common working hours. The geofencing feature will work by identifying when the user leaves their home using the phone's GPS signal. Users will also be able to view security events, which include when a door or window has been previously opened or closed.

Settings page:



6. Conclusion of Phase III

We have now successfully established a robust conceptual design for Honeywell's new Smart Home Security System by iterating through FAST, HOQ, and CDP. After critically reviewing an initial implementation that lacked mobile integration and detailed sub-functions, the team developed six distinct product concepts, ultimately selecting the "Emergency Ready Lock" (Concept 6) as the most viable solution due to its superior utility score of 3.035 and emphasis on detection reliability, tamper resistance, and emergency responsiveness.

The final design proposal is further strengthened by a newly conceptualized mobile application, "Home Security By Honeywell," which ensures intuitive remote management, geofencing capabilities, and direct emergency escalation, positioning the product to effectively meet modern consumer security demands.

7. Revised Phase III Table

Issues needed to be addressed	Changes to address the issue	Team Member
Executive summary and introduction were not explicitly labeled	Revising the heading to be an executive summary.	Jesus Ortiz Rivas
HOQ details were placed in the main body instead of the appendix	Moved HOQ detailed to the appendix.	Mike Melendez
Critical review appeared to be at the end of the report instead of between the initial and final implementations	Condensed critical review between the initial and final implementations.	Jesus Ortiz Rivas Mike Melendez

Project Conclusions and Next Steps

The entire project successfully simulated a research and development cycle, going from market identification to a finalized conceptual design for Honeywell.

Specific and Meaningful Conclusions:

- Strategic Differentiation: Honeywell's strategy in the Smart Home Security market differentiated through high reliability and system integration, rather than competing on low cost. This approach uses the company's existing scale, technological expertise, and sensor portfolio.
- Core Product Design: The conceptual design phase resulted in the selection of the "Emergency Ready Lock" (Concept 6) as the most viable solution, achieving the highest utility score. The product's core guidelines are to prioritize:
    - Detection reliability: Using advanced components like MEMS vibration sensors for anti-tamper detection
    - Safety and Redundancy: Incorporating redundant power systems and stepper motor actuation to ensure fail-safe operation and emergency responsiveness
- Critical Engineering Focus: The HOQ analysis revealed that the foundation of customer trust depends on reliable security and minimal false alarms. So, the most critical engineering characteristics for the proposed system are maximizing sensor accuracy and wireless signal reliability.
- Software Ecosystem: The product's success depends on a robust, user-centric software platform. The conceptualized "Home Security by Honeywell" mobile application is essential for providing remote management, geofencing capabilities, and direct emergency escalation, effectively positioning the product to meet modern customer security demands

Next Steps (Future Work)

- The group's projection for the next steps focuses on transitioning the conceptual design into a market-ready product, requiring actions related to prototyping (product platform and product lines), testing, financial modeling, etc.
- Prototyping: Product Platform and Product Lines
    - Product platform: Creating a modular internal core that can be reused across different external product components and functionalities. Core would include the MEMS sensor module, redundant battery system, Z-wave/Wifi communication chip, and stepper motor logic.
    - Product Lines: Some possible product lines are: A base "Emergency Ready Lock" for common residential use. A higher-end version with enhanced features like facial recognition, biometric fingerprint access, etc. A cost-optimized lock that

retains the core security and remote management capabilities suited for property managers.

- Testing and Validation:
    - Would include the core value proposition derived from the HOQ
    - Rigorously test the fail-safe mechanisms to ensure the lock can always be opened in an emergency
    - Subject the lock's mechanical components to a lifecycle test to ensure a minimum expected lifespan under heavy use
    - Test the lock's compatibility with a wide range of existing smart home protocols to ensure maximum market reach
    - Test on wireless communication protocols and measure end-to-end latency to ensure a rapid response time
- Financial Modeling:
    - Necessary to ensure profitability and resource allocation.
    - Finalize the cost of every component, like the MEMS sensors, stepper motor, casing materials, etc.
    - Determine the maximum allowable cost of goods sold and identify areas for cost reduction without compromising quality.
    - Model the required investment in equipment for the chosen manufacturing location.

Project Appendix

1. Project Proposal Appendix

Initial Company List

To arrive at our final three companies, the team conducted its own research and brainstormed a list of ten potential companies. These were evaluated against our constraints and innovation potential.

Company	Reason for Consideration
Thermo Fisher Scientific	Produces analytical instruments and lab equipment with cloud-based data platforms
Honeywell	Diverse industry presence; broad scope for tech integration and worker safety applications
Stryker	Medical technology innovator with scope for automation
General Electric	Leader in energy and industrial automation
Zebra Technologies	Tech company focused on data capture and automation for logistics and healthcare
3M	Manufacturer of industrial and consumer products with strong innovation capability
Johnson & Johnson	Major player in robotic medical technologies
Ford	Vehicle manufacturer exploring EV and other technologies, with room for safety and accessibility innovation
Garmin	Expert in GPS and wearable technology
Whirlpool	Leader in smart home appliances

2. Phase I Appendix

Integration and Meeting 2 Notes

During meeting 2, our team met to integrate and finalize all individual components of the phase 1 assignment. Each member shared their ideas for the company and industry analysis, then we reviewed how they aligned with the SPSP and Porter's framework. We planned the structure to ensure a cohesive report and made adjustments where sections overlapped.

- After reviewing each member's sections, the team collaboratively merged all content into a single document. We used a shared file to edit in real time, ensuring consistent formatting, tone, and framework alignment.
- Each member cross-checked data and citations and updated the tables for visual appeal.
- The group confirmed that the company and industry analysis connected logically before assigning final editing and proofreading roles.

Sources

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"Investor Relations." *Honeywell Investor Relations*, Honeywell International Inc., n.d., https://investor.honeywell.com/investor-relations.

3. Phase II Appendix

Integration and Meeting Records
Meeting 1 Summary: Conducted on October 31, 2025. The team shared individual product analyses and finalized the plan for FAST and HOQ development. Key outcomes included competitor selection, role confirmation, and agreement on deliverables and deadlines.

Meeting 2 Summary
Conducted on November 2, 2025. Members reviewed individual FAST diagrams, consolidated them into the unified product FAST, which captured the most important functions, and completed the HOQ matrix. Peer review focused on consistency, logic, and clarity of the relationship. Final edits and validation were completed collaboratively in Google Docs.

Meeting 3 Summary

Conducted on November 6, 2025, the team met to review the feedback provided by Professor Desa and TA John. We incorporated their critiques directly into our annotated document, using them to refine our research direction and determine key areas for improvement. In response, we expanded our discussion of pre- and post-processing considerations to more clearly convey our design rationale and analytical approach. We also assigned tasks equally and across every section to ensure that we would have adequate work done before the next planned meeting.

Meeting 4 Summary

Conducted on November 10, 2025. We reviewed previously assigned tasks and conducted a peer review to determine which sections to retain or remove. We discussed improvements to the overall report, agreeing on which areas should be expanded or refined based on each member's contributions. Additionally, we checked for grammatical inconsistencies and data errors to ensure the document maintained clear and professional formatting.

Sources

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4. Phase III Appendix

Meeting 1 Summary: Framework Development and Initial Implementation

Meeting Date: 11/19/25
Location: SNE

Meeting Objectives: Realize the team's shared design process

1. Establish Primary Functions of the Smart Home System
2. Begin initial implementation of HOQ
3. Update FAST analysis for competitive products
4. Generate initial product concepts

The team adopted a framework and finalized the primary function, to reliably secure all household entry points by detecting unauthorized access and alerting the homeowner for further action. A hierarchical customer needs structure was created spanning primary (secured entry points), secondary (detect access, alert homeowner, smart-home integration), and tertiary (minimize false alarms, emergency response, easy installation) levels. Three competitive products were selected: SimpliSafe Smart Lock ($99-$120), Ring Alarm Pro ($799.94), and Ecobee Smart Security System (~$200-$600). Initial technical requirements included door/window sensor actuation, system uptime, false alarm, and battery life.

Meeting 2 Summary: Fast Review & Morphological Analysis

Meeting Date: 11/21/25
Location: SNE

Meeting Objectives:

1. Review FAST diagrams for competitive products
2. Finalize HOQ with benchmarking
3. Develop a morphological matrix
4. Generate six alternative product concepts

FAST analysis examined SimpliSafe Smart Lock ($99-$120), Ring Alarm Pro ($799 with smart lock), and Ecobee Smart Security System ($200-$600). Analysis revealed SimpliSafe lacks context-aware detection, Ring has high complexity, and Ecobee offers the best event recognition. The market gap identified: no product combines SimpliSafe's affordability with Ring's multi-sensor sophistication and Ecobee's event awareness. Honeywell's existing sensor expertise provides a competitive advantage. HOQ analysis determined that sensor precision drives top customer needs, event accuracy is critical but difficult to achieve, and target MSRP should range from $249-$279. A morphological matrix was developed with five sub-functions (sense entry point, process signal, determine state, communicate status, power system) and four solution principles each. Six alternative concepts were generated: Optical Detection LE, Magnetic Detection LE, Hall Effect Detection, Reed + Wi-Fi Cloud, Mechanical Microswitch (Wired), and Reed + Zigbee Mesh. A utility function framework was established with three weighted criteria levels: Critical Safety (0.40), Core Performance (0.35), and Usability & Lifecycle (0.25). Key

gaps identified for refinement: mobile app UX design, expanded morphological matrix with keypad and lock control options, standardized FAST diagram detail levels, solution-neutral function structure diagram, improved concept sketches, and use case validation.

Meeting 3 Summary: Refinements and Concept Selection

Meeting Date: 11/22/25
Location: SNE

Meeting Objectives:

1. Address Gaps/Problems from initial implementation and meeting 1 & 2
2. Refine CDP outputs
3. Apply Utility Function scoring
4. Select 2 concepts for phase 4

The team addressed identified gaps by creating mobile app mockups (home screen with arm/disarm toggle, settings with scheduling/geofencing), expanding the morphological matrix to 7 sub-functions with lock control and UI options, standardizing FAST diagrams, creating a function structure diagram, developing professional concept sketches, and creating a use case scenario. Utility function scoring ranked Concept 5 (Outdoor Optimized Solar Lock) and Concept 6 (Emergency Ready Lock) as the top two performers, both selected for Phase 4 advancement. Concept 5 features a Hall-effect sensor, a stepper motor, and solar+Li-ion power for maintenance-free operation targeting property managers. Concept 6 features a MEMS vibration sensor, redundant power (hard-wire + battery), and emergency response capabilities for security-conscious homeowners. These two concepts represent distinct market strategies and demonstrate Honeywell's ability to serve diverse customer needs while leveraging core sensor and lock technologies. Key technical risk identified: event accuracy for false alarm reduction requiring resource allocation in Phase 4.

Meeting 4 Summary: Final Iteration + Revisions

Meeting Date: 11/24/25
Location: SNE

Meeting Objectives:

1. Final Review of all Phase 3 Work
2. Confirmations of improvements between iterations
3. Verification of concept selection

The team conducted a comprehensive review of all Phase 3 work, confirming that the conceptual design process successfully evolved through two distinct iterations. Initial implementation produced six alternative concepts derived from a morphological matrix with 5 sub-functions, while final implementation expanded to 7 sub-functions and incorporated critical refinements including mobile app UX design, standardized FAST diagrams, function structure, professional concept sketches, and validated use case scenarios. Key improvements realized between

iterations included the addition of mobile app mockups addressing a critical customer need, expansion of the morphological matrix to include lock control mechanisms and multiple UI/authentication options, standardization of competitive product analysis across all three FAST diagrams, and quantitative utility function scoring replacing subjective concept evaluation. The team verified that Concepts 5 and 6 selection was data-driven and well-justified, representing two distinct market segments (maintenance-focused property managers and security-conscious homeowners) that leverage Honeywell's core sensor and lock technologies. Final House of Quality analysis confirmed target MSRP of $249-$279 aligns with top-of-the-line sensor investment strategy, and critical success factors were identified: sensor precision, event accuracy for false alarm reduction, and intuitive mobile app UX. The team documented all deliverables, confirmed participation requirements were met, and established that Phase 4 should prioritize ML model development for false alarm reduction and detailed CAD design for selected concepts.

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5. Final Report Appendix

Collaboration

Our group's work process can best be demonstrated by our efforts in Phase 2, particularly in creating the House of Quality and the structured steps leading up to it. This phase effectively displayed how our unified process enabled us to translate individual preparations into a cohesive team analysis. Each member began by independently researching related products, completing FAST diagrams, and compiling both customer and technical needs. Once we reconvened as a group, we agreed upon a unified workflow and a consolidated structure for completing FAST diagrams, function structures, and the HoQ, in which our best insights and inputs would be incorporated. We first finalized a set of related products to reverse engineer and confirmed that they reflected market expectations relevant to our Smart Lock System. We then used these to create a standardized FAST structure aligned with our agreed-upon definitions. We then made the accompanying customer needs list, technical requirements, and determined the benchmarking criteria and correlation rules necessary to create the HOQ. With the assignment of roles and responsibilities, we were able to further ensure a supported and efficient workflow that allowed for equal member contributions throughout the process. This collaborative approach helped us effectively identify and synthesize all the information necessary to produce a fully developed HOQ.

Despite our strong process, we identified several areas of improvement. Although we followed the unified process closely, we occasionally struggled with keeping our metrics terse and did not always explore potential edge cases. As a result, certain sections of our report were less detail-oriented and precise than intended. Moving forward, we would benefit from incorporating a dedicated time to clarify assumptions and refine metrics before integrating them into the final diagrams and tables. Overall, the HoQ task illustrates our group's ability to collaborate effectively using a structured and repeatable design process. We demonstrated alignment on shared methodologies, consistent terminology, and coordinated division of tasks. What could be improved is the thoroughness of our early-stage metric refinement and our proactive identification of gaps before submission. Ultimately, this experience strengthened our team's ability to approach complex problems systematically, allowing us to lay a solid foundation for future phases of this project.